

暨南大学博士学位论文

题名（中英对照）：大型海藻龙须菜栽培生态系统微生物群落结构及功能研究

The structure and function of microbial communities in a seaweed cultivation ecosystem of *Gracilaria lemaneiformis*

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摘 要

海洋微生物多样性高,分布广泛,既是初级生产者又是分解者,与浮游植物、浮游动物共同组成“微食物环”,在海洋生态系统物质运输与能量传递中起到承上启下作用,是参与海洋生物地球化学循环不可或缺的一环。因此,了解海洋微生物的多样性、群落结构及其功能显得尤为重要。已有的研究主要集中在自然海域微生物的群落特征,对海水养殖区尤其是大型海藻栽培区微生物群落特征所知甚少。为揭示大型海藻栽培区微生物群落结构及功能,本研究以广东省汕头市南澳岛龙须菜栽培区和相邻自然海区(对照区)为调查海域,通过采集不同季节、不同水深的样品,提取 DNA 并进行 16S rRNA MiSeq 高通量测序及功能基因芯片(GeoChip)杂交,比较了大型海藻龙须菜栽培区和对照区微生物群落差异,分析了龙须菜体表附生微生物组成及功能,旨在探讨龙须菜规模栽培对微生物群落的影响及其分子机理。所获结论对海洋环境保护和生物资源可持续利用具有重要科学意义。

主要研究结果如下:

1. 于 2014 年 4 月,在南澳岛深澳湾龙须菜栽培区和对照区各设置 3 个采样点。每个采样点均采集表层水体、底层水体和沉积物,提取 DNA 并纯化后,进行 16S rRNA 基因 MiSeq 测序。结果显示,18 个样品共产生超过 530,000 条高质量序列,生成 3454 个 OTU (Operational Taxonomic Units),分别属于 28 个门,74 个纲,134 个目,275 个科,618 个属。群落差异性检验(adonis)结果表明,龙须菜栽培区和对照区表层水体微生物群落结构差异不明显,但底层水体和沉积物中微生物群落结构差异显著($p < 0.05$)。在表层和底层水体中,拟杆菌、 α -变形菌和 γ -变形菌均为优势菌,其相对丰度在栽培区和对照区中有差异。沉积物中优势菌为拟杆菌、 γ -变形菌和 ϵ -变形菌。在属水平上,栽培区表层水体中 *Aestuariibacter*, *Crocinitomix*, *Leucothrix* (亮发菌属)及 *Tenacibaculum* (黄杆菌属) 的相对丰度高于对照区($p < 0.05$)。底层水体中,栽培区 *Aestuariibacter*, *Bacillariophyta* (硅藻门) 和 *Croceitalea* 的相对丰度高于对照区 ($p < 0.05$)。沉积物中, *Arenibacter*, *Glaciecola*, *Halomonas* (盐单胞菌属), *Leisingera* (亮发菌属), *Maribacter*,

Peredibacter 和 *Sulfitobacter* (亚硫酸杆菌)在龙须菜栽培区的相对丰度高于对照区 ($p < 0.05$)。大型海藻栽培区相对丰度高的微生物种属与利用海藻多糖的微生物密切相关,例如 *Arenibacter*, *Croceitalea*, *Glaciecola*, *Leucothrix* 和 *Maribacter*。响应比分析表明,栽培区表层和底层水体中显著变化的 OTU 分别有 89.9%和 72%增加了相对丰度;栽培区沉积物中显著变化的 OTU 有 84.5%降低了相对丰度。CCA 结果显示,栽培区水体微生物群落结构变化与温度、DO、pH 和 Chl-a 相关,而对照区水体微生物群落结构变化与盐度和总氮密切相关。Pearson 相关性分析结果表明,表层及底层水体中显著变化的 OTU 与温度、pH 和 DO 正相关($p < 0.01$),与盐度和总氮负相关($p < 0.01$)。沉积物中显著变化的 OTU 与总氮、总硫正相关($p < 0.01$),与总碳负相关($p < 0.01$)。对上述结果综合分析,提出了龙须菜栽培对微生物影响的概念模型:龙须菜栽培可以提高水中溶解氧,降低营养盐,从而改善水质,对微生物群落结构演变产生影响。该模型为研究大型海藻与微生物相互作用奠定了科学基础。

2. 为进一步研究龙须菜栽培对微生物群落结构的影响,于 2014 年 12 月至 2015 年 7 月在龙须菜栽培区和对照区进行每月一次的连续采样调查,对微生物样品进行 16S rRNA 基因 MiSeq 测序,探讨龙须菜生长对微生物群落动态变化的影响。利用 adonis 进行群落差异性分析,结果发现采样时间、龙须菜生物量变化及水深均对微生物群落结构有显著影响($p < 0.01$);同样,龙须菜栽培、采样时间对不同水层微生物群落结构亦有显著影响($p < 0.05$)。Adonis 分析发现,龙须菜栽培区与对照区水体微生物群落结构在 1-2 月($p < 0.05$)和 4-5 月($p < 0.01$)均有显著差异,尤其是 1 米深和底层水体。NMDS 分析结果表明,微生物群落不同月份形成不同聚类,说明龙须菜栽培时间导致微生物群落结构差异。利用线性回归分析 α -多样性与栽培时间的关系,发现栽培区表层水体微生物群落随着龙须菜栽培生物量的增加而降低,其余水层及对照区均无线性关系。而且,微生物群落 α -多样性与龙须菜生物量呈负相关,说明龙须菜对栽培区表层水体微生物群落具有一定的选择作用,使得栽培区的微生物群落结构趋于简单。基于 Bray-Curtis 距离分析了龙须菜栽培区和对照区微生物群落结构随着龙须菜栽培时间的变化,结果表明,表层水体中两者微生物群落结构差异随栽培时间而增加($p < 0.05$),1 米深及底层水体无线性增加关系。线性回归分析结果表明,龙须菜栽培区和对照

区表层、1米和底层水体微生物群落 β -多样性均随着龙须菜生物量的增加而增加,且龙须菜栽培区表层和1米深水体微生物群落 β -多样性增加得更快,说明龙须菜栽培加快了微生物群落的演替速率。CCA 结果显示,龙须菜生长初期(1-3月)微生物群落结构与 DO 、 NO_3^- 、 NO_2^- 、 TP 正相关,龙须菜生物量较大时(4-5月)微生物群落结构与 pH 、 TN 正相关,说明龙须菜栽培改变了微生物群落结构与环境因子,龙须菜栽培对微生物群落结构及组成具有选择性,并驱动了微生物群落演替加快。

3. 为研究龙须菜规模栽培对微生物群落功能的影响,于2014年12月-2015年7月分别采集了龙须菜栽培区和对照区1米水体,利用功能基因芯片 GeoChip 5.0 分析微生物群落功能随龙须菜栽培时间的变化情况。同时,于2015年5月采集了龙须菜栽培区和对照区表层、1米、水深一半、水深75%及底层水体,利用 GeoChip 5.0 分析龙须菜栽培对微生物群落功能的影响。结果显示,龙须菜栽培区和对照区的微生物群落功能随着龙须菜生物量的增加而出现显著差异,尤其是在龙须菜生物量较大的四月和五月。栽培区氮循环中 *nrfA*, *nir*, *nasA* 基因丰度增加,基因 *amoA* 丰度降低,提高了生态系统中 NH_4^+ 的含量,有利于龙须菜的吸收利用。同时,龙须菜栽培改变了不同水层微生物群落功能特征,尤其是1米及底层水体。1米深水体微生物群落参与碳降解及有机污染物修复的大多数基因在龙须菜栽培区的信号强度低于对照区。环境因子在影响微生物群落功能上起到重要作用。功能基因的改变、龙须菜生物量、环境因子及其相互作用可能反映了大型海藻-微生物-环境之间的复杂关系,龙须菜栽培能够调控微生物群落功能,微生物群落功能的时空变化为揭示海水养殖对微生物群落功能影响提供了科学依据。

4. 为研究龙须菜体表附生微生物群落的结构及功能,于2015年4月采集南澳岛深澳湾新鲜龙须菜藻体,利用16S rRNA 和 ITS 扩增子测序、功能基因芯片 GeoChip 5.0 等高通量技术对龙须菜体表和龙须菜周围水体微生物群落进行了比较。群落差异性检验表明两者在细菌群落结构、真菌群落结构及群落功能上均有显著差异($p < 0.05$)。龙须菜体表微生物群落的多样性(Shannon 指数和 Simpson 指数)均低于周围水体,但均匀度(Simpson 均匀度)高于周围水体,说明龙须菜体表微生物群落结构更加单一。测序结果表明,与藻类相关、具有溶藻特性及降解

海藻多糖能力的微生物类群在龙须菜体表远多于周围水体，如 *Croceitalea*, *Formosa*, *Granulosicoccus* (颗粒状球菌属), *Leucothrix*, *Polaribacter* (极地杆菌属), *Aureobasidium* (短梗霉)。GeoChip 5.0 分析表明，多数参与生物地球化学循环的基因在龙须菜体表微生物群落丰度较低，但是同化氮还原，硫/硫化物氧化及除草剂相关的基因在体表群落有较高丰度。此外，重金属解毒、植物激素、干旱胁迫、生物膜形成等与海藻表面生活相关的基因在体表微生物群落中表现出较高丰度，说明为适应海藻体表生活，龙须菜体表微生物群落保留或选择了特定种类和功能类群。

综上所述，龙须菜规模栽培对微生物群落结构具有重要影响。栽培区水体、沉积物及龙须菜体表中有关海藻及可利用海藻多糖的微生物种属明显多于自然海区，龙须菜对微生物群落结构及功能具有较强的选择性。通过本研究，揭示了微生物在龙须菜栽培生态系统生物地球化学循环中的作用，论证了龙须菜体表附生微生物对龙须菜正常生长的重要性，提出了龙须菜规模栽培对微生物群落影响的概念模型，阐释了其相互作用的分子机理。所获结论为揭示海水养殖生态系统结构和功能提供了基础数据和科学依据。

关键词：龙须菜；规模栽培；水体/沉积物/龙须菜体表微生物群落；高通量测序；功能基因芯片(GeoChip)

Abstract

As both primary producers and degraders, marine microbes are highly diverse, widely spread in the ecosystem. Together with phytoplankton and zooplankton, microbes link all organisms by forming the “microbial loop”, and play indispensable roles in material cycling and energy transfer as well as biogeochemical cycling. Therefore, it is crucial to understand the diversity, structure and function of marine microbial communities. Previous studies mainly focused on microbial communities in natural sea waters, so little is known about the characteristics of microbial communities in aquaculture zones, especially in seaweed cultivation ecosystems. To study the structure and function of microbial communities in *Gracilaria lemaneiformis* zones (GZ), water samples in different depths and growth times were collected from the seaweed cultivation zones and control zones (CZ), respectively. DNA isolated from these water samples was then analyzed using 16S rRNA gene MiSeq sequencing and GeoChip 5.0 technologies to reveal the effects of *G. lemaneiformis* cultivation on water/sediment microbial communities and ascertain the dissimilarity of microbial communities between the two study zones. In addition, DNA isolated from the surface of fresh thalli of *G. lemaneiformis* was analyzed to understand the epiphytic microbial community. The main results are as follows.

1. Effects of large-scale cultivation of *G. lemaneiformis* on the structure of microbial communities near the Nan’ao Island.

The major objectives of this study were to test the hypotheses that (i) changes in environmental properties resulting from seaweed cultivation would shift the diversity, composition and structure of water and sediment microbial communities; and (ii) selective microbial groups would be enriched in the seaweed cultivation ecosystem. To test these hypotheses, water and sediment samples were collected in April 2014 from a seaweed cultivation zone (the *G. lemaneiformis* cultivation zone) and a similar marine zone without seaweed for this study. DNA isolated from water and sediment

samples was analyzed using Illumina 16S rRNA gene amplicon sequencing to examine the shift of microbial community structure under seaweed cultivation conditions. Over 533,000 sequences were obtained from all 18 samples and 3,454 OTUs were generated at 97% sequence identity classified into 28 phyla, 74 classes, 134 orders, 275 families and 618 genera. Significant dissimilarities ($p < 0.05$) of microbial community structure were observed between GZ and CZ in both bottom water and sediment samples although the community structure in the surface water did not show obvious differences between the two zones. The dominant phyla or classes (relative abundance $> 1\%$) in both surface and bottom water were Bacteroidetes, α -Proteobacteria and γ -Proteobacteria, but their proportions differed at the two study zones. In the sediment, the dominant phyla at both zones were Bacteroidetes, γ -Proteobacteria and ε -Proteobacteria. At the genus level, higher abundances of *Aestuariibacter*, *Crocinitomix*, *Leucothrix* and *Tenacibaculum* were observed in the surface water at GZ than those at CZ ($p < 0.05$). In the bottom water, *Aestuariibacter*, *Bacillariophyta* and *Croceitalea* showed higher percentages ($p < 0.05$) at GZ in comparison with CZ. In the sediment, *Arenibacter*, *Glaciecola*, *Halomonas*, *Leisingera*, *Maribacter*, *Peredibacter* and *Sulfitobacter* exhibited a higher abundance at GZ than at CZ ($p < 0.05$). Most of the above genera were associated with seaweed, such as *Arenibacter*, *Croceitalea*, *Glaciecola*, *Leucothrix* and *Maribacter* were enriched at the cultivation zone. The response ratio analysis further indicated the effect of *G. lemaneiformis* cultivation on microbial community composition at the OTU level, revealing significant changes in the surface water, bottom water and sediment at GZ. In the surface and bottom water, the relative abundance of significantly changed OTUs increased at GZ by 89.9% and 72%, respectively. The relative abundance of significantly changed OTUs in sediment decreased by 84.5% at GZ. Canonical correspondence analysis (CCA) and Pearson correlation analysis were applied to detect the environmental factors on shaping the microbial community structure. The result of CCA showed the microbial community structure was strongly correlated with temperature, DO, pH and Chl-*a* at GZ, while it was most closely associated with salinity and TN at CZ. Pearson correlation analysis showed that

significantly changed OTUs exhibited a generally positive correlation with temperature, pH and DO ($p < 0.01$), and a negative correlation with TN and salinity ($p < 0.01$) in both surface and bottom water samples. In addition, the abundance of the most significantly changed OTUs in the sediment showed a strongly positive relationship with TN and TS concentrations ($p < 0.01$), and a negative correlation with TC ($p < 0.01$). Based on the above results, a conceptual model was proposed for future studies. *G. lemaneiformis* cultivation increased pH and DO, and decreased nutrients by photosynthesis and secreting specific compounds (e.g., algal polysaccharides) and producing detritus in the ecosystem. Those changes might not only lead to improved water quality, but also alter the microbial community in terms of α -diversity, composition (e.g., enriched *Arenibacter*, *Croceitalea*, *Glaciecola*, *Leucothrix* and *Maribacter* groups) and structure. This conceptual model can be used as a basic molecular mechanism to explain the interaction between seaweed and microbial communities.

2. Temporal and spatial dynamics of microbial communities in *G. lemaneiformis* cultivation ecosystem.

The major objectives of this study were to address the following questions: (i) how diversity and structure of microbial communities responded to the 5-months' seaweed growth; (ii) whether the diversity of microbial communities decreased along with the water depths; and (iii) did seaweed have a selection effect on microbial communities assembly. To answer the above questions, a field investigation was conducted at *G. lemaneiformis* cultivation zones and natural sea zones near the Nan'ao Island, South China Sea. Water samples were collected at different depths (surface, 1-m depth and near the bottom) at the two study zones from December 2014 to July 2015. The dynamics of microbial communities responses to seaweed growth and different water depths was analyzed by sequencing 16S rRNA gene amplicons. Large-scale cultivation of *G. lemaneiformis* improved water quality at seaweed cultivation zones. Adonis tests revealed that the structure of microbial communities was significantly influenced by seaweed biomass changes, time and water depths ($p < 0.01$). Also,

seaweed cultivation, time and the interaction between cultivation and time all affected the microbial community structure at all water depths ($p < 0.05$). Our results showed that significantly divergent community composition was detected during the seaweed cultivation period (January to May 2015) between cultivation zones and control zones, especially at 1-m depth and bottom water (Adonis test, $p < 0.05$), indicating that *G. lemaneiformis* cultivation did shift the microbial community assembly. Nonmetric multidimensional scaling (NMDS) analyses confirmed that seaweed cultivation time was the most important factor driving the microbial community structure. Only α -diversity of surface water in the cultivation zones showed a negative linear correlation with cultivation time, revealing that *G. lemaneiformis* may select certain microbial groups. Bray-Curtis distance based β -diversity of microbial communities in all water depths increased and it increased faster in cultivation zones compared to control zone in the surface and 1-m depth water by the linear regression. The results of CCA demonstrated that DO, NO_3^- , NO_2^- and TP showed a positive correlation with the structure of microbial communities during the primary stage of seaweed growth, whereas pH and TN were positively correlated with microbial communities when the biomass of *G. lemaneiformis* reached a high level. The major results on the temporal turnover of microbial communities implied that seaweed cultivation has a selection effect on microbial community assembly.

3. The effects of *Gracilaria* cultivation on microbial functional potential

To understand the temporal and spatial dynamics of functional microbial community structure and potential in response to seaweed cultivation, water samples at 1-m depth throughout the growth cycle of *Gracilaria lemaneiformis* (December 2014 to July 2015) and water samples at different depths (i.e. surface water, 1-m depth, half of water depth, 75% of water depth and near the bottom) were collected at the *Gracilaria* cultivation zones (GZ) and the control zones (CZ), respectively. Samples were analyzed using a functional gene array (GeoCip 5.0) to test the following hypotheses that: (i) *Gracilaria* cultivation would alter the microbial functional structure with the increasing biomass of seaweed; and (ii) the overall structure or

functional gene composition in various categories would shift at different water depths due to seaweed cultivation. Our results showed that significant variances of community functional potentials were observed between GZ and CZ along with the growth of seaweed, especially in April and May when seaweed biomass was high. Higher intensities of gene *nrfA*, *nir* and *nasA* and lower intensity of *amoA* were detected at GZ, indicating an accumulation of NH_4^+ , which is the favorable nutrient to *Gracilaria*. In addition, seaweed cultivation shifted the microbial functional structure between the two study zones at different depths in May, especially at 1-m depth and the bottom water. Most of genes involved in C degradation and organic remediation had lower signal intensities at GZ than at CZ. Environmental variables were also key factors shaping functional composition of microbial communities. The close relationships among functional gene changes, seaweed biomass and environmental factors could reflect the intricate interactions between seaweed-microbe-environment. These results indicate that micro-environment affected by *Gracilaria* cultivation may mediate the effects of seaweed cultivation and environmental changes on the functional community composition. The temporal and spatial dynamics of functional microbial community structure in seaweed cultivation ecosystems can shed light on the future work of microbial community functional potentials in mariculture ecosystems.

4. Interactions between *G. lemaneiformis* and epiphytic microbes

Seaweed surface provides suitable substrates and organic matters for epiphytes to live on. As one of the important roles in epiphytes, epiphytic microbes of *G. lemaneiformis* are rarely reported. Here, we hypothesized that: (i) the structure and function of epiphytic microbial communities would be significantly different from the free living planktonic microbial communities; and (ii) certain seaweed associated microbial groups and functional genes would be enriched in epiphytic microbial communities; and (iii) *G. lemaneiformis* would select epiphytic microbes to perform stable interactions between the host seaweed and their epiphytic microbes. To test the above hypotheses, fresh seaweed was collected in April 2015 and analyzed using 16S

rRNA and ITS gene amplicon sequencing as well as GeoChip 5.0 to compare the epiphytic community with the free-living microbial community in the surrounding water. Significant divergences in the composition and function of microbial communities were observed between epiphytes and free-living groups according to a dissimilarity test. More similar structure of microbial communities was detected in the epiphytic community with a lower Shannon index and Simpson index, and higher Simpson evenness compared to the free living community. Some epiphytic genera, such as *Croceitalea*, *Formosa*, *Granulosicoccus*, *Leucothrix*, *Polaribacter*, *Aureobasidium*, *Penicillium*, in high relative abundances are known to specialize in an alga-associated lifestyle with agarolytic or algal polysaccharides degrading potential. Genes associated with adaptation or responses to the seaweed surface were enriched in the epiphytic microbial community. The abundances of most genes involved in the biogeochemical cycling decreased, while those affiliated with assimilatory N reduction, sulfide/sulfur oxidation, chlorinated solvents and herbicide related compounds were enriched in the epiphytic community compared to the free-living community. Therefore, the distinct structure and function of epiphytic microbial communities generally support our hypotheses that seaweed can select microbes especially those seaweed-associated microbial groups. Understanding the interactions between *G. lemaneiformis* and epiphytic microbes can help us to explore the community assembly mechanism between seaweed and microbes in the coastal environment.

In summary, a large-scale cultivation of *G. lemaneiformis* has significant effects on the diversity, structure and function of water and sediment microbial communities in the *G. lemaneiformis* cultivation ecosystem through changing environmental factors such as increases in DO and pH, and decreases in nutrients. The α -diversity of microbial communities decreased in the seaweed cultivation zone, but the β -diversity increased with the growth of *G. lemaneiformis*. Seaweed-associated genera were enriched in the water column and sediment of cultivation zones, and epiphytes of *G. lemaneiformis*, indicating that a selection effect may occur in the *G. lemaneiformis*

cultivation ecosystem. For example, higher abundances of seaweed-associated genera *Croceitalea*, *Formosa*, *Leucothrix*, *Polaribacter*, *Arenibacter*, *Glaciecola*, and *Maribacter* were observed in the *G. lemaneiformis* cultivation ecosystem. Also, genes *cqsS* involved in quorum sensing, *amsG* for biofilm formation, *tre-arc* associated with desiccation tolerance, and many genes in the stress response category were enriched in epiphytic communities. Thus, *G. lemaneiformis* cultivation altered the diversity, structure and function of microbial communities in seaweed cultivation zone by changing environmental factors and selecting seaweed-associated microbes. This study provides possible molecular mechanisms by which *G. lemaneiformis* cultivation influences microbial communities, revealing the role of microbes in the biogeochemical cycling of seaweed cultivation ecosystems and importance of epiphytes to the host seaweed. The above results provide basic data and scientific basis to uncover the structure and functional potential of microbial communities in mariculture ecosystems.

Keywords: *Gracilaria lemaneiformis*, large-scale cultivation, water/sediment/epiphytic microbial communities, high-throughput sequencing, functional gene array (GeoChip)

学位论文中文提要

一、研究背景

海洋能够为地球提供将近一半的初级生产力,海洋生态系统在全球的物质循环及能量流动中起到至关重要的作用。海洋微生物多样性高,分布广泛,既是初级生产者又是分解者,与浮游植物、浮游动物共同组成“微食物环”,是参与海洋生物地球化学循环不可或缺的一环。其中,细菌、古菌及真菌在海洋生态系统中拥有很高的生物量及生产力,尤其是细菌,具有很高的物种多样性、丰度和代谢活力。基于细菌上述特征,其广泛应用于环境生物修复等领域。此外,在深海海底、深海火山口等极端环境中也存在着细菌,说明细菌对环境具有极强大的适应能力。由于海洋微生物对环境变化非常敏感,它们成为研究生物对气候变化、全球变暖、海洋酸化、富营养化等环境变化响应的模式生物之一。因此,了解海洋微生物群落的多样性、结构及其功能显得尤为重要。

2000 年以前,大多数研究海洋微生物的学者是利用传统方法进行研究,包括纯培养法、BioLog 碳源分析法、PLFA 和 FAME 生物标记法以及荧光显微镜、扫描电镜及透射电镜等方法。上述方法无论是否基于微生物培养,均表现出较大的局限性,很难全面揭示特定环境中微生物群落的真实情况。目前,微生物分子生态学的相关研究技术与方法逐渐成为研究海洋微生物群落的常用方法。例如,基于 PCR 的 RFLP、T-RFLP 和 DGGE/TGGE,基于 DNA 杂交的 BrdU 标记技术、FISH、CARD-FISH 及基因芯片。在分子生态学研究方法中,最近 10 年发展最为迅速的是高通量技术,包括高通量测序和功能基因芯片(Functional Gene Array, FGA)。高通量技术的最大优点是需要的 DNA 样品少,测试周期短,能快速得到数据量大的序列,且能保证序列的高准确率,结合生物信息学分析即可分析出微生物群落的多种信息,如群落多样性、优势种、群落结构及其与环境的关系。随着高通量测序成本的不断降低,越来越多研究海洋微生物生态的学者开始选择该方法对样品进行分析,其中 Illumina 公司的 MiSeq 是比较常用的测序平台。MiSeq 测序平台要求测试者提供的 DNA 不多,前期建库、上机等工作比较简单,可以在同一批次测序中完成对细菌、真菌和真核生物的分析。此外,为进一步研究海洋微生物的群落功能,可以利用功能基因芯片如 GeoChip 5.0。这种基因芯片主

要含有生物地球化学循环的基因, 以及胁迫、次级代谢、金属稳态等功能基因, 能较全面地反映特定环境中微生物群落功能信息。

过去几十年对海洋微生物群落的研究主要集中在自然海域, 对海水养殖区尤其是大型海藻栽培区微生物群落特征所知甚少。大型海藻作为海洋初级生产者, 在海洋生态系统中具有十分重要的地位和作用。大型海藻包括红藻门, 褐藻门及绿藻门, 其中常见的海藻有海带、紫菜、江蓠、龙须菜、裙带菜等。多数海藻具有食用、药用等经济价值, 在世界上很多国家与地区均有栽培。中国的大型海藻栽培面积及产量在全球海藻栽培产业中位列第一, 大多数沿海省份都有大型海藻栽培产业, 主要栽培的种类包括海带、紫菜和龙须菜等。作为食物、优质的琼脂来源、鱼类的饵料及良好的生物修复材料, 龙须菜在广东省汕头市南澳海域被广泛栽培, 已成为当地海水养殖业的支柱之一。

本研究以汕头市南澳岛龙须菜栽培区和相邻自然海域(对照区)为调查海域, 通过采集不同水深样品, 提取 DNA 并进行了 16S rRNA 基因 MiSeq 测序和 GeoChip 杂交, 比较了龙须菜栽培区和对照区微生物群落的差异, 分析了龙须菜体表附生微生物组成, 研究了龙须菜规模栽培对微生物群落的影响及其分子机理。所获结论对海洋环境保护和生物资源可持续利用具有重要科学意义。

二、主要结果

(一)龙须菜栽培对微生物群落结构的影响

为了研究龙须菜栽培对水体及沉积物中微生物群落的影响, 于 2014 年 4 月在南澳岛深澳湾龙须菜栽培区和对照区各设置 3 个采样点, 标注为 G1, G2, G3 和 C1, C2, C3。每个采样点均采集表层水体、底层水体和沉积物, 提取 DNA 并纯化后, 进行 16S rRNA 基因 MiSeq 高通量测序。同时, 在采样时记录现场理化因子, 水化学样品低温运送至实验室后立即进行测定。结果表明, 龙须菜栽培区 pH 和 DO 在表层及底层水体均高于对照区($p < 0.05$), 总氮低于对照区($p < 0.05$)。栽培区沉积物中的总氮和总硫含量低于对照区($p < 0.05$)。龙须菜规模栽培可以提高溶氧、降低营养盐浓度和提高水质, 具有重要的生物修复效应。

16S rRNA 基因 MiSeq 测序结果显示, 18 个样品共产生超过 530,000 条高质量序列, 生成 3454 个 OTU (Operational Taxonomic Units), 分别属于 28 个门, 74 个纲, 134 个目, 275 个科及 618 个属。群落差异性检验(adonis)结果表明, 龙须菜

栽培区和对照区表层水体微生物群落结构差异不明显,但底层水体及沉积物中微生物群落结构差异显著($p < 0.05$)。在表层及底层水体中,拟杆菌门, α -变形菌和 γ -变形菌均为优势菌,但其相对丰度在栽培区和对照区有差异。沉积物中的优势菌为拟杆菌, γ -变形及 ϵ -变形菌。在属水平上,栽培区表层水体中 *Aestuariibacter*, *Crocinitomix*, *Leucothrix* (亮发菌属)及 *Tenacibaculum* (黄杆菌属)相对丰度高于对照区($p < 0.05$)。底层水体中,栽培区 *Aestuariibacter*, *Bacillariophyta* (硅藻门)及 *Croceitalea* 的相对丰度高于对照区 ($p < 0.05$)。沉积物中, *Arenibacter*, *Glaciecola*, *Halomonas* (盐单胞菌属), *Leisingera*, *Maribacter*, *Peredibacter* 及 *Sulfitobacter* (亚硫酸杆菌)在龙须菜栽培区的相对丰度高于对照区($p < 0.05$)。龙须菜栽培区相对丰度高的种属多与利用海藻多糖的微生物密切相关,如 *Arenibacter*, *Croceitalea* 及 *Maribacter*。利用 MEGA 5 对不同水层的样品数据进行系统进化树构建,结果与上述群落种属特征一致。响应比分析(Response ratio)结果显示,栽培区表层和底层水体显著变化的 OTU 分别有 89.9%及 72%增加了相对丰度;栽培区沉积物中显著变化的 OTU 有 84.5%降低了相对丰度。CCA 结果显示,栽培区水体微生物群落与温度、DO、pH 及 Chl.a 相关,而对照区水体微生物群落与盐度及总氮密切相关。Pearson 相关性分析表明,表层和底层水体中显著变化的 OTU 与温度、pH 和 DO 呈正比($p < 0.01$),与盐度和总氮成反比($p < 0.01$)。沉积物中显著变化的 OTU 与总氮、总硫呈正相关($p < 0.01$),与总碳含量呈负相关($p < 0.01$)。

综合分析以上研究结果,提出了龙须菜栽培影响微生物群落结构的概念模型。龙须菜在生长过程中通过光合作用固定 CO_2 , 释放 O_2 , 提高了水体中的 pH 及 DO; 吸收氮磷降低水体和沉积物的营养盐, 改善水质; 在生长过程中龙须菜分泌的海藻多糖及其脱落的残枝均可供微生物分解利用。龙须菜栽培改变了水体和沉积物环境因子,使得与海藻多糖利用等相关的微生物类群增多,如 *Arenibacter*, *Croceitalea*, *Glaciecola*, *Leucothrix* 及 *Maribacter*, 从而改变了龙须菜栽培区微生物群落结构。该模型为研究大型海藻与微生物相互作用奠定了科学基础。

(二) 深澳湾海域微生物群落结构动态变化

2015 年龙须菜的生长期为 1-5 月。为进一步研究龙须菜栽培对微生物群落结构的影响,于 2014 年 12 月至 2015 年 7 月(采样期涵盖了龙须菜栽培前、栽培期和收获后)在汕头南澳龙须菜栽培区和自然海区(对照区)进行每月一次的连续采

样, 采样水层分别为表层、1 米和距底 0.5 米。在野外采样的同时, 现场记录各采样点的环境因子。所有采集的水样依次经过 10 μm 和 0.2 μm 滤膜抽滤, 富集有生物的滤膜封存于液氮罐中, 样品送到实验室后转移滤膜至 -80°C 冰箱直至 DNA 提取。所有样品均利用试剂盒提取并纯化样品 DNA, 并用引物 515F 和 806R 对 DNA 16S rRNA V4 区基因进行扩增。建库及上机完成后, 将样品进行 Miseq 高通量测序, 分析微生物群落结构随着龙须菜生长的动态变化。

龙须菜栽培期间(2015 年 1 月-5 月), 栽培区表层水体 pH 和 DO 高于对照区, TP、TN 和 NO_3^- 低于对照区 ($p < 0.05$); 栽培区 1 米深水体 DO 高于对照区, PO_4^{3-} 低于对照区($p < 0.05$); 底层水体 DO 高于对照区, NH_4^+ 低于对照区($p < 0.05$)。以上结果表明, 龙须菜栽培能有效提高水质。

利用 adonis 进行群落差异性分析, 结果显示采样时间、龙须菜栽培及水深均对微生物群落结构有显著影响($p < 0.01$); 同样, 龙须菜栽培时间对不同水层微生物群落结构亦有显著影响($p < 0.05$)。分析不同时期不同水层栽培区与对照区微生物群落结构差异, 结果表明, 1-2 月和 4-5 月栽培区与对照区 1 米深和底层水体微生物群落结构差异显著($p < 0.05$)。NMDS 分析结果表明, 微生物群落分别在龙须菜栽培前(2014 年 12 月)、栽培期间(2015 年 1-5 月)及收获后(2015 年 6-7 月)形成明显不同的聚类, 且每个月样品的聚类中, 龙须菜栽培区样品均与对照区样品有所区别, 说明龙须菜栽培时间长短导致微生物群落结构差异。利用线性回归分析 α -多样性与栽培时间的关系, 发现栽培区表层水体微生物群落多样性随龙须菜栽培时间而降低, 其余水层及对照区均无线性关系。微生物群落 α -多样性与龙须菜生物量呈负相关, 说明龙须菜对栽培区表层水体的微生物群落具有一定的选择作用, 导致龙须菜栽培区的微生物群落结构单一化。

基于 Bray-Curtis 距离分析了龙须菜栽培区和对照区两者微生物群落结构差异随着龙须菜栽培时间的变化情况。结果表明, 表层水体中两区域微生物群落结构差异随栽培时间而增加($p < 0.05$), 1 米深及底层水体无变化。在门水平上对不同水层微生物群落组成进行分析, 发现随着龙须菜栽培时间的变化, 各水层的优势菌群由变形菌门逐渐转变为拟杆菌门, 且表层水体中随着栽培时间发生变化的菌群门类最多。基于 Bray-Curtis 距离的 β -多样性分析表明, 龙须菜栽培区和对照区表层、1 米和底层水体微生物群落 β -多样性均随着龙须菜生物量的增加而增

加,且龙须菜栽培区表层和1米深水体微生物群落的 β -多样性增加得更快,说明龙须菜栽培加快了微生物群落的演替速率。CCA结果显示,在表层和1米深水体中,pH、TN与4-5月微生物群落结构呈正相关,DO、 NO_3^- 、 NO_2^- 与1-3月微生物群落结构呈正相关;在底层水体,pH、TN、 NO_3^- 对4-5月微生物群落结构有显著影响,DO、TP、 NO_2^- 对1-3月微生物群落结构有显著影响。上述结果表明,龙须菜生物量较大时(4-5月)与生长初期(1-3月)影响微生物群落结构的环境因子不同,龙须菜规模栽培随着环境因子变化持续改变着微生物群落组成,龙须菜驱动了微生物群落演替和环境因子的改变。

(三) 深澳湾海域微生物群落功能特征的时空变化

为研究龙须菜规模栽培对微生物群落功能的影响,于2014年12月-2015年7月分别采集了龙须菜栽培区和对照区1米深水样,利用功能基因芯片GeoChip 5.0分析微生物群落功能随龙须菜栽培时间的变化情况。同时,于2015年5月采集了龙须菜栽培区和对照区表层、1米、水深一半、水深75%及底层水体,利用GeoChip 5.0分析龙须菜栽培对微生物群落功能影响随水深的变化。研究显示,龙须菜栽培区和对照区微生物群落功能随着龙须菜生物量的增加而出现显著差异,尤其是在龙须菜生物量较大的四月和五月。栽培区参与碳降解和有机污染物修复的基因信号低于对照区;栽培区氮循环中*nrfA*、*nir*和*nasA*基因丰度增加($p < 0.05$),基因*amoA*丰度降低($p < 0.01$),提高了生态系统中 NH_4^+ 的含量,有利于龙须菜的吸收利用。同时,龙须菜栽培改变了不同水层微生物群落的功能特征,尤其是1米及底层水体。1米深水体微生物群落参与碳固定、磷循环及金属稳态的基因在龙须菜栽培区的信号强度高于对照区($p < 0.05$)。然而,大多数参与碳降解和有机污染物修复的基因在龙须菜栽培区的信号强度低于对照区($p < 0.05$)。环境因子同样在影响微生物群落功能结构上起到关键作用,5月龙须菜栽培区1米水体微生物群落功能与温度、pH、DO、 NO_3^- 及 PO_4^{3-} 显著相关,5月对照区1米水体微生物群落功能与TN含量呈正相关。功能基因的改变、龙须菜生物量、环境因子及其相互作用可能反映了海藻-微生物-环境之间的复杂关系。上述结果表明受龙须菜栽培影响的微环境通过调控栽培区环境变化,从而影响微生物群落功能。研究龙须菜栽培生态系统中微生物群落功能的时间和空间变化对了解海水养殖对微生物群落功能影响提供了科学依据。

(四) 龙须菜体表附生微生物群落结构及功能初步研究

为研究龙须菜体表附生微生物群落的结构及功能,于2015年4月采集南澳岛深澳湾新鲜龙须菜藻体,密封于无菌自封袋,低温保存带回实验室。同时,采集原位海水(每样品5.0升)并测定现场理化因子。称取2克龙须菜藻体,并用无菌海水冲洗干净,浸泡在100毫升无菌海水中,用超声波分离龙须菜体表微生物。每个体表附生样品(2.5升)依次经过10 μm 及0.2 μm 滤膜抽滤,待提取DNA。用试剂盒对所有样品DNA进行提取及纯化后,采用16S rRNA及ITS测序、功能基因芯片GeoChip 5.0等高通量技术对龙须菜体表微生物群落与龙须菜周围水体中微生物群落进行比较。

Adonis 分析显示,龙须菜体表及周围水体微生物在细菌群落结构、真菌群落结构及群落功能均有显著不同($p < 0.05$)。DCA 分析结果与 adonis 一致。龙须菜体表微生物群落多样性(Shannon 指数和 Simpson 指数)均低于周围水体,但均匀度(Simpson 均匀度)高于周围水体,说明体表微生物群落结构更加单一。通过对不同分类水平优势类群的分析,门水平上拟杆菌门(40.5%), α -变形菌(32.3%)及 γ -变形菌(22.1%)在体表细菌群落的相对丰度更高;科水平上,黄杆菌门(38.8%),红杆菌(29.5%),交替单胞菌(2.4%)及假交替单胞菌(1.96%)是体表细菌群落的优势类群;体表细菌群落在属水平上的优势类群为 *Formosa* (8.57%), *Methylophaga* (嗜甲基菌属, 6.73%), *Meridianimaribacter* (4.84%), *Croceitalea* (3.86%), *Granulosicoccus* (颗粒状球菌属, 3.73%)。真菌群落中, Ascomycota (子囊菌门, 13.92%), Basidiomycota (担子菌门, 18.37%), Chytridiomycota (壶菌门, 23.48%)及 Zygomycota (接合菌门, 41.88%)是体表微生物群落的优势类群;科水平上 Mucoraceae (毛霉菌科, 16.2%)及 Tremellales-unidentified (1.8%)在体表群落中更丰富;体表群落的优势属为 *Pilaira* (16.2%)。测序结果表明,与藻类相关、具有溶藻特性及降解海藻多糖能力的种属在龙须菜体表微生物群落中远多于周围水体群落,如 *Croceitalea*, *Formosa*, *Granulosicoccus*, *Leucothrix*, *Polaribacter*, *Aureobasidium*, *Penicillium*。

GeoChip 5.0 结果表明,多数参与生物地球化学循环的基因在龙须菜体表微生物群落丰度较低,但是同化氮还原,硫/硫化物氧化和除草剂相关的基因在体表群落有较高丰度。更重要的是,与海藻表面生活相关的基因在体表微生物群落

表现出较高丰度, 如与重金属解毒、植物激素、干旱胁迫、生物膜形成等相关的基因。其中基因 *cqsS* 负责微生物群体应答系统, *amsG* 参与生物膜形成, 上述两个基因在体表微生物群落中具有很高的丰度, 说明龙须菜体表微生物群落保留或加强了相关基因以适应附生于海藻表面附生的生活。

三、结论

1. 龙须菜规模栽培能提高水体 DO, 降低营养盐含量, 改善水质。龙须菜通过改变周围水体和沉积物环境影响微生物组成, 对微生物群落结构具有重要影响, 通过综合分析提出了龙须菜栽培对微生物影响的概念模型。

2. 龙须菜栽培区水体、沉积物及龙须菜体表中有关海藻及可利用海藻多糖的微生物种属明显多于自然海区。龙须菜栽培期间, 微生物多样性随着海藻生物量的增加而增加, 且龙须菜栽培区增加速率高于自然海区。龙须菜栽培加快了微生物群落的演替速率, 对微生物群落的结构及功能具有较强的选择性。

3. 通过对龙须菜栽培生态系统中水体、沉积物及体表附生微生物群落的研究, 发现龙须菜体表微生物群落保留或选择了特定种类和功能类群, 龙须菜体表微生物群落结构更加单一, 这一结论提供了龙须菜规模栽培对微生物群落影响的分子证据, 阐释了其相互作用的可能分子机理。

所获结论为揭示海水养殖生态系统结构和功能提供了基础数据和科学依据。

四、创新点

1. 以大型海藻龙须菜栽培生态系统为研究对象, 利用高通量技术, 深入地研究了龙须菜规模栽培对微生物群落结构及功能的影响, 揭示了大型海藻与微生物相互作用的可能分子机制。

2. 提出了龙须菜栽培对微生物群落结构影响的概念模型, 阐释了龙须菜规模栽培对环境和微生物影响的生态过程及其机理。

3. 综合利用 16S、ITS 和 GeoChip 5.0 等高通量技术研究了龙须菜体表附生微生物群落的结构及功能, 发现龙须菜对体表微生物群落组成及功能均有选择作用, 初步揭示了微生物对大型海藻附着适应的机理。

五、展望

本文通过研究广东汕头南澳岛龙须菜栽培区与自然海区微生物群落结构及功能特征,初步揭示了大型海藻规模栽培对微生物群落影响的分子机制,探讨了海水养殖对微生物群落结构和功能的影响。虽然采用了高通量技术对样品进行分析,但是由于本研究采集样品数量等限制,仍有一些问题尚待解决:

1. 由于时间及人力资源限制等原因,本研究野外调查的时间跨度虽有 2 年,但总调查时间仅 8 个月。野外调查最好采集多年(3-5 年)连续的样品,以全面揭示不同季节龙须菜栽培和非栽培期对微生物群落影响的差异及其机制。

2. 本研究仅研究了一种大型海藻规模栽培对微生物群落的影响,今后的工作中,应扩大研究对象范围,比较不同大型海藻,同时对同一种大型海藻进行不同海区差异的分析。

3. 本研究显示大型海藻对微生物群落组成及功能具有选择性,但研究证据有待进一步补充。今后应在数据分析中引入群落聚集机制(Community assembly mechanism)、分子生态网络(Network)等生物信息学方法对数据进行深入挖掘,以便较全面地揭示龙须菜栽培对微生物群落影响的分子机制。

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Chapter 1: Introduction

1.1 The ecological significance of marine microbes

Nearly half of global primary production occurs in marine ecosystems (Field, 1998) and the production situation by different producers is one of the fundamental questions in marine studies. Scientists often pay most attention to small and large animals but ignored microbes largely (Williams, 1981), while this phenomenon may be caused by traditional and limited study methods before. With the development of high throughput technologies, such as sequencing and microarrays, we have been able to extensively explore marine microbes, which are considered as a large biomass and productivity in marine ecosystems (Azam et al., 1983). Furthermore, bacteria and archaea can utilize particulate organic matters (POM) and dissolved organic matters (DOM) dissolved from primary production (Pomeroy, 2007) and provides energy to zooplankton through “microbial loop”. Thus, marine microbes like bacteria and archaea can link primary producers like phytoplankton and seaweed, viruses and zooplankton, which form a complicated marine food web (Polis and Strong, 1996). As primary producers and degraders, marine microbes play important roles in carbon (C) cycling and nitrogen (N) cycling and are indispensable in marine ecosystems.

Due to its high diversity, large abundance and various metabolic types, marine microbes have been largely utilized as marine bioremediation, such as petroleum degradation (Lu et al., 2012), harmful algal bloom (HAB) control (Yang et al., 2013). Marine microbes are wide spread, including those extreme environments such as deep sea and volcanic vents (Raulf et al., 2015). Also, marine microbes are very sensitive to environmental changes, so they are good study models of microbial responses to climate change, global warming, ocean acidification, eutrophication, HABs and volcanic eruption (Leibold, 2004). High diversity of microbial communities in different marine ecosystems can also provide research datasets of origin of life and evolution. In addition, we may extract many valuable organic matters from marine microbes and find high pressure, temperature and other stress resistant mechanisms.

1.2 Methods for marine microbial ecology studies

Many methods and technologies can be applied to study marine microbes including pure culture and carbon utilization. To overcome the limitation of traditional methods, molecular biology methods have been developed to make more microbes detected and studied easier with high accuracy and resolution. Among all molecular methods, high-throughput technologies, especially next-generation sequencing, are becoming more and more popular to examine the diversity, structure and functional potential of microbial communities.

1.2.1 Traditional methods

To study the diversity, abundance and activity of microbial communities, many traditional methods were applied. The most commonly used method is isolation and functional characterization of pure cultures, which is often conducted with specific culture media. However, it is believed that only less than 1% microorganisms can be cultured in the laboratory (Dykhuizen, 1998). Such a method takes long time and cannot get a full picture of microbial communities.

Biochemical analysis of microbial communities is also largely used in microbial ecology studies. One of the commonly used methods is carbon utilization, such as BioLog, which can identify pure species or compare different microbial communities using various carbon sources in one microplate to get the community-level physiological profiling (CLPP) (Hill et al., 2000). The most commonly used BioLog microplate types are GN, ECO, FF and YT to analyze Gram-negative bacteria, biological profile, yeast and filamentous bacteria, respectively (Choi and Dobbs, 1999; Wang, 2011). In addition, since biomarkers are generally positively correlated with microorganism biomass, testing variety and abundance of specific biomarkers can reflect the composition and potential/activity of microbial communities. Phospholipid fatty acids (PLFA) and fatty acid methyl ester (FAME) are often used as biomarkers (Scow, 2000). Both PLFA and FAME can analyze the diversity of microbial communities, but it is difficult to keep high extractability and stability of all biomarkers.

Other traditional methods including fluorescence microscope observation, scanning electron microscope (SEM), transmission electron microscope (TEM), and microbial quinone profiling can observe abundances of microbes, identify the morphologic characteristics and study the diversity and structure of microbial communities (Tong, 2002).

1.2.2 Molecular methods

Molecular methods can study more microorganisms including those not-yet cultured microbes. One of the DNA-based methods is to incorporate BrdU with DNA of active microbial cells to examine the metabolic changes of microbial communities in various environments (Tada and Grossart, 2014). Fluorescence *in situ* hybridization (FISH) and CARD-FISH (catalyzed reporter deposition fluorescent *in situ* hybridization) are also powerful techniques to identify the diversity of microbial communities (Kubota, 2013). In addition, DNA/RNA-based stable isotope probing (DNA /RNA-SIP) is a molecular ecological method to explore active microbes in a wide range of terrestrial and aquatic environments (Dunford and Neufeld, 2010).

PCR based techniques are widely applied in microbial ecology, such as restriction fragment length polymorphism (RFLP), and terminal restriction fragment length polymorphism (T-RFLP) (Schutte et al., 2008), and DGGE/TGGE (Muyzer, 1999). These methods have great advantages over traditional methods, which take long time but give limited information of microbial communities. However, DGGE only reflects a small part of community diversity and often generates false positive results.

1.2.3 High-throughput technologies

High-throughput sequencing, also called next-generation sequencing, with enough sequence read lengths and accuracy has enabled large numbers of samples to be examined at a greater depth, which is becoming the major approach in microbial ecology and environmental genomics studies. The most popular sequencing platforms include Roche 454 pyrosequencing and Illumina HiSeq/MiSeq sequencing platforms (Mardis, 2008). Due to their low cost, MiSeq sequencing is one of the most widely

used sequencing methods, which can even enable multiplexing of samples including bacteria, fungi and protozoa to be sequenced in one run. High-throughput sequencing takes less time and uses less amount of DNA samples to get high quality reads, deeper sequencing and more information about microbial communities.

Another powerful approach of high-throughput technology is functional gene array (FGA) technology, which can reveal the functional profile of microbial communities through DNA hybridization with specific functional gene probes (Waldron et al., 2010). One of such FGAs, GeoChip has been invented to study the biogeochemical cycling and environmental processes of microbial communities (He et al., 2007). For example, to identify different functional potential and activity of microbial communities, GeoChip 4 was designed containing 82,000 probes and covering 410 functional gene families from 11-major functional categories (Tu et al., 2014). GeoChip can be applied to profile microbial community structure and function in many various environments, such as soil, waste water, ocean, sediment, and sewage sludge (He et al., 2012b).

1.3 Environmental effects of seaweed cultivation on marine ecosystems

Seaweeds are largely cultivated in China coastal areas and make great contributions to mariculture yield, food market, pharmacy and other industries. Also, seaweed cultivation is environmental beneficial to marine ecosystems since they can improve water quality and control HABs by increasing pH and DO, and up-taking redundant nutrients of coastal waters. Thus, seaweed cultivation is one of the important bioremediation methods and should be applied in more coastal areas.

1.3.1 Status of seaweed cultivation

As macroscopic, multicellular, marine alga, seaweeds live near the seabed and conduct photosynthesis by Chl.a. Seaweeds belong to Phaeophyta, Rhodophyta or Chlorophyta. The most common seaweeds are *Laminaria*, *Undaria*, *Sargassum*, *Gracilaria*, *Porphyra*, *Enteromorpha* and *Ulva*. Most of the above seaweeds have economic values and many of them can be used as food. Seaweeds contribute a major part of primary production, serve as one of the most important blue-carbon sink, and

play an important role in the structure and biodiversity of coastal communities (Duarte et al., 2013). However, many seaweed habitats have been degraded or contaminated due to the development of mariculture, fishing, tourism, entertainment, and/or other anthropogenic perturbations (Nogales et al., 2011). Thus, seaweed cultivation has been developed for nearly 100 years to supplement human food and explore other utilization values. Nowadays, more than 10 genera, 20 species seaweeds have been all around the world including countries in Asia, Africa, America, Europe and Oceania. The yield of cultivated seaweeds is the largest in Asia, accounting for nearly 90% of the world's gross production, followed by America. In Asia, the mainly cultivation countries are China, India, Japan and Korea (FAO, 2014). With the development of cultivation techniques and market requirements, China has become the largest seaweed cultivation country in the world. For example, the total yield of cultivated seaweed in China was 23.8×10^6 t (fresh weight) in 2012, accounted for 54% of global total yield, and ranked the first among all the 33 countries followed by India, accounting for 27%. From 2000 to 2012, the global yield of cultivated seaweeds has doubled to meet the demands of food market and industrial applications.

In China, most coastal provinces such as Guangdong, Fujian, Shandong, Zhejiang, Liaoning, Hainan, and Jiangsu have cultivated seaweeds. According to the China Fishery Statistical Yearbook of 2016, the cultivation area of seaweed was 0.13 million hm^2 with 4.46% increasing rate compared to last year, and the annual yield was 2.09×10^6 t, 8.4×10^4 t more than the yield of last year (CFAMOA, 2016). The mainly cultivated seaweeds are *Porphyra*, *Gracilaria*, and *Laminaria*.

Porphyra has the highest economic value with high protein and nutrition compared to other cultivated seaweeds. In China, *Porphyra* is mainly cultivated in Fujian, Zhejiang and Jiangsu (CFAMOA, 2016). For example, with the improvement of cultivation techniques and managements, the cultivation area of *Porphyra* in Jiangsu keeps around 2.3×10^4 hm^2 and the yield and industrial scale account for more than 97% of national market. Furthermore, these *Porphyra* have been exported to 65 countries and regions with more than 60% of trade share in the international market of *Porphyra*. Thus, the cultivation of *Porphyra* and relevant industry have provided jobs

to more than 45,000 fishermen and settled lives of 100,000 fishermen, which have made a great contribution to the development of local fishery and economic improvement of fishermen (Lu, 2016).

Laminaria japonica is a kind of cold-water seaweed and widely spread in Shandong and Liaoning China since 1950s. With the development of cultivation techniques, *Laminaria* were widely cultivated in more and more sea areas including southern sea areas in Fujian province (Zeng, 1958). Many Chinese scientists dedicate to the genetic breeding and tried to cultivate *Laminaria* in more coastal areas. The development of new cultivars improves the yield and quality of *Laminaria* in China, making a great contribution to the industry of seaweed cultivation. *Laminaria* cultivation is one of the crucial industries in mariculture and plays important roles in the life of citizens (Lüning and Pang, 2003; Tseng, 2001).

1.3.2 *Gracilaria* cultivation development in China

Gracilaria is a good resource of agar, and can improve water quality. The annual yield and cultivation area were increasing almost every year. For example, the cultivation area of *Gracilaria* in 2015 was 9,912 hm², and increased 2.22% compared to the area in 2014; the yield in 2015 was 270,149 t with 3.02% growth rate compared to that in 2014. Pond scattering and floating raft cultures are the two commonly used methods for *Gracilaria* cultivation in China. As one representative of *Gracilaria*, *G. lemaneiformis* has been largely cultivated in China from northern to southern coastal areas and brings great ecological and economic efficiencies. However, the cultivation of *G. lemaneiformis* was only performed in the spring and autumn of northern sea area because it is sensitive to high temperature and cannot live in the summer of coastal waters. The limitation of temperature makes it difficult to accumulate biomass of *G. lemaneiformis* and hard to develop the cultivation industry. To solve this issue, researchers such as Xiugeng Fei dedicate to solve technical problems and cultivate *G. lemaneiformis* in southern coastal waters since 1988. After several years' work, strain 981 was selected and cultivated successfully at the Nan'ao Island, Guangdong (Zhang et al., 2009). Nowadays, *G. lemaneiformis* has been cultivated in many coastal

provinces such Guangdong, Fujian, Zhejiang, Shandong, and Liaoning, and has been developed to one of the most widely cultivated seaweed except *Laminaria*, *Porphyra* and *Undaria* (Yang et al., 2015).

Especially in the Nan'ao Island, Guangdong, *G. lemaneiformis* cultivation is a pillar industry and provides many job opportunities to local fishermen. The cultivation area has increased from 0.13 hm² in 2000 to 1500 hm² in 2011. Except for the food and agar source, *G. lemaneiformis* is a good bioremediation material and widely used in industry, medicine and cosmetics. Field investigation reported that *G. lemaneiformis* can remove redundant nutrients such as NH₄-N, NO₃-N, NO₂-N and PO₄-P, and improve water quality and mitigate eutrophication (Du et al., 2010). In addition, *G. lemaneiformis* are rich in bioactive polysaccharides which have multiple biological active and medicinal effects, such as modulate immune activity and anticancer (Kang et al., 2016). The medical applications of *G. lemaneiformis* could explore more values of *G. lemaneiformis* in the future.

1.3.3 Bioremediation effects of seaweed cultivation

Cultivation of economic seaweeds has been promoted worldwide to protect the seaweeds' habitats and utilize their environmental and economic benefits, especially in Asia (Loureiro, 2015; Taelman et al., 2015). As one of the most important seaweed cultivation countries, China has developed the cultivation of economic seaweeds since 1950s and has obtained remarkable gains in the past decades (Yang et al., 2015). More and more studies showed that seaweed cultivation has bioremediation effects in marine ecosystems. Firstly, seaweeds can improve the surrounding environment by its physiologic activities and metabolism. For example, *G. lemaneiformis* uptake nitrogen and phosphorus to decrease nutrients, and fix CO₂ to release O₂ into water. As the results of the above physiology of *G. lemaneiformis*, pH and DO increase and nutrients decrease in the cultivation zone. Thus, the water quality is improved and eutrophication is alleviated due to seaweed cultivation.

For example, the red alga (Rhodophyta), *G. lemaneiformis*, is widely cultivated along the coast of China and has been proved to be a powerful bioremediator to

alleviate eutrophication and improve water quality in coastal ecosystems (Fei, 2004; Yang et al., 2015) due to its ability of up-taking excessive nutrients, releasing oxygen and relieving ocean acidification by photosynthesis (Chen et al., 2016; Koch et al., 2013). Also, it can serve as human daily diets and be used as a cost effective feed of abalone. In addition, *G. lemaneiformis* is good agar material (Chirapart et al., 1995; Li et al., 2008), used as food (Wen et al., 2006), industrial and pharmaceutical products, and serves as an economical feedstuff for mariculture such as fish and abalone (Qi et al., 2010; Xu et al., 2011; Xuan et al., 2013).

1.4 The research progress in interactions between seaweeds and microbial communities

Since seaweed has environmental beneficial effects on marine ecosystems, it is interesting to study the interaction between seaweeds and microbes living in the seaweed cultivation ecosystem. For examples, few studies were conducted to study the influences of seaweed cultivation on water and sediment microbial communities. However, more studies were focused on the epiphytic microbial communities of seaweed and their interaction.

1.4.1 Effects of seaweed cultivation on water and sediment microbial communities

Given the importance of microbes in marine ecosystems, one of the key questions of microbial ecology is how microbial communities respond to anthropogenic activities like mariculture. As mariculture has been one of the predominant human activities in marine ecosystems and caused eutrophication and fish pathogenesis world-wide, more attention has been focused on the response of microbial communities to mariculture areas including seaweed cultivation. During the growth and metabolism of seaweeds, they can secrete many organic matters like algal polysaccharides to the surrounding water and attract microbes to utilize them. Also, seaweeds can increase DO and decrease nutrients by up-taking nitrogen, phosphorus and CO₂ and releasing O₂ into the water. By improving the water quality, seaweed cultivation can alter the sediment

environment by reducing precipitate of nutrients and release seaweed detritus into water. Thus, significantly changed water and sediment environments have become a new habitat for marine microbes to live. Due to the quick response and adaption of microbes to their habitats, the diversity, composition and structure of water and sediment microbial communities will change correspondingly to seaweed cultivation (Xie, 2017). Certain seaweed-associated groups are enriched in the microbial community of seaweed cultivation area. Zhang reported that *G. lemaneiformis* cultivation decreased the abundances of heterotrophic bacteria (e.g., *coli*, *vibrio* species), indicating that *G. lemaneiformis* could regulate the structure of microbial communities and control the abundance of harmful groups (Zhang et al., 2010). However, the effect of seaweed cultivation on microbial communities still needs further studies.

1.4.2 Diversity of epiphytes on the seaweed surface

Seaweed surfaces provide nutrients and living substrates to many epiphytic organisms including bacteria, fungi, protozoa, diatoms, spores, and larvae of marine invertebrates (Bulleri et al., 2002; Haywood et al., 1995) and can secrete various organic matters that help multiplication of microbes and formation of biofilms (Staufenberger et al., 2008). The interaction between epiphytic organisms and the host seaweed is critical for the completion of their life cycles. For example, epiphytic bacteria induce liberation of carpospores in the red alga, *Acrochaetium* (Weinberger et al., 2007) and settlement of zoospores in the green algae, *Enteromorpha* and *Ulva* (Joint et al., 2002; Joint et al., 2007). Also, they produce the plant growth-promoting substances (auxin, cytokinin and kinetin) (Maruyama et al., 1986; Provasoli, 1958; Singh et al., 2011); quorum sensing signaling molecules (Williams, 2007) and bioactive molecules (Goecke et al., 2010) that are responsible for the normal morphological development and growth of the host seaweed. Epiphytic bacterial communities, therefore, are indispensable both to the host seaweed and coastal ecosystems where they live.

To understand the diversity of epiphytic microbial communities, many technologies

have been used to reveal this important group, such as DGGE, 16S rRNA gene amplicon sequencing, and T-RFLP. Previous studies have shown that Proteobacteria and Firmicutes are generally the most abundant phyla detected on seaweed surfaces (Singh and Reddy, 2014). Also, the diversity, composition and structure of epiphytic bacterial communities are species-specific and function-driven in different host seaweeds across seasons and habitats (Burke et al., 2011a; Burke et al., 2011b; Lachnit et al., 2009; Lachnit et al., 2011). The host seaweed and associated microbes form a stable mutualism as a defense against further colonization (Steinberg et al., 2002). Furthermore, the same epiphytic microbes could also produce various compounds that promote growth and photosynthesis, as well as act as antifouling and antifungal agents (Singh and Reddy, 2014). Given the critical interaction between host seaweed and their epiphytes, an analysis of microbial community composition and function provides us with a better understanding of this intricate interaction in coastal ecosystems.

1.5 Research significance

As the indispensable primary producer, seaweeds have been cultivated in China for a long time. Most cultivated seaweeds can serve as human food, while some can be used by industry, medicine and ecological management. Considering the bioremediation effect and economic benefit, we chose a *G. lemaneiformis* cultivation ecosystem. Since microorganisms play crucial roles in marine ecosystems, how seaweed cultivation influences water and sediment microbial communities of marine ecosystems has become very important. However, previous studies mainly focused on revealing the diversity of seaweed epiphytes, little is known about free living microbial communities in seaweed cultivation ecosystems. Also, few reports were found about relationships between *G. lemaneiformis* cultivation and free living microbial communities.

The major objectives of this study were to address the following questions: (i) how would water, sediment and epiphytic microbial communities respond to large-scale cultivation of *Gracilaria*; (ii) would seaweed cultivation increase the succession rate

of microbial communities during a seaweed growth cycle; and (iii) would *Gracilaria* have a selection effect on assembly of water, sediment and epiphytic microbial communities. In order to understand the above questions, we collected water samples in consecutive 8 months near the Nan'ao Island and analyzed those samples using high throughput technologies, such as 16S rRNA and ITS gene amplicon MiSeq sequencing, functional gene array (GeoChip 5.0), which were first applied in the study of *G. lemaneiformis* cultivation ecosystems. In addition, we also used the same high-throughput technologies to examine the diversity and function of epiphytic microbial communities on *G. lemaneiformis* surface. In this study, we have tested three general hypotheses that: (i) large-scale cultivation would alter the diversity, structure and function of microbial communities; (ii) certain seaweed-associated microbial populations would be enriched in *Gracilaria* cultivation zone; and (iii) *Gracilaria* would select the structure and function of microbial communities to perform stable interactions between seaweed and microbes.

The results of our studies supported the above hypotheses. *Gracilaria* cultivation decreased the α -diversity of water, sediment and epiphytic microbial communities and enriched seaweed-associated taxa/functional microbial groups, suggesting that seaweed cultivation has selection effect of microbial communities. This study provides new insights into our understanding of water and sediment microbial communities in mariculture systems and the impact of seaweed cultivation on coastal ecosystems. We explored molecular mechanisms of interactions between *G. lemaneiformis* and associated microbial communities using high throughput technology, which could provide data for the future study regarding microbial assembly mechanisms. A conceptual model was built in this study to explain how *G. lemaneiformis* cultivation affects microbial communities through changes in the environment. In addition, the structure and function of epiphytic microbial communities revealed a selection effect of *G. lemaneiformis* to certain seaweed-associated populations.

Chapter 2: Cultivation of *Gracilaria* affects the microbial community structure

2.1 Introduction

As land resources decrease and the human population booms, mariculture has increasingly grown to exploit the sea to meet global food demands. For example, China has promoted large-scale mariculture of fish, shellfish, shrimps and seaweeds since the 1980s (Tseng, 1993). However, the flourishing industry of marine animal culture has caused severe eutrophication and other environmental problems in coastal areas (Cao et al., 2007). To ameliorate the water quality of contaminated zones with mariculture of economical animals (such as fish and shell fish), a large scale cultivation of seaweed, *Gracilaria lemaneiformis*, was introduced as a bioremediation approach at Nan'ao Island, Guangdong Province in 2000 with a cultivation area reaching approximately 1,500 ha in 2014 (Yang et al., 2015). Although the cultivation of seaweed significantly improved water quality (Fei, 2004; Yang et al., 2015) and the local economy, little is known about the impact of large scale *G. lemaneiformis* cultivation on water and sediment microbial communities as well as microbial feedback responses to *G. lemaneiformis* cultivation.

Since microorganisms play important ecological roles in coastal ecosystems through recycling dissolved nutrients, and mediating global biogeochemical flux of C, N, phosphorus (P), sulfur (S) and iron as well as other metals (Azam et al., 1983; Azam and Malfatti, 2007), it is essential to understand how microbial communities respond to environmental changes such as perturbations from anthropogenic activities (e.g., fishing, mariculture) (Halpern et al., 2007; Halpern et al., 2008; Nogales et al., 2011), severe weather, and climate change (Giovannoni and Vergin, 2012). Previous studies mainly focused on the dynamics of microbial communities across time and space in natural marine ecosystems (Fuhrman et al., 2015; Ghiglione et al., 2012). The response of microbes to mariculture, especially seaweed cultivation, has not been well studied (Blancheton et al., 2013; Tamminen et al., 2011). *G. lemaneiformis* has been cultivated in Chinese coastal areas to alleviate eutrophication and balance the effects of mariculture of economical animals (Yang et al., 2006b; Zhou et al., 2006) and

proved to be an efficient nutrient pump by removing large amounts of N and P to support its growth. Also, biopolymers and carbon-rich nutrients such as polysaccharide and alginate, secreted by *G. lemaneiformis*, add organic substrates to surface water, which can be utilized by microbes (Egan et al., 2013; Lachnit et al., 2013). The changes of environmental properties due to *G. lemaneiformis* cultivation, such as decreased inorganic nutrients, increased pH, dissolved oxygen (DO), polysaccharides and seaweed detritus (Yang et al., 2015; Yang et al., 2006b) may promote changes in the composition, structure and function of microbial communities in both water column and sediment environments.

In this study, we specifically aimed to understand the impact of seaweed cultivation on water and sediment microbial communities as well as microbial feedback responses to seaweed cultivation, and we hypothesized that (i) changes in environmental properties such as decreased nutrients and increased dissolved oxygen resulting from seaweed cultivation would shift the diversity, composition and structure of water and sediment microbial communities; and (ii) certain microbial groups would be enriched in the seaweed cultivation ecosystem. To test those hypotheses, we sampled surface water, bottom water and sediment samples from a cultivation zone (GZ) and a control zone (CZ) at Nan'ao Island in the South China Sea, and analyzed the microbial community diversity, composition and structure by MiSeq sequencing of 16S rRNA gene amplicons. This study provides useful information for our understanding the impact of seaweed cultivation on microbial communities as well as their feedbacks to ecosystems, which will promote our future studies on microbial communities, seaweed, and their environments in the *G. lemaneiformis* cultivation system.

2.2 Materials and methods

2.2.1 Sample collections

This study was conducted in April 2014 on the coastal area of Shen'ao Bay, Nan'ao Island, where a large scale cultivation of *G. lemaneiformis* was carried out and marine

fish were cultured in cages for decades (Yang et al., 2006a; Zhou et al., 2006). Three sites each were chosen from a seaweed cultivation zone, GZ (the *G. lemaneiformis* cultivation zone, 117° 07'05" E, 23° 28'05"N) and a natural sea area, CZ (control zone, 117° 05'33" E, 23° 29'17"N) for this study (Fig. 2-1). Surface (0.5 m depth) or bottom water (0.5 m above sediment) was collected at each site in 4.0 L and filtered immediately on a 0.2 μ m polycarbonate membrane (EMD Millipore GTPP04700, USA). The filters were then stored in a tank filled with liquid nitrogen for DNA extraction. Surface sediments from an area of 0.2 m $\times$ 0.2 m and a depth of 0.1-m were sampled with a Peterson grab sampler. Therefore, a total of 18 samples (three samples each site for surface water, bottom water, and sediment at both GZ and CZ) were taken and used for physicochemical and molecular analyses.

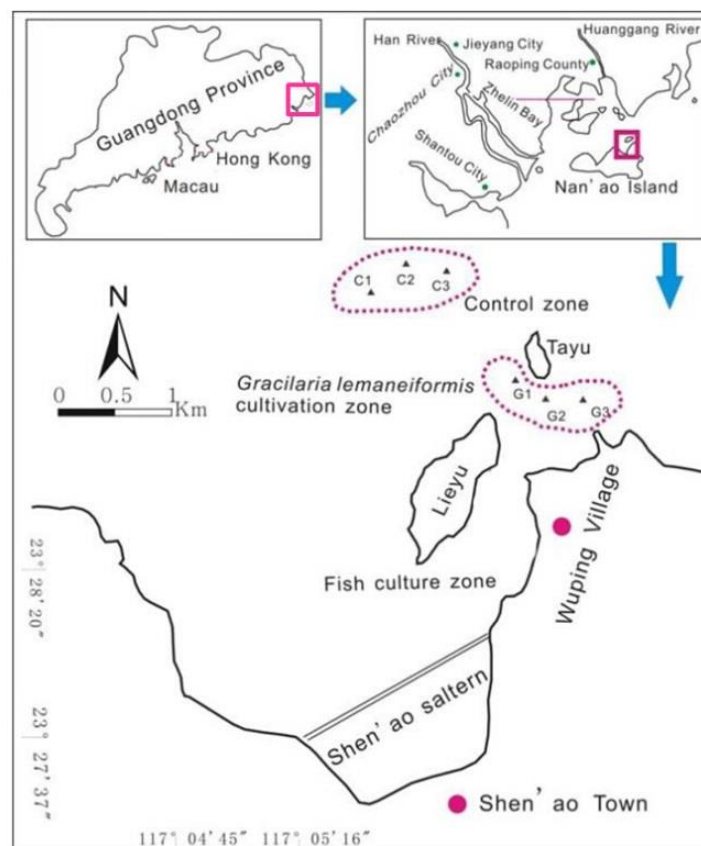


Fig. 2-1 Map of sampling sites at Nan'ao Island, Guangdong Province, South China Sea. GZ: *G. lemaneiformis* cultivation zones (G1, G2, G3); CZ: control zones (C1, C2, C3)

图 2-1 广东省汕头市南澳岛采样站点图。龙须菜栽培区(GZ): G1, G2, G3, 对照区(CZ): C1, C2, C3

2.2.2 Determination of environmental factors

Physicochemical factors including temperature, salinity, pH, DO and water depth were obtained from the *in-situ* data using the Potable Water Quality Analyzer-YSI 556 systems (YSI Incorporated, USA), a Secchi disc and a Digital Depth Sounder (Speedtech, USA), respectively. Total nitrogen (TN), total phosphorus (TP), chlorophyll a (Chl-*a*) of water samples and TP of sediment samples were analyzed according to the standard methods (AQSIQ, 2007). Total carbon (TC), total nitrogen (TN) and total sulfur (TS) of sediments were determined using Element Tester (Perkin Elmer, USA) at Analytical and Testing Center of Jinan University, China.

2.2.3 DNA extraction, amplification and MiSeq sequencing of 16S rRNA gene amplicons

Microbial community DNA of water and sediment was isolated from the polycarbonate membrane or 0.5 g of sediment (wet weight) using Omega E.Z.N.A.® Water DNA Kit and Soil DNA Kit (Omega Bio-Tek, USA) based on the manufacturer's instructions. The extracted DNA was purified using the E.Z.N.A.® MicroElute DNA Clean-Up Kit (Omega Bio-Tek, USA) and met the quality criteria: $A_{260\text{nm}}/A_{280\text{nm}} \sim 1.8$, $A_{260\text{nm}}/A_{230\text{nm}} \sim 1.8$, and then quantified with a spectrophotometer (NanoDrop ND-2000, USA).

The V4 regions of microbial 16S rRNA genes were amplified using the primer pair 515F (5'-GTGCCAGCMGCCGCGGTAA-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3'), with a unique barcode sequence added to the reverse primer for identifying mixed samples in one run (Caporaso et al., 2012). PCR amplification of the V4 region was performed in triplicate using a Thermal Cycler S1000 (Bio-Rad, USA) in a total volume of 25 μL , containing 12.5 μL PCR mix (pre-mixture of 10 $\times$ PCR Buffer, *Taq* DNA Polymerase and dNTP), 9.5 μL ddH₂O, 1 μL forward primer, 1 μL reverse primer and 1 μL DNA template (20~30 ng). Thermal cycling conditions of amplification were: an initial denaturation at 94°C for 5 min, followed by 28 cycles at 94°C for 45 s, 62°C for 45 s and 72°C for 1-min, with a

final extension at 72°C for 10 min. Triplicate PCR amplicons of each sample were combined and examined by 1% agarose gel electrophoresis and then purified using the Omega Gel Purification Kit (Omega Bio-Tek, USA). Purified amplicons were quantified by μ DropTM Plate reader (Thermo Scientific, USA) and an equal amount (e.g. 200 ng) of DNA from each sample was combined for a sequencing run. The pooled mixture was diluted and denatured to obtain the 8 pM sample DNA library mixed with an equal volume of 8 pM PhiX according to the MiSeq Reagent Kit Preparation (Illumina, San Diego, CA, USA). As a final step, 600 μ L of the library was loaded with read 1, read 2 and the index sequencing primers (Caporaso et al., 2012) on a 500 cycles (2 $\times$ 250 bp paired ends) kit and run on a MiSeq machine at the School of Minerals Processing and Bioengineering, Central South University, China.

2.2.4 Data processing and statistical analysis

After assigning each read to its corresponding sample by the unique barcode, 533,871 sequences were obtained from both ends for all 18 samples (NCBI accession number SRP082338). The sequencing data were trimmed by Btrim with an average quality score greater than 20 and a minimum length of 100 bp (Kong, 2011) and the paired reads were joined using FLASH (Magoc and Salzberg, 2011). After removing sequences contained N, sequences with length between 245 and 258 bp were subjected to chimeras detection using U-Chime (Edgar et al., 2011). Operational taxonomic units (OTUs) were generated by UCLUST (Edgar, 2010) and then assigned to microbial taxonomy by the RDP Classifier (Wang et al., 2007a). Singletons were removed and resampled at 10,000 sequences per sample for downstream analyses. The abovementioned steps were performed on the Galaxy pipeline (<http://zhoulab5.rccc.ou.edu/>) developed by Qin *et al.* (unpublished) at the Institute of Environmental Genomics, University of Oklahoma, USA.

The alpha-diversity and beta-diversity of microbial communities were analyzed using multiple indices (Shannon Index, Pielou Evenness, observed numbers of OTU and Chao1) and the Bray-Curtis distance between samples. Nonmetric multidimensional scaling (NMDS) was applied to visualize the overall difference in

the distribution patterns of microbial communities at each site (Kruskal, 1964). To calculate the similarity of the microbial community structure between GZ and CZ, the dissimilarity tests were performed by the permutational multivariate analysis of variance (PERMANOVA) with the Adonis function. Canonical correspondence analysis (CCA) and Pearson's correlation were used to identify the effect of environmental factors on microbial community structure. All the above analyses were conducted by R (version 3.2.2; [http:// www.r-project.org/](http://www.r-project.org/)) using vegan (Oksanen et al.), gplots (Warnes et al., 2013) and ggplot2 packages. A circular phylogenetic tree was constructed based on sequences of dominant OTUs using a neighbor-joining method with a bootstrap value of 1000 displayed by MEGA5 (Tamura et al., 2011) and the plot was generated on iTOL (Letunic and Bork, 2011).

2.3 Results

2.3.1 Effects of *G. lemaneiformis* cultivation on the water and sediment properties

We measured the environmental variables including temperature, pH, DO, total nitrogen (TN) and total phosphorous (TP) of water and sediment samples (Table 2-1). In surface water, significantly ($p < 0.05$) higher pH, DO and TP were observed at GZ, while significantly ($p < 0.05$) higher TN and Chl-*a* were detected at CZ. In bottom water, GZ showed higher pH and DO and lower TN ($p < 0.01$) compared to CZ. Generally, the nutrient concentrations were lower, and pH and DO were higher at GZ compared to CZ, which indicated a better water quality at GZ due to *G. lemaneiformis* cultivation. For the environmental factors in sediments, the concentrations of TN and total sulfur (TS) were significantly ($p < 0.05$) lower at GZ compared to CZ. Such divergences in environmental properties in water columns and sediments were expected to affect the diversity and composition of microbial communities at GZ.

Table 2-1 Physical and chemical parameters of water column and sediment at the two study zones. GWS: surface water of GZ; GWB: bottom water of GZ; CWS: surface water of CZ; CWB: bottom water of CZ; GS: sediment of GZ; CS: sediment of CZ. Different letters (a, b, c) represent significant differences ($p<0.05$) in mean value between GZ and CZ in water and sediment samples by ANOVA. DO: dissolved oxygen (mg/L); TN: total nitrogen in water ($\mu\text{mol/mL}$) and sediment (%); TP: total phosphorus in water ($\mu\text{mol/mL}$) and sediment (mg/kg); TC (%): total carbon; TS (%): total sulfur; ND: not determined.

表 2-1 龙须菜栽培区 and 对照区水体及沉积物理化因子

(GWS: 龙须菜栽培区表层水体, GWB: 龙须菜栽培区底层水体, CWS: 对照区表层水体, CWB: 对照区底层水体, GS: 龙须菜栽培区沉积物, CS: 对照区沉积物; 不同字母(a, b, c) 表示 ANOVA 分析龙须菜栽培区 and 对照区水体及沉积物的理化因子存在显著差异 ($p<0.05$); DO:溶解氧(mg/L); TN: 水体及沉积物总氮($\mu\text{mol/mL}$); TP: 水体及沉积物总磷($\mu\text{mol/mL}$); TC: 沉积物总碳(%); TS: 沉积物总硫(%); ND: 未检测)

	GWS	CWS	GWB	CWB	GS	CS
Temperature ($^{\circ}\text{C}$)	21.76 $\pm$ 0.31 ^a	21.26 $\pm$ 0.15 ^a	20.60 $\pm$ 0.26 ^b	20.03 $\pm$ 0.06 ^b	ND	ND
Salinity (‰)	29.71 $\pm$ 0.43 ^a	29.01 $\pm$ 0.31 ^a	29.71 $\pm$ 0.81 ^a	31.16 $\pm$ 0.35 ^b	ND	ND
pH	8.68 $\pm$ 0.08 ^a	7.73 $\pm$ 0.09 ^b	8.43 $\pm$ 0.19 ^a	7.66 $\pm$ 0.03 ^b	ND	ND
DO (mg/L)	14.14 $\pm$ 1.14 ^a	10.83 $\pm$ 1.11 ^b	11.27 $\pm$ 0.24 ^b	8.49 $\pm$ 2.03 ^c	ND	ND
Chl- <i>a</i> ($\mu\text{g/L}$)	26.39 $\pm$ 1.45 ^a	28.66 $\pm$ 0.11 ^b	23.58 $\pm$ 1.79 ^a	21.60 $\pm$ 3.74 ^a	ND	ND
TN	14.67 $\pm$ 1.92 ^a	28.31 $\pm$ 0.76 ^b	15.31 $\pm$ 9.96 ^a	26.74 $\pm$ 3.23 ^b	0.03 $\pm$ 0.01 ^a	0.08 $\pm$ 0.01 ^b
TP	2.74 $\pm$ 1.36 ^a	1.87 $\pm$ 0.22 ^b	1.91 $\pm$ 1.61 ^a	2.69 $\pm$ 1.74 ^a	620.56 $\pm$ 232.08 ^a	737.07 $\pm$ 194.29 ^a
TC (%)	ND	ND	ND	ND	1.42 $\pm$ 0.12 ^a	1.33 $\pm$ 0.04 ^a
TS (%)	ND	ND	ND	ND	0.06 $\pm$ 0.00 ^a	0.14 $\pm$ 0.02 ^b
Water content (g/g)	--	--	--	--	0.37 $\pm$ 0.14 ^a	0.46 $\pm$ 0.01 ^a

2.3.2 Responses of water and sediment microbial communities to *Gracilaria* cultivation

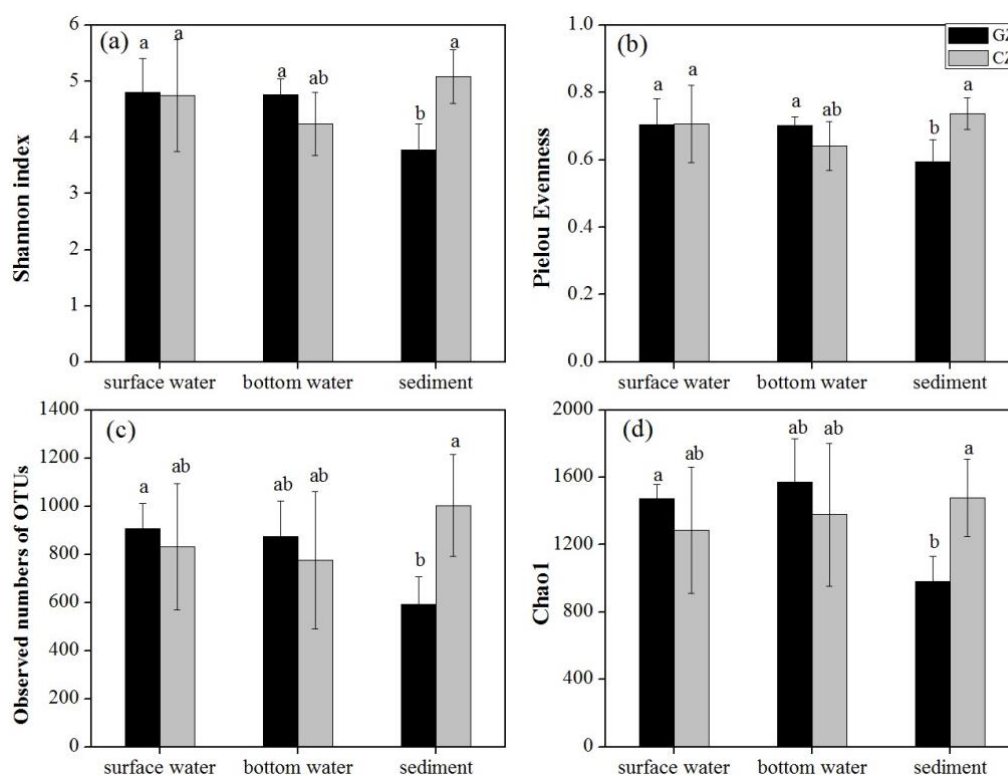


Fig. 2-2 Effects of *G. lemaneiformis* cultivation on the alpha-diversity indices of microbial communities. Shannon index (a), Pielou Evenness (b), the observed number of OTUs (c), and Chao1 (d) showed α -diversity of microbial communities across different layers at the two study zones. Different letters (a, b) represent significant differences ($p < 0.05$) in mean value between GZ and CZ by ANOVA.

图 2-2 龙须菜栽培对微生物群落 α -多样性的影响

Shannon 指数(a)、Pielou 均匀度(b)、OTUs 数(c)以及 Chao1 (d) 表明龙须菜栽培区和对照区不同层微生物群落的 α -多样性. 不同字母(a, b)说明栽培区和对照区的微生物群落 α -多样性有显著差异 ($p < 0.05$).

A total of 18 microbial communities (12 water samples from two layers, and 6 sediment samples) were analyzed by MiSeq sequencing of 16S rRNA gene amplicons. Over 0.53 million high quality sequences were obtained across all 18 samples. After resampling of 10,000 reads per sample, 3,454 OTUs were generated at 97% sequence identity. Overall, these OTUs were classified into 28 phyla, 74 classes, 134 orders,

275 families and 618 genera, indicating taxonomically diverse microbial communities at both zones. Among 2,705 OTUs detected from 12 water samples, 1,472 (54.4%) were shared by GZ and CZ, and 1,233 unique OTUs were detected with 639 OTUs at GZ (23.6%) and 594 OTUs at CZ (22.0%). Also, 1,941 OTUs were generated for 6 sediment samples, and among those, 216 OTUs (11.1%) were unique to GS, 873 OTUs (45.0%) were unique to CS, and 852 OTUs (43.9%) were shared. The Shannon index, OTU richness (observed numbers of OTU and Chao1 diversity estimator) and evenness were lowest ($P < 0.05$) in the sediment samples at GZ compared with surface water at GZ and sediment at CZ (Fig 2-2).

Table 2-2 Dissimilarity tests of microbial communities of surface water, bottom water and sediment between GZ and CZ by ADONIS. p value < 0.05 was highlighted in bold. GWS: surface water of GZ; GWB: bottom water of GZ; GS: sediment of GZ; CWS: surface water of CZ; CWB: bottom water of CZ; CS: sediment of CZ.

表 2-2 龙须菜栽培区和对照区表层水体、底层水体和沉积物微生物群落差异性检验(adonis) $p < 0.05$ 用粗体标注. GWS: 龙须菜栽培区表层水体, GWB: 龙须菜栽培区底层水体, CWS: 对照区表层水体, CWB: 对照区底层水体, GS: 龙须菜栽培区沉积物, CS: 对照区沉积物

p	GWS	GWB	GS
CWS	0.398	0.001	0.001
CWB	0.099	0.013	0.001
CS	0.009	0.001	0.001

To determine the differences in microbial community structure between GZ and CZ in response to *G. lemaneiformis* cultivation, dissimilarity test were performed using PERMANOVA (adonis) across different layers at both GZ and CZ. Significant dissimilarities ($p < 0.05$) of community structure were observed between GZ and CZ both in bottom water and sediment although the community structure in surface water did not show obvious difference between the two zones (Table 2-2). Non-metric multidimensional scaling (NMDS) based on the Bray-Curtis distance also showed the

divergence in microbial community structure between GZ and CZ in water and sediment samples (Fig. 2-3). The close clustering of samples in surface and bottom water at GZ indicated similar origin of microbial communities compared to CZ samples. As expected, the samples of sediment at GZ and CZ were distinctly separated from each other in Quadrant 1 and Quadrant 2. This shows a clear dissimilarity of microbial community structure in the sediments of the two study zones.

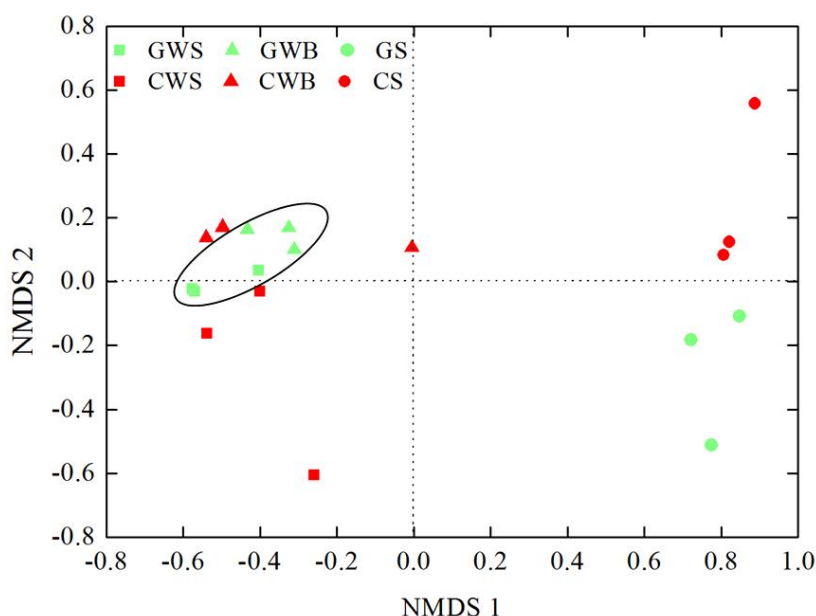


Fig. 2-3 Non-metric multidimensional scaling (NMDS) showed the differences of microbial communities between GZ and CZ both in surface water, bottom water and sediment. Green symbols represent samples at GZ and red symbols represent samples at CZ. Square symbols represent samples in surface water, triangles represent samples in bottom water, and circles represent samples in sediments.

图 2-3 龙须菜栽培区和对照区表层、底层水体和沉积物微生物群落结构的 NMDS 分析

绿色标志代表龙须菜栽培区样品，红色标志代表对照区样品；方形标志代表表层水体，三角形代表底层水体，圆形标志代表沉积物

The dominant phyla or classes (relative abundance > 1%) in both surface and bottom waters were Bacteroidetes, α -Proteobacteria and γ -Proteobacteria, but their proportions differed at the two study zones (Fig. 2-4). In the bottom water, the relative abundance of Cyanobacteria ($p < 0.05$) was higher at GZ than that at CZ. In the

sediment, the dominant phyla at both zones were Bacteroidetes, γ -Proteobacteria and ϵ -Proteobacteria, and lower ($p < 0.05$) abundances of Chloroflexi were observed at GZ compared to CZ.

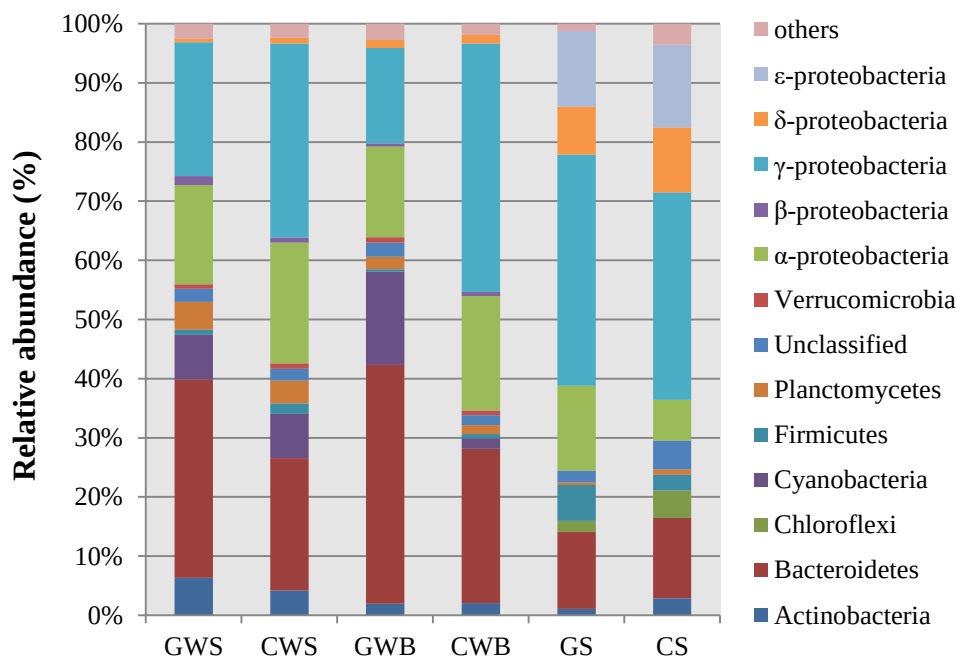


Fig. 2-4 Microbial community composition in water column and sediment samples at the phylum level. The phylum Proteobacteria was represented by Alpha-, Beta-, Gamma-, Delta- and Epsilon-divisions.

图 2-4 水体及沉积环境中门水平上微生物群落组成的比较(变形菌门由 Alpha-, Beta-, Gamma-, Delta-和 Epsilon-变形菌亚门表示)

At the genus level, higher abundances of *Aestuariibacter*, *Crocinitomix*, *Leucothrix* and *Tenacibaculum* were observed in the surface water at GZ ($p < 0.05$) than those at CZ (Fig. 2-5a). In the bottom water, *Aestuariibacter*, *Bacillariophyta* and *Croceitalea* showed higher percentages ($p < 0.05$) at GZ in comparison with CZ (Fig. 2-5b). In the sediment, *Arenibacter*, *Glaciecola*, *Halomonas*, *Leisingera*, *Maribacter*, *Peredibacter* and *Sulfitobacter* exhibited a higher abundance at GZ ($p < 0.05$) than they were at CZ (Fig. 2-5c).

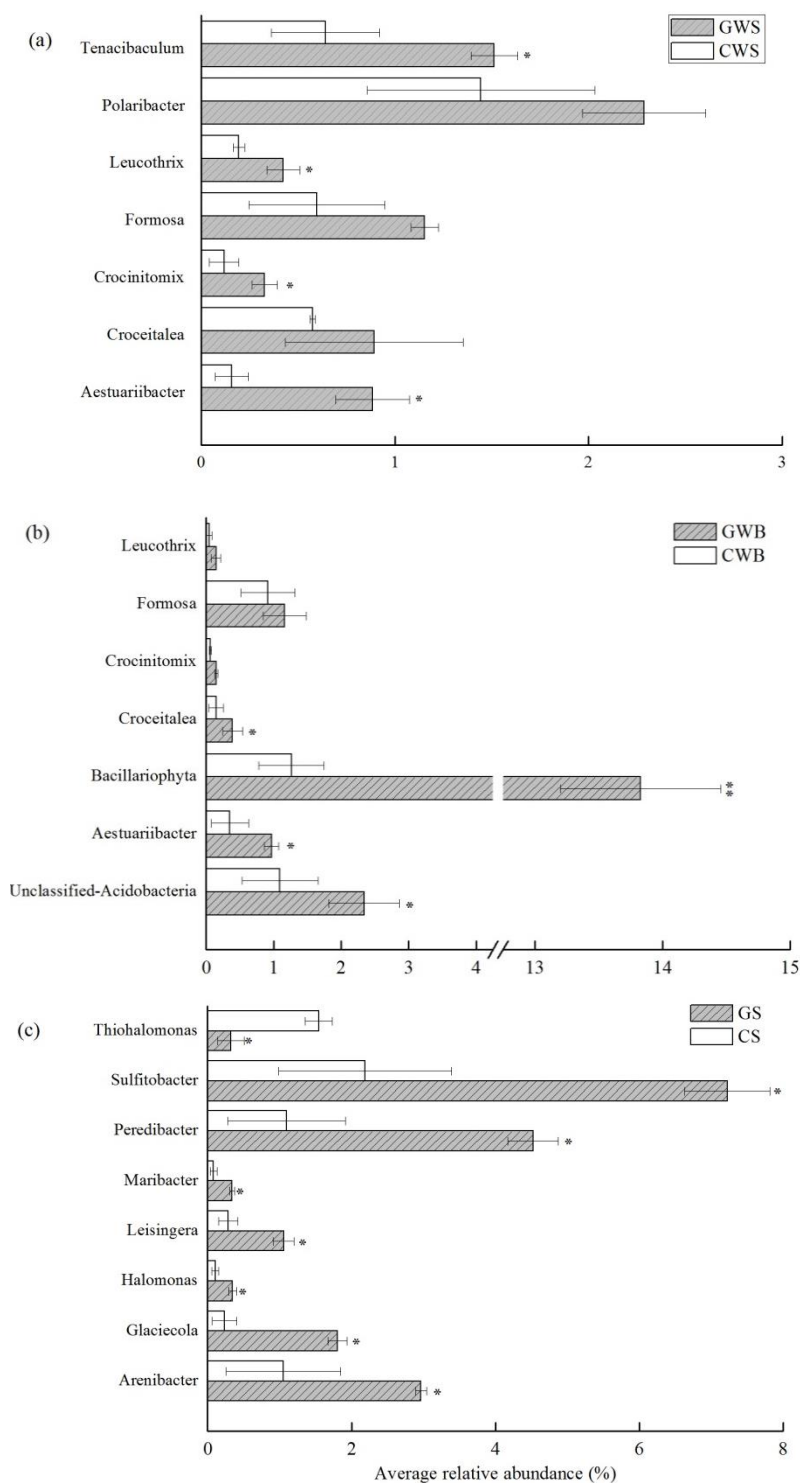


Fig. 2-5 Microbial community composition in surface water (a), bottom water (b) and sediment (c) at genus level. *: $p < 0.05$, **: $p < 0.01$ by ANOVA test.

图 2-5 表层水体(a)、底层水体(b)及沉积物(c)中微生物群落组成在属水平的差异分析(*: $p < 0.05$, **: $p < 0.01$)

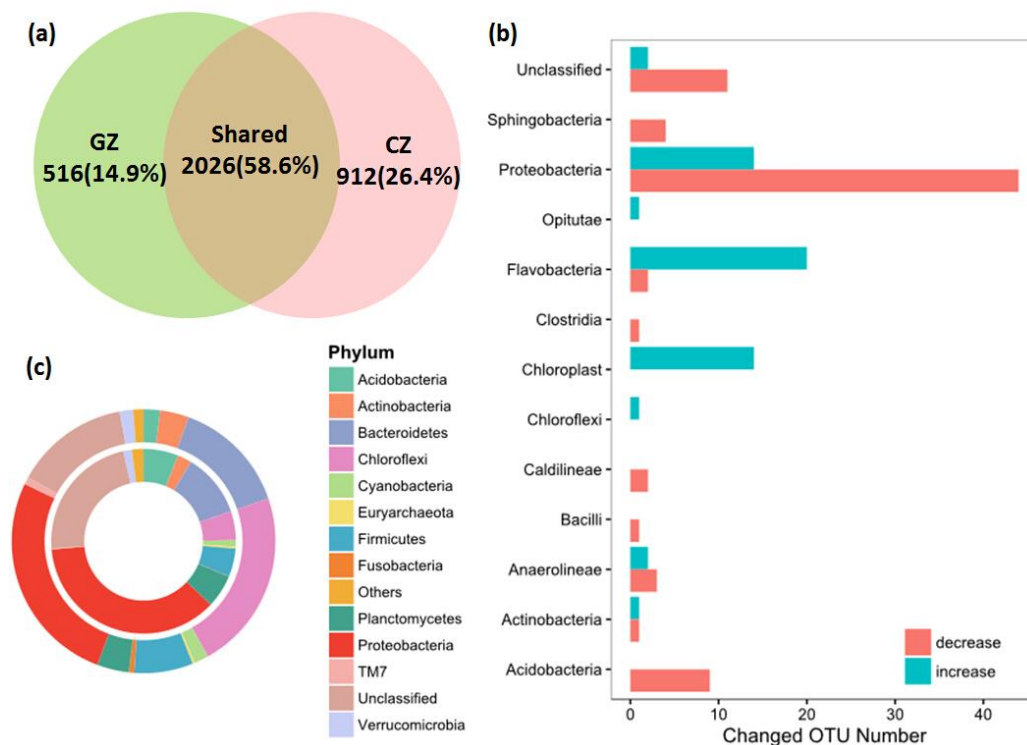


Fig. 2-6 Shared and unique OTUs detected at both GZ and CZ. (a) Venn diagram shows the number of shared and unique OTUs; (b) Shared numbers of OTUs with significant changes at the phylum level at GZ; (c) The phyla those unique OTUs belonged to; Outside circle presents the unique OTUs at GZ, and inner circle presents the unique OTUs at CZ.

图 2-6 龙须菜栽培区及对照区共有及独有的 OTU 情况

(a) 韦恩图表明共有及独有的 OTU 数目; (b) 在龙须菜栽培区门水平显著变化的共有 OTU; (c) 独有 OTU 所属的门: 外环代表龙须菜栽培区的独有 OTU, 内环代表对照区的独有 OTU

To demonstrate the effects of *G. lemaneiformis* cultivation on microbial community composition, further analyses were performed to identify the significantly changed shared OTUs and unique OTUs in the water and sediment samples at GZ and CZ (Fig. 2-6). Of all these shared OTUs, *G. lemaneiformis* cultivation caused an increase in relative abundances of 20 OTUs and a decrease of two OTUs in the surface water, an increase of 23 OTUs and decrease of seven OTUs in the bottom water, and an increase of 12 OTUs and a decrease of 69 OTUs in the sediment. For those significantly increased OTUs at GZ, higher abundances of OTU-10, OTU-93, OTU-421, OTU-28, OTU-164, OTU-492 and OTU-1281 associated with genus *Polaribacter*, *Crocinitomix*, *Litorimonas*, *Arenibacter* or *Maribacter* were observed.

Also, increased OTUs at GZ such as OTU-8, OTU-73, were affiliated with genera *Sulfitobacter* and *Leisingera*, respectively. For those OTUs decreased at GZ, OTU-84 and OTU-2807 were associated with *Thiopfundum*, and OTU-517 was associated with *Pelobacter*. For those unique OTUs, higher abundances of OTU-17 and OTU-56 associated with *Sulfurimonas*, OTU-2071 associated with *Desulfovibrio* and more than 10 OTUs affiliated with genus *Lewinella* were observed at GZ (Table 2-3), while for unique OTUs at CZ, many of them were associated with *Alcanivorax* (OTU-146), *Idiomarina* (OTU-146), *Thiohalophilus* (OTU-1301, OTU-2930, OTU-1697 and OTU-3412) and *Desulfopila* (OTU-1937) (Table 2-4).

Table 2-3 Unique OTUs and their affiliated metabolic potentials in the *Gracilaria* cultivation zone

表 2-3 龙须菜栽培区独有 OTU 及其相关的代谢特征

Genus	GZ unique OTUs	Metabolic potential of affiliated microorganisms	References
		Using sulfur, H ₂ or reduced sulfur compounds to grow chemolithoautotrophically	
<i>Sulfurimonas</i>	OTU 17, OTU 56		(Grote et al., 2012)
<i>Sulfurospirillum</i>	OTU 962, OTU 2283	Sulfur-reducing bacteria	(Kodama et al., 2007a)
<i>Desulfovibrio</i>	OTU 2071	Sulfate-reducing bacteria	(Heidelberg et al., 2004)
<i>Thiohalomonas</i>	OTU1567	Sulfur-oxidizing bacteria	(Mori et al., 2015)
	OTU574, OTU1013, OTU1019, OTU2409, OTU2321, OTU2326, OTU2777, OTU2882, OTU3397, OTU3678, OTU3852	Chemo-organotrophic and able to degrade starch and cellulose	(Khan et al., 2007a)
<i>Lewinella</i>			
	OTU 845, OTU 1778, OTU 2390, OTU 3716	Isolated from a seaweed	(Fukunaga et al., 2009)
<i>Phycisphaera</i>			
	OTU3273	Having the ability to oxidize organic compounds and metals	(Cologgi et al., 2014)
<i>Geobacter</i>			

Table 2-4 Unique OTUs and their affiliated metabolic potentials in the control zone

表 2-4 对照区独有 OTU 及其相关的代谢特征

Genus	CZ unique OTUs	Metabolic potential of affiliated microorganisms	References
<i>Alcanivorax</i>	OTU146	Alkane-degrading marine bacteria	(Kasai et al., 2002)
<i>Idiomarina</i>	OTU 3630	Use amino acid catabolism to obtain carbon and energy	(Hou et al., 2004)
<i>Thiohalophilus</i>	OTU 1301, OTU 2930, OTU 1697, OTU 3412	Obligatorily chemolithoautotrophic sulfur-oxidizing bacteria Chemoheterotrophic	(Sorokin et al., 2007)
<i>Sphingomonas</i>	OTU 533	strictly aerobic bacteria with some species as coral and/ human pathogen	(Maragakis et al., 2009a; Richardson et al., 1998)
<i>Desulfopila</i>	OTU 1937	Sulfate-reducing bacteria with the Wood–Ljungdahl pathway for CO ₂ fixation.	(Gittel et al., 2010)
<i>Planctomyces</i>	OTU1195	Many planctomycetes for anammox metabolism	(Callaghan et al., 2009b)

The response ratio analysis further indicated the effect of *G. lemaneiformis* cultivation on microbial community composition at the OTU level, revealing significant changes in surface water, bottom water and sediment at GZ. In surface and bottom water, the relative abundance of significantly changed OTUs increased at GZ by 89.9% and 72%, respectively (Fig. 2-7a, b). The relative abundance of significantly changed OTUs in sediment decreased by 84.5% at GZ (Fig. 2-7 c).

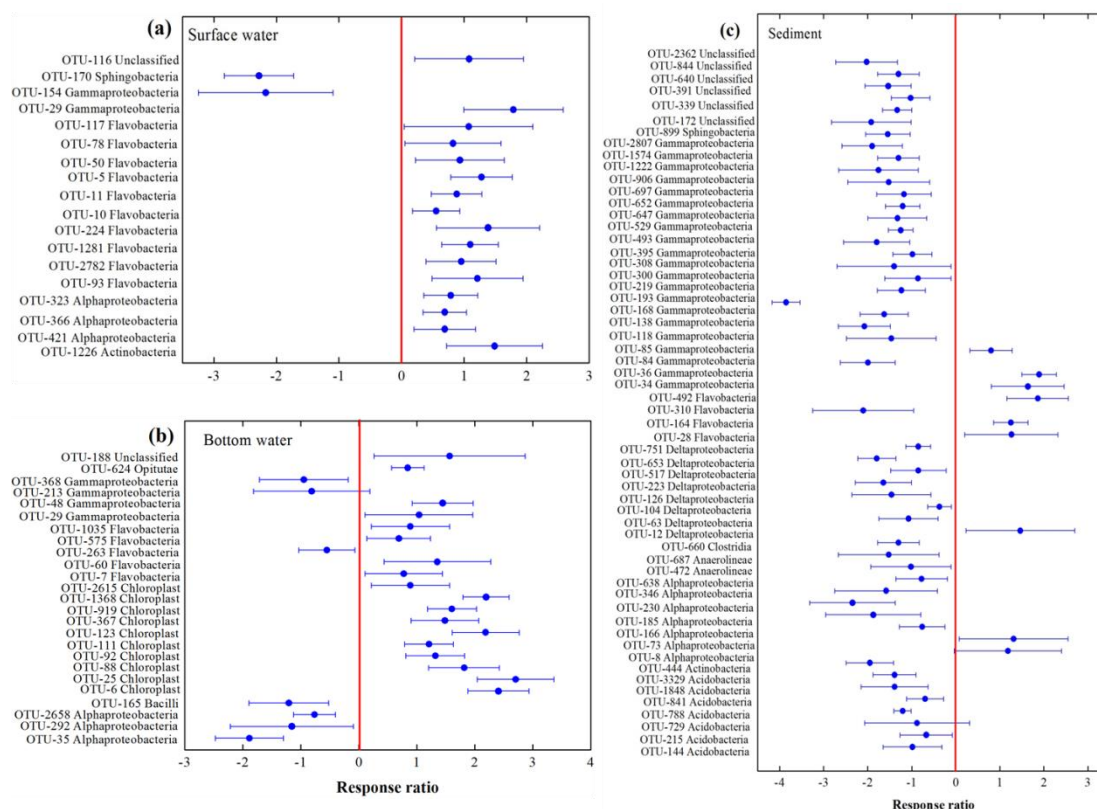


Fig. 2-7 Significantly changed OTUs in surface water (a), bottom water (b) and sediment (c) at GZ analyzed by response ratio at 95% confidence interval.

图 2-7 龙须菜栽培区表层水体(a)、底层水体(b)和沉积物(c)中显著变化 OTU 的响应比分析

The phylogenetic tree revealed the characteristics and distribution patterns of dominant microbial groups (relative abundance > 1%) in the water column and sediment at both zones using the neighbor-joining method of MEGA 5. In the surface water, OTUs belonging to *Aestuariibacter*, *Crocinitomix*, *Polaribacter*, *Tenacibaculum*, unclassified-Flavobacteriaceae and unclassified-Microbacteriaceae exhibited higher abundances ($p < 0.05$) at GZ than at CZ (Fig. 2-8). In the bottom water, OTUs belonging to *Aestuariibacter*, *Bacillariophyta* and *Haliea* showed higher abundances ($p < 0.05$) at GZ compared to CZ (Fig. 2-9). The OTUs in sediment affiliated with *Arenibacter*, *Glaciecola*, *Leisingera*, *Sulfitobacter* and *Peredibacter* had higher abundance ($p < 0.05$) at GZ than at CZ, while lower abundant OTUs in sediment were associated with *Desulfobulbus*, *Thiohalomonas* and unclassified-Gammaproteobacteria at GZ compared to CZ (Fig. 2-10). Therefore, our analyses of the microbial community composition from different taxonomic levels

indicated that *G. lemaneiformis* cultivation significantly changed the phylogenetic diversity and composition as well as the distribution of microbial communities in both water column and sediment.

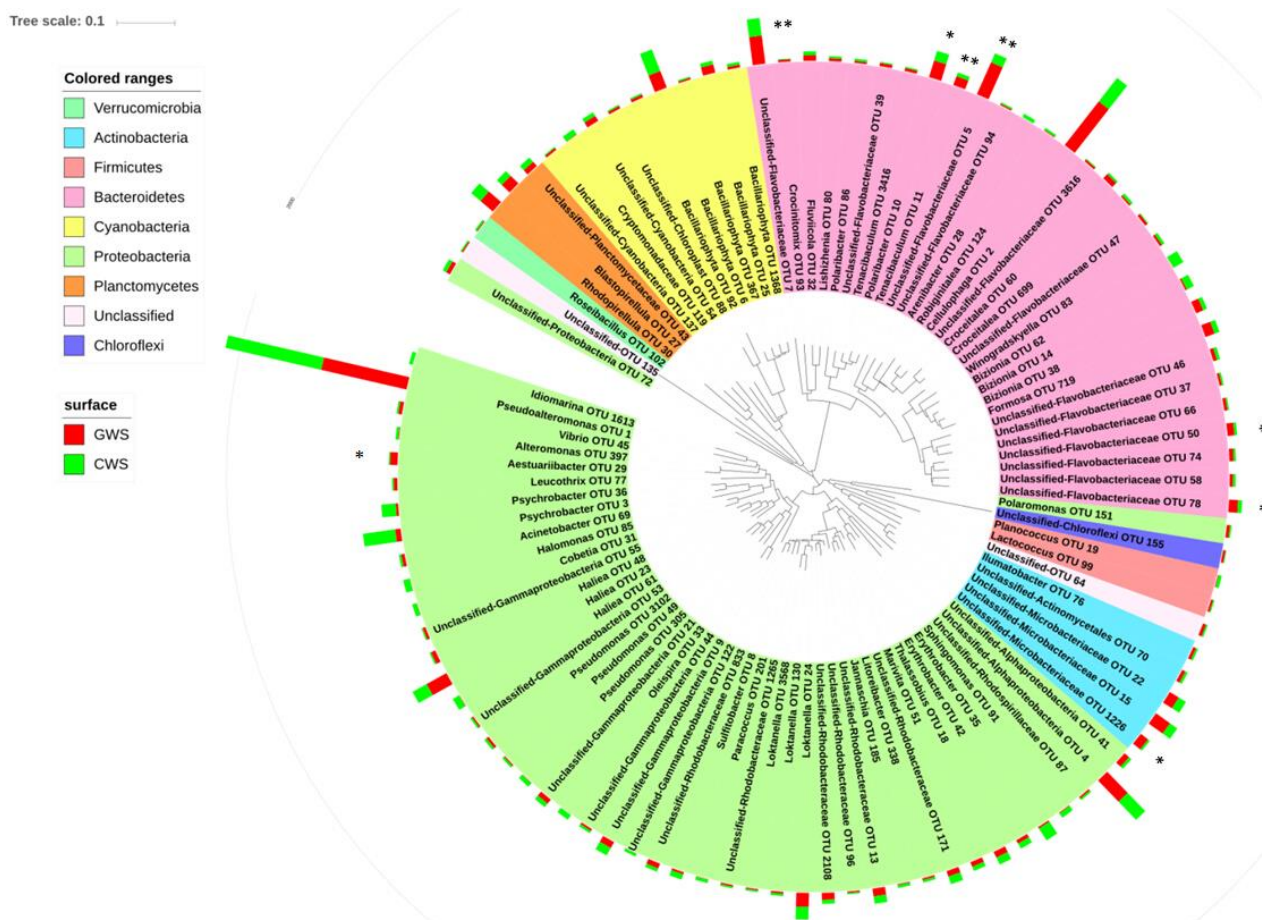


Fig. 2-8 Circular phylogenetic tree based on dominant OTUs (relative abundance >1%) examined in the surface water. The color ranges of OTUs represent different phyla. The relative abundance of OTUs in the surface water at GZ and CZ was shown as red bars and green bars, respectively. *: $p < 0.05$, **: $p < 0.01$ by ANOVA test.

图 2-8 基于表层水体优势 OTU (相对丰度>1%)构建的环形系统进化树(不同颜色代表 OTU 属于不同门: 红色表示龙须菜栽培区 OTU 的相对丰度, 绿色代表对照区表层水体 OTU 的相对丰度;龙须菜栽培区和对照区 OTU 相对丰度变化的 ANOVA 检验: * $p < 0.05$, ** $p < 0.01$)

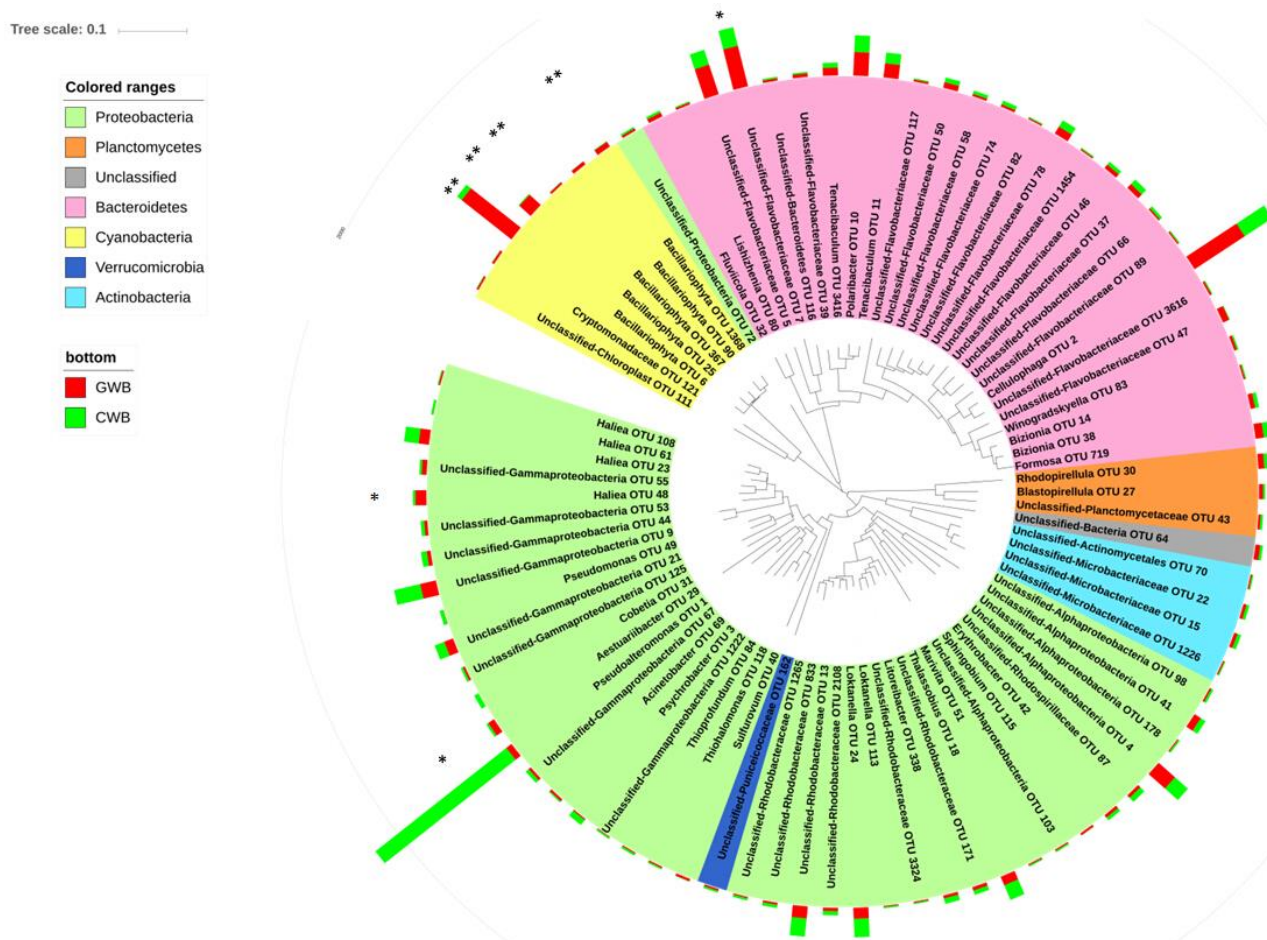
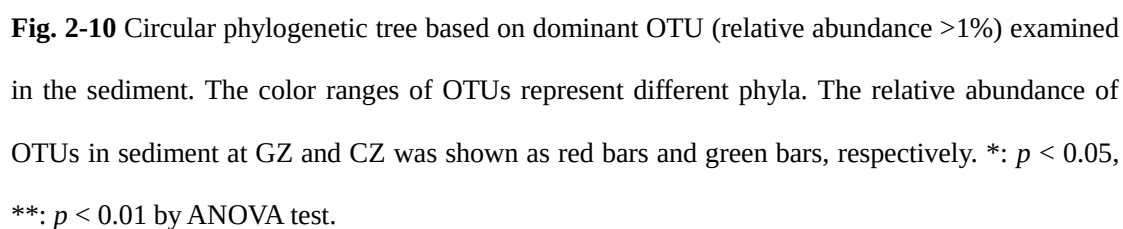


Fig. 2-9 Circular phylogenetic tree based on dominant OTU (relative abundance >1%) examined in the bottom water. The color ranges of OTUs represent different phyla. The relative abundance of OTUs in bottom water at GZ and CZ was shown as red bars and green bars, respectively. *: $p < 0.05$, **: $p < 0.01$ by ANOVA test.

图 2-9 基于底层水体优势 OTU (相对丰度>1%)构建的环形系统进化树(不同颜色代表 OTU 属于不同门: 红色表示龙须菜栽培区 OTU 的相对丰度, 绿色代表对照区表层水体 OTU 的相对丰度;龙须菜栽培区和对照区 OTU 相对丰度变化的 ANOVA 检验: * $p < 0.05$, ** $p < 0.01$)



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2.3.3 Linkages between microbial community structure and environmental factors

To reveal effects of environmental factors on the microbial community composition and structure, Pearson correlation analysis and canonical correspondence analysis (CCA) were applied. The linkage between significantly changed OTUs and environmental variables was well demonstrated by heat map plots based on the Pearson correlations between OTU abundance and environmental factors (Fig. 2-11).

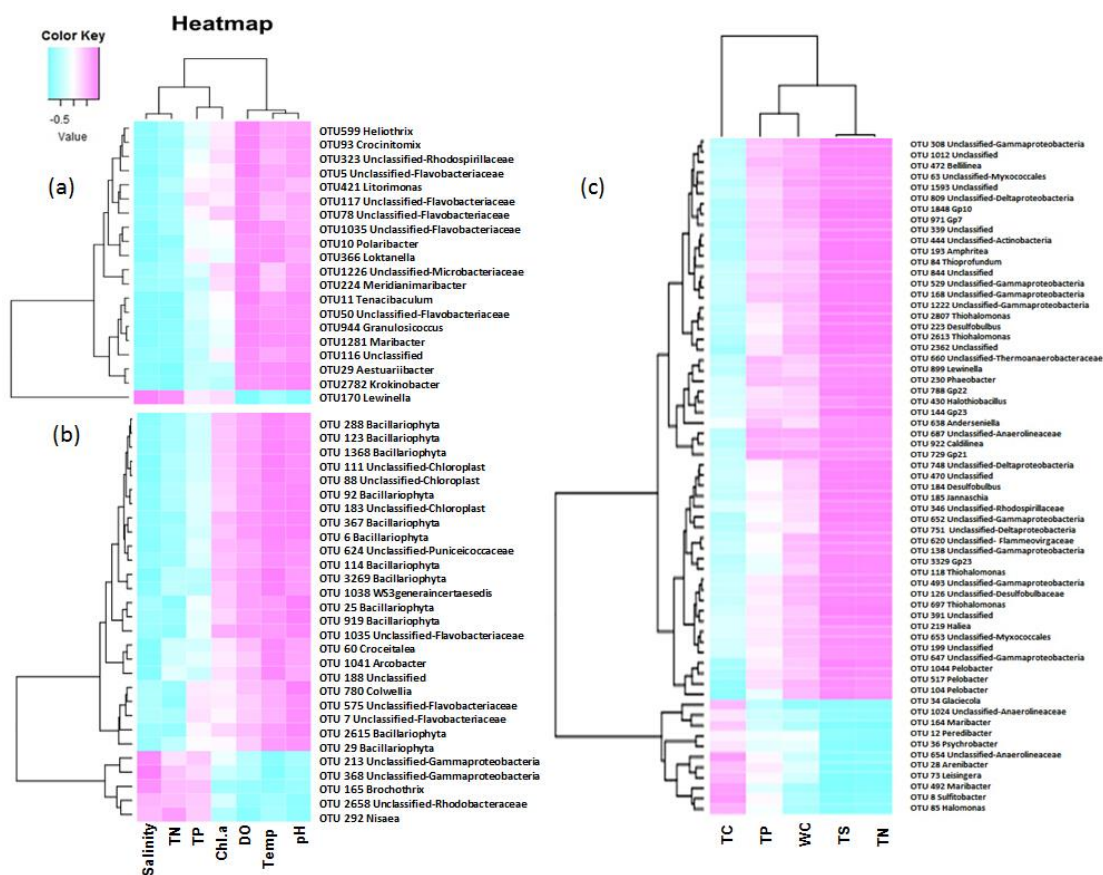


Fig. 2-11 Heat maps indicated correlation between significantly changed OTUs and environmental factors in surface water (a), bottom water (b) and sediment (c). Values of Pearson's correlation coefficients are color coded in the heat map legends. Temp, temperature; TN, total nitrogen; TP, total phosphorus; Chl-a, chlorophyll-a; DO, dissolved oxygen; TC: total carbon; TS: total sulfur; WC: water content.

图 2-11 表层水体(a)、底层水体(b)及沉积物(c)中显著变化的 OTU 与环境因子的关系
热图中的颜色图例表示 Pearson's 相关系数值。环境因子缩写如下: Temp, 温度; TN, 总氮; TP, 总磷; Chl-a, 叶绿素-a; DO, 溶解氧; TC: 总碳; TS: 总硫; WC: 含水率。

Significantly changed OTUs exhibited a generally positive correlation with temperature, pH and DO ($p < 0.01$, $r > 0.6$), and a negative correlation with TN and salinity ($p < 0.01$, $r < -0.8$) in both surface and bottom water samples. In addition, the abundance of most significantly changed OTUs in sediment showed a strongly positive relationship with TN and TS concentrations ($p < 0.01$, $r > 0.8$), and a negative correlation with TC ($p < 0.01$, $r < -0.6$). Also, the result of CCA showed the microbial community structure was strongly correlated with temperature, DO, pH and Chl-*a* at GZ, while it was most closely associated with salinity and TN at CZ. Furthermore, the CCA results showed that temperature, salinity and pH ($p < 0.05$) were key environmental variables in shaping microbial community structure (Fig. 2-12), which is consistent with the Pearson correlation analysis.

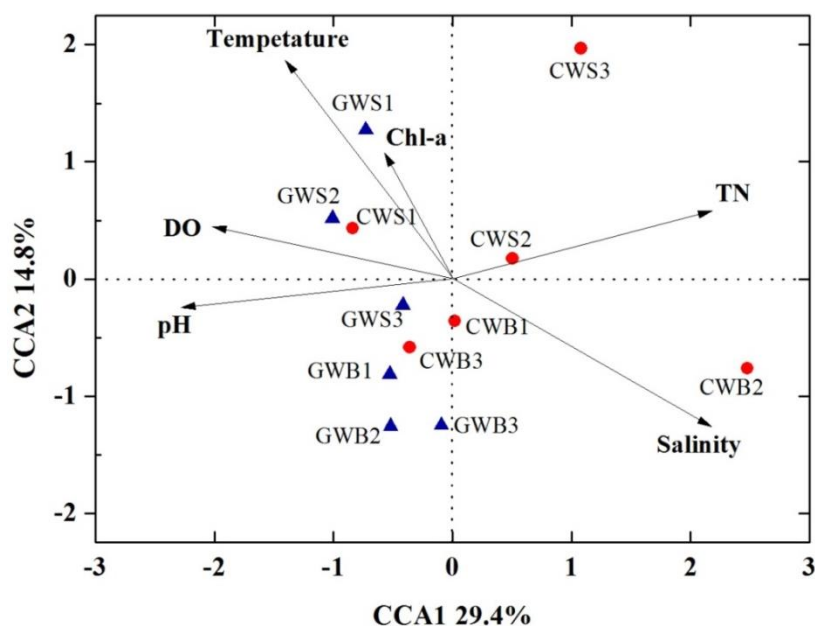


Fig. 2-12 CCA correlated microbial communities and environmental factors in water column. TN, total nitrogen; TP, total phosphorus; Chl-*a*, chlorophyll-*a*; DO, dissolved oxygen; GWS1, GWS2, GWS3: surface water of GZ; GWB1, GWB2, GWB3: bottom water of GZ; CWS1, CWS2, CWS3: surface water of CZ; CWB1, CWB2, CWB3: bottom water of CZ.

图 2-12 水体微生物群落与环境因子的 CCA 分析 (龙须菜栽培区表层水样: GWS1, GWS2, GWS3; 底层水样: GWB1, GWB2, GWB3; 对照区表层水样: CWS1, CWS2, CWS3; 底层水样: CWB1, CWB2, CWB3)

2.4 Discussion

2.4.1 Cultivation of *Gracilaria* benefits the environment

As an essential part of mariculture, seaweed cultivation has developed as a healthy mariculture pattern, which can bring ecological, environmental and economic benefits simultaneously. For example, *G. lemaneiformis* cultivation improves water quality by removing large amounts of nutrients related to eutrophication (e.g., $\text{NH}_4^+\text{-N}$, $\text{NO}_3^-\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{PO}_4^{3-}\text{-P}$) (Xu et al., 2008; Yang et al., 2006b), increasing dissolved oxygen levels and buffering ocean acidification (Chen et al., 2016; Zou and Gao, 2009). Also, *G. lemaneiformis* cultivation can inhibit the growth of phytoplankton and harmful algae (Lu et al., 2011; Wang et al., 2007b). Because of the high bioremediation efficiency (Wu et al., 2015), *G. lemaneiformis* has been introduced to co-culture with animals as an integrated multi-trophic aquaculture (IMTA) (Han et al., 2012). In addition, *G. lemaneiformis* is good agar material (Chirapart et al., 1995; Li et al., 2008) used as food (Wen et al., 2006), industrial and pharmaceutical products, and serves as an economical feedstuffs for mariculture such as fish and abalone (Qi et al., 2010; Xu et al., 2011; Xuan et al., 2013). Due to great ecological and economic efficiencies brought by large-scale cultivation of *G. lemaneiformis*, residents at Nan'ao Island have cultivated this seaweed since early 21th century.

2.4.2 *Gracilaria* cultivation altered the microbial community structure through changes in environmental factors

In this study, we first hypothesized that the cultivation of the seaweed *G. lemaneiformis* would significantly alter the diversity, composition and structure of microbial communities in both the water column and sediment through changes in environmental properties such as increased pH and DO and decreased inorganic nutrients. Species-energy theories predict that low concentrations of oxygen, nutrients and solar energy could restrict primary production along the water column, which could lead to declines in species richness (Mason et al., 2008; Wright, 1983). Since many microbes in the ocean require oxygen for survival and growth, especially in

important redox processes necessary for basic metabolism, DO in the water column is crucial to microbial communities in coastal ecosystems. Thus, higher concentrations of DO in oceans could relate to increased phylogenetic and functional diversity of microbial communities (Bryant et al., 2012). Also, DO concentration is one of the universal indices for water quality assessment, indicating that high DO concentration may be a characteristic of a good living environment for aquatic organisms (Kannel et al., 2007; Sánchez et al., 2007). Additionally, pH is one of the most important chemical environmental factors that affect microbial communities in many habitats (Kroeker et al., 2013). Anthropological activity has caused a decline in pH of seawater, termed ocean acidification, which has effected on microbial community diversity and microbe-driven biogeochemical cycling (Liu et al., 2010; Witt et al., 2011). Because seaweed has high rates of CO₂ fixation through photosynthesis (Gao and McKinley, 1994; Iñiguez et al., 2015), the pH in seaweed surrounding water is higher than other natural sea area (Suzuki et al., 2015). In addition, the pattern and concentration of nutrients have direct influence on microbial metabolism such as respiration and growth in marine ecosystems (Stelzer et al., 2003; Zeigler Allen et al., 2012). For example, dissolved organic matters (DOM) are released by algae and many heterotrophic organisms and can be utilized directly by microbes, which together constitute a microbial loop in aquatic ecosystems (Anderson and Ducklow, 2001; Azam et al., 1994; Jiao et al., 2010). The gradient of DOM in marine ecosystem could affect the diversity, structure and succession patterns of microbial communities through changes in microbial cells, rRNA and DOM-responsive gene transcripts (McCarren et al., 2010; Stocker, 2012). Those previous studies indicated that seaweed cultivation affected microbial communities in terms of diversity and function in aquatic ecosystems.

Our results of this study showed significant differences in environmental factors between the two study zones even though they were geographically adjacent and experienced the same currents and climate regime. The seaweed cultivation zones showed a better water quality with higher DO and pH, and lower Chlorophyll-*a* (Chl-*a*) and TN compared to the control zone due to physiological activities and

metabolism of *G. lemaneiformis*. First, photosynthesis of *G. lemaneiformis* could uptake CO₂ and release O₂ into water so that pH and DO greatly increased in water column at GZ. Second, the concentration of inorganic nutrients decreased at cultivation zone due to nutrient uptake in *G. lemaneiformis*, impacting the diversity and composition of microbial communities (Donachie et al., 2001; Kirchman, 1994; Meyer-Reil and Köster, 2000). Third, highly abundant algal-rich or agar-digesting potential microbial groups were observed at GZ. Additionally, the analyses of correlation between environmental factors and microbial communities indicated that DO, pH, TN, TS, and Chl-*a* as well as salinity and temperature had strong impacts on the microbial community structure. Therefore, our results indicate that *G. lemaneiformis* cultivation alters environmental parameters with a decrease in nutrition and an increase in oxygen and pH, which may shape the microbial community structure in both water and sediment between GZ and CZ.

Meanwhile, our results could be further explained by the species-sorting perspective of the meta-communities framework, which emphasizes that local microbial communities adapt to new environmental conditions by adjusting their composition (Leibold et al., 2004; Van der Gucht et al., 2007). This phenomenon is more often observed in aquatic ecosystems (Langenheder and Szekely, 2011; Logue and Lindstrom, 2010; Van der Gucht et al., 2007). Divergences in alpha- and beta-diversity, composition and dominant genera between GZ and CZ indicated the different sensitivities of microbial taxa in response to perturbations caused by *G. lemaneiformis* cultivation. Some of the altered environmental properties, such as decreased inorganic nutrients, increased DO and pH introduced by cultivation could be the driving force in changing the microbial communities in water column and sediment at GZ. Thus, our results support the first hypothesis that seaweed cultivation would alter the diversity, composition and structure of microbial communities through changes in environmental properties (e.g., increased pH and DO, decreased nutrients). However, as the shifts in microbial community composition are often linked with changes in functional redundancy, further studies are needed to understand the changes of ecological functional processes associated with certain microbial taxa.

2.4.3 Certain microbial groups were enriched in response to *Gracilaria* cultivation

Our second hypothesis is that certain microbial groups would be stimulated in response to seaweed cultivation. Previous studies on marine microbial communities showed that Alpha-proteobacteria, Gamma-proteobacteria, Bacteroidetes and Cyanobacteria were dominant groups in seawater of different seasons all around the world (Fuhrman et al., 2015; Gilbert et al., 2009; Tinta et al., 2015). By linking the diversity, composition and structure of microbial communities with their ecological functions, research was conducted using metagenomic sequencing (Yooseph et al., 2010), COG and SEED annotations (Burke et al., 2011a), GeoChip (Lu et al., 2012) and network analysis (Fuhrman, 2009). Those studies showed that microbial community assembly was largely based on functions rather than phylogenetic similarity and could adapt to new environments by changing community structure and composition (Langenheder and Szekely, 2011; Liang et al., 2015; Logue and Lindstrom, 2010). In this study, our results showed that the most dominant phylum is Proteobacteria, including δ - Proteobacteria and γ - Proteobacteria, which exhibited higher abundance at CZ and generally function in sulfate-reduction, sulfur-oxidization and nitrate assimilation (Allen et al., 2001; Asami et al., 2005; Castine et al., 2009; Kawahara et al., 2008). The second dominant phylum Bacteroidetes was in a higher proportion in water column of GZ, which is widely distributed in marine waters and has the potential of degrading biopolymers (Asami et al., 2005; Bissett et al., 2006). Other common phyla such as Cyanobacteria, Actinobacteria, Planctomycetes and Acidobacteria, account for about 12% in all examined phyla. Such results are generally consistent with previous studies (Kawahara et al., 2009; Michelou et al., 2013; Staufenger et al., 2008).

Also, we found a significant discrepancy of genera in surface water, bottom water and sediment samples between two study zones. In surface water, GZ had larger percentages of *Crocinitomix* (Bowman et al., 2003), *Leucothrix* (Brock, 2006a), *Polaribacter* (Fukui et al., 2013) and unclassified-*Flavobacteriaceae* (Sakai et al.,

2002), which are considered as the potential microorganisms for degrading seaweed agar. In the bottom water, seaweed-associated genera *Croceitalea* (Lee et al., 2008) exhibited higher abundances at GZ. In sediment, agar-digesting bacteria such as *Glaciecola* (Romanenko et al., 2003), *Arenibacter* and *Maribacter* (McBride, 2014) showed higher abundances at GZ probably due to *G. lemaneiformis* detritus in sediment (Burke et al., 2011b; Hehemann et al., 2012). Genera *Desulfobulbus* (Kuever, 2014), *Pelobacter* (Lovley et al., 1995) and *Thiohalomonas* (Shao et al., 2011) had higher abundances at CZ, and they play important roles in cycling of S, N and organic compounds in marine sediment and may be correlated with high concentrations of TS and TN in the CZ sediments. In addition, some specific abundant genera at GZ may have environmentally beneficial potentials. For example, *Peredibacter* (Lu and Cai, 2010) was shown to control the growth of one kind of fish pathogen *Typhimurium*. Also, some potential pathogens, such as *Vibrio* (Austin and Zhang, 2006; Hubbard et al., 2016) and *Arcobacter* (Lehner et al., 2005) were abundant at CZ but not GZ sediments. Such a divergence in abundant genera could be explained by different environmental properties such as lower nutrients and higher algal polysaccharide at GZ, implying a healthier community composition at the seaweed cultivation zone.

With the respect to those enriched OTUs at GZ, their affiliated microorganisms were isolated from seaweed or have the ability to digest algal polysaccharides of seaweed, such as genera *Polaribacter*, *Crocinitomix* (Bowman et al., 2003), *Litorimonas* (Nedashkovskaya et al., 2013), *Arenibacter* and *Maribacter* (McBride, 2014). Also, increased OTUs at GZ are affiliated with microorganisms for sulfite-oxidization (*Sulfitobacter* OTU 8) (Pujalte et al., 2014) and methyl halides-oxidization (*Leisingera* OTU 73) (Schaefer, 2002). For those OTUs decreased at GZ, OTU 84 and OTU 2807 affiliated with obligatorily chemolithoautotrophic sulfur-oxidizing bacteria *Thiopfundum* (Mori et al., 2011), and OTU 517 associated with *Pelobacter* (Lovley et al., 1995) have the ability to reduce Fe (III) and S (0) with bioremediation potential in contaminated environments. A large number of unique OTUs affiliated with genus *Sulfurimonas* were highly abundant at GZ, which use S, H₂ or reduced S compounds to grow chemolithoautotrophically in marine sediments

(Grote et al., 2012). In addition, many unique OTUs at GZ have the metabolic potential for sulfur/sulfate reduction (Kodama et al., 2007b) and organic compounds degradation (e.g., starch, cellulose) (Khan et al., 2007b), or their affiliated microorganisms were isolated from seaweed (Fukunaga et al., 2009). For unique OTUs at CZ, many of them exhibited potentials in alkane-degrading (Kasai et al., 2002), anammox metabolism (Callaghan et al., 2009a) or pathogenicity to human (Maragakis et al., 2009b). The results confirmed our second hypothesis that certain microbial groups associated with the seaweed at the genus and OTU levels are responsive to *G. lemaneiformis* cultivation.

2.5 Chapter Summary

Based on the above results and current knowledge, we have developed a conceptual model (Fig. 2-13) for future studies, especially on relationships among seaweed processes, microbial communities and their environments in the *G. lemaneiformis* cultivation ecosystem. *G. lemaneiformis* cultivation utilized nutrients (e.g., N, P) for growth and took up CO₂ for photosynthesis, resulting in increased pH and DO, and decreased nutrients. Also, the seaweed could secret specific compounds (e.g., algal polysaccharides) and produce detritus in the ecosystem. Those changes might not only lead to improved water quality, but also alter the microbial community in terms of α -diversity, composition (e.g., enriched *Arenibacter*, *Croceitalea*, *Glaciecola*, *Leucothrix* and *Maribacter* groups) and structure. Therefore, this study not only provides new insights into our understanding the effect of seaweed cultivation on microbial communities, but also guides future studies on intricate interactions among seaweed processes, microbial communities and their environments in this mariculture system.

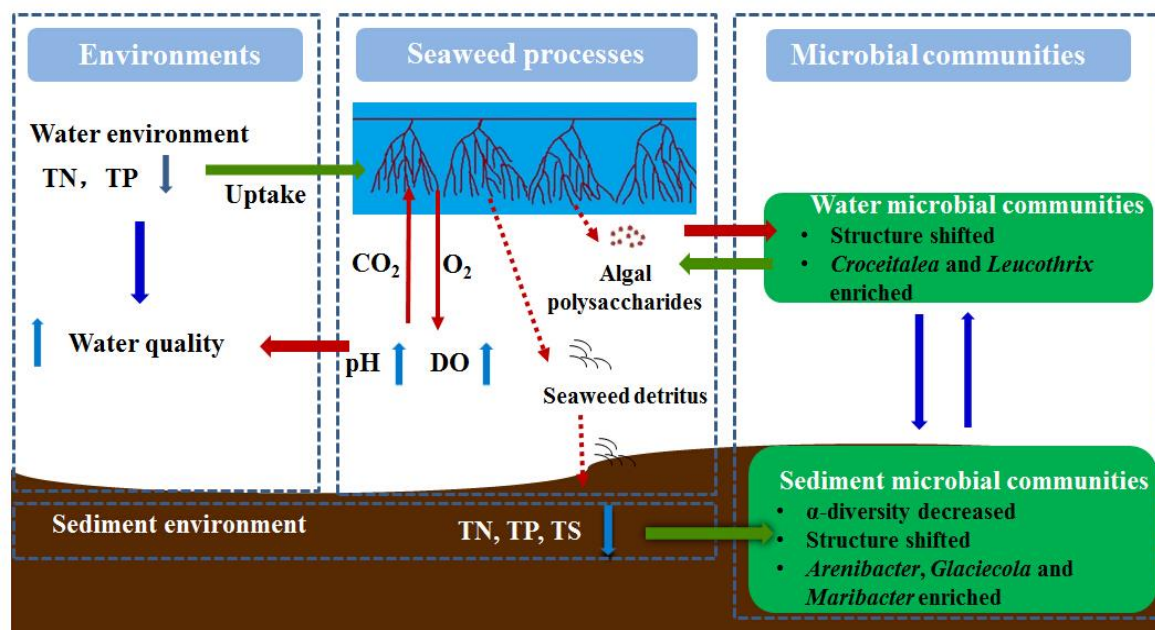


Fig. 2-13 A conceptual model for summarizing major results in this study and proposing future studies on relationships among seaweed processes, microbial communities and their environments in the *G. lemaneiformis* cultivation ecosystem.

图 2-13 龙须菜栽培对微生物群落结构影响的概念模型

Chapter 3: The dynamics of microbial community structure in Shen'ao Bay

3.1 Introduction

Seaweeds play an important role in the structure and biodiversity of coastal communities. They are not only a major primary producer, but also serve as one of the most important blue-carbon sinks (Duarte et al., 2013). Large-scale cultivation of economic seaweed has been promoted worldwide, especially in Asia in order to protect seaweed habitats and utilize their economic and ecological values (Taelman et al., 2015). In China, useful seaweeds such as *Laminaria*, *Undaria*, *Sargassum*, *Gracilaria*, *Porphyra*, *Enteromorpha* and *Ulva*, have been cultivated along the coasts. For example, the red alga (Rhodophyta), *Gracilaria lemaneiformis*, is documented as a powerful bioremediator to alleviate eutrophication and improve water quality in coastal ecosystems (Fei, 2004; Yang et al., 2015). It is able to take up excessive nutrients from the environment, release oxygen and decrease ocean acidification in the process of photosynthesis (Chen et al., 2016; Koch et al., 2013). The site of this investigation was in Nao'an county, where the large-scale cultivation of *G. lemaneiformis* is a key industry to the local economy (Yang et al., 2015; Zhang et al., 2009).

Since microbes play a major role in marine ecosystems, understanding seaweed cultivation on the microbial communities is fundamental. Previous studies were mainly focused on investigating bacterioplankton and the microbial community in sediment and on seaweed surfaces. Little is known about microbial communities and their responses to large-scale seaweed cultivation throughout the growth cycle. Recently, a study of microbial communities in *Gracilaria* cultivation ecosystems showed that seaweed decreased the α -diversity of microbial communities and enriched the associated microbial taxa by releasing O₂ and reducing nutrients and thereby improving water quality (Xie, 2017).

In this study, we investigated the diversity and structure of the microbial community in both the *Gracilaria* cultivation zone and a natural sea zone at Nan'ao Island for consecutive eight months in order to understand the dynamics and

succession within those communities. Water samples were collected at different depths (surface water, 1-m depth and near the bottom) through the growth cycle of the seaweed *Gracilaria*. The following questions were addressed: (i) how the diversity and structure of microbial communities responded to 5-months seaweed growth? (ii) whether did the diversity of microbial communities change with depths? (iii) did seaweed cultivation affect the microbial communities assembly? DNA extracted from water samples was analyzed by sequencing 16S rRNA gene amplicons with the Illumina MiSeq technology (San Diego, CA, USA) to test the following hypotheses: (i) *G. lemaneiformis* cultivation would significantly alter the microbial community assembly; and (ii) seaweed growth would lead to higher succession rates of the microbial communities. Our results showed an obvious pattern of changes in microbial communities between the *Gracilaria* cultivation and control zones. Seaweed cultivation led to lower α -diversity and higher temporal microbial community turnover.

3.2 Materials and methods

3.2.1 Sampling site and sample collections

For this study three sampling sites in each of the two study zones near Shen'ao Bay, Nan'ao Island coastal area ($23^{\circ} 28' N$, $117^{\circ} 06' E$) were chosen. One zone was in the *Gracilaria lemaneiformis* cultivation area (GZ) and the other serve as control (CZ). 2.5 L of seawater were collected from the surface water (0.5-m depth), 1m depth and the bottom water (0.5-m above sediment) monthly from December 2014 to July 2015. Water samples were immediately serially filtered through 10 and 0.2 μm pore-size membrane filters. Geochemical variables (temperature, salinity, pH, DO, water depth and transparency) were measured *in situ* using the Potable Water Quality Analyzer-YSI 556 systems (YSI Incorporated, USA), a Secchi disc and a Digital Depth Sounder (Speedtech, USA). Chemical properties (total nitrogen, total phosphorus, NO_3^- , NO_2^- , NH_4^+ , PO_4^{3-} , Chl. a) were measured in the lab immediately after sampling (AQSIQ, 2007). All filters were preserved in liquid nitrogen onsite and

then transferred to a -80°C freezer for processing.

3.2.2 DNA extraction, amplification and MiSeq sequencing of 16S rRNA gene amplicons

Microbial genomic DNA was extracted from each filter using Omega E.Z.N.A.® Water DNA Kit (Omega Bio-Tek, USA) as per the manufacturer's instructions. The DNA quality and quantity were measured using PicoGreen through a FLUOstar Optima fluorescence plate reader (BMG LABTECH, Jena, Germany) and purified using the MicroElute® Cycle-Pure Kit (Omega Bio-Tek, USA) to meet the criteria as follows: $A_{260nm}/A_{280nm} > 1.8$, $A_{260nm}/A_{230nm} > 1.8$. The dynamics and assembly of microbial communities were analyzed using MiSeq sequencing of the 16S rRNA gene amplicons. The V4 of 16S rRNA genes was amplified in triplicate for each sample with the primer pair 515F (5'-GTGCCAGCMGCCGCGGTAA-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3') according to the following PCR conditions: an initial denaturation at 94 °C for 5 min, followed by 28 cycle at 94 °C for 45 s, 62 °C for 45 s, and 72 °C for 1-min, with a final extension at 72 °C for 10 min. The triplicate PCR products of each sample were pooled together and quantified by μ DropTM Plate reader (Thermo Scientific, USA). Then an equal amount (e.g. 200 ng) of DNA from each sample was combined and purified by Omega Gel Purification Kit (Omega Bio-Tek, USA). The purified DNA was loaded on a 500 cycles (2×250 bp paired ends) kit and run onto a Miseq machine (Illumina, San Diego, CA, USA) at School of Minerals Processing and Bioengineering, Central South University, China.

Raw sequences were assigned to their corresponding samples each by the unique barcode and trimmed by Btrim with an average quality score greater than 20 and a minimum length of 100 bp (Kong, 2011). The paired reads were joined using FLASH with at least a 10 bp overlap and lower than 5% mismatches (Magoc and Salzberg, 2011). Sequences with length between 245 and 258 bp were subjected to chimeras detection using U-Chime (Edgar et al., 2011) after removing sequences containing N. Operational taxonomic units (OTUs) were generated by UCLUST (Edgar, 2010) and

assigned to microbial taxonomic groups by the RDP Classifier (Wang et al., 2007a). At last singletons were removed and the resample was rarefied at 13,113 sequences for downstream data processing. Finally, 521,143 sequences were obtained from both ends for all 180 samples. The above-mentioned steps were performed on the Galaxy pipeline (<http://zhoulab5.rccc.ou.edu/>) developed by Qin *et al.* (unpublished) at the Institute of Environmental Genomics, University of Oklahoma, USA.

3.2.3 Statistical analysis

Statistical analysis was carried out by R software (version 3.2.2; <http://www.r-project.org/>) using package *vegan* (Oksanen et al., 2011). The α - and β -diversity of microbial communities were analyzed using multiple indices and the Bray-Curtis distance between samples. Non-metric multidimensional scaling ordination (NMDS) based on Bray-Curtis distances were made to visualize the microbial distribution patterns of the two study zones (Kruskal, 1964). Non-parametric multivariate (Adonis) (Zapala and Schork, 2006) was performed to test the variances of the microbial community assembly between GZ and CZ. Linear regression was used to determine the relationships between sampling time/depth and similarity in microbial composition based on the Bray-Curtis distance. Scatterplots were applied to modeling the time-decay relationship using the equation: $\ln(S) = \text{constant} - w \ln(T)$, where S is the pairwise similarity in microbial community composition, w is the rate of community turnover and T is the time interval. Because some similarity values were equal to zero, we coded the similarity data by adding 0.00001 before log transformation (Horner-Devine et al., 2004). The significance of w values was calculated using one-sample t -test by bootstrapping (permuted 999 times) followed by a pairwise t -test (John Harte, 1999; Zhou et al., 2008). Canonical correspondence analysis (CCA) was used to identify the effect of environmental factors on microbial community assembly. Environmental factors with variance inflation factors less than 20 were selected for analyses. The significance of the CCA model and each environmental variable was tested by analysis of variance (ANOVA).

3.3 Results

3.3.1 Environmental characteristics in Shen'ao Bay

In the surface water (Table 3-1), DO and pH at GZ were significantly higher ($p < 0.05$) than those at CZ during the seaweed cultivation period (Jan.-May 2015). The concentrations of TP, TN and NO_3^- at GZ were lower during the cultivation period ($p < 0.05$), and higher after harvest (Jun.-Jul. 2015) than those in CZ. At 1-m depth (Table 3-2), DO at GZ was also higher and PO_4^{3-} was lower than those at CZ during the cultivation period ($p < 0.05$). TP was lower during the cultivation period and higher after harvest at GZ than that at CZ ($p < 0.05$). In the bottom water (Table 3-3), DO at GZ was higher and NH_4^+ was lower than those at CZ during the cultivation period ($p < 0.05$). Concentrations of TN, NO_2^- and NH_4^+ were higher at GZ after harvest compared to CZ ($p < 0.05$).

Linear regressions were performed to examine the relationships between environmental factors and seaweed cultivation time. In the surface water, positive relationships between environmental factors and cultivation time were observed for PO_4^{3-} , temperature and salinity, and a negative relationship was detected between DO and cultivation time. With respect to the positive relationships between environmental factors and cultivation time, however, the slope of GWS was steeper than that of CWS for PO_4^{3-} , while the slopes of CWS were steeper than GWS for the regression between temperature, or salinity and time ($p < 0.05$). The negative regressions between DO and cultivation time revealed that DO of CWS decreased faster than that of DO of GWS (Fig. 3-1).

Table 3-1 Environmental factors of surface water column at the two study zones. $p < 0.05$ was in bold. Abbreviation: GWS, surface water in *G. lemaneiformis* cultivation zones (GZ); CWS, surface water in control zones (CZ).

表 3-1 龙须菜栽培区和对照区表层水体主要环境因子的比较 ($p < 0.05$ 的环境因子加粗显示; GWS, 龙须菜栽培区表层水体; CWS, 对照区表层水体)

		Dec. 2014	Jan. 2015	Feb. 2015	Mar. 2015	Apr. 2015	May 2015	Jun. 2016	Jul. 2015
Temperature	GWS	18.60±0.35	14.53±0.06	14.40±0.17	15.30±0.10	18.60±0.00	23.23±0.15	26.93±0.21	29.53±0.12
	CWS	18.30±0.00	13.93±0.06	14.17±0.12	15.30±0.00	18.63±0.06	23.30±0.10	27.23±0.23	30.27±2.29
Salinity	GWS	30.73±0.01	29.66±0.08	31.00±0.02	32.09±0.05	33.68±0.01	34.14±0.02	30.41±0.19	31.71±0.02
	CWS	30.75±0.01	29.38±0.17	31.00±0.05	32.11±0.01	33.64±0.03	34.10±0.03	30.97±0.48	31.82±0.01
DO	GWS	8.41±0.30	10.95±0.71	9.58±0.62	8.94±0.20	5.61±0.45	9.27±1.60	8.56±0.50	11.83±0.36
	CWS	8.09±0.13	8.96±0.11	9.22±0.13	8.47±0.41	3.34±0.41	6.60±0.23	8.23±0.57	12.31±0.68
pH	GWS	8.14±0.11	8.13±0.07	8.40±0.11	8.18±0.05	8.21±0.07	8.25±0.02	7.77±0.12	8.15±0.04
	CWS	8.09±0.11	8.23±0.11	8.24±0.10	8.02±0.06	8.15±0.04	8.19±0.02	7.95±0.01	8.20±0.03
TN	GWS	41.28±5.14	23.71±4.86	37.06±2.97	16.77±2.02	21.36±3.80	24.58±0.33	29.79±3.50	27.79±8.00
	CWS	42.15±2.67	25.94±4.02	40.87±4.40	16.35±1.49	25.73±6.37	38.50±5.33	27.68±3.57	13.28±1.06
TP	GWS	3.30±0.38	2.28±0.17	4.64±0.44	2.87±0.45	0.79±0.39	3.83±0.43	2.41±0.27	0.92±0.25
	CWS	4.04±1.31	4.11±0.73	6.09±0.70	2.36±0.33	0.98±0.70	3.72±0.65	1.14±0.17	1.07±0.20
NO ₃ ⁻	GWS	19.72±2.64	14.40±5.20	8.95±1.22	8.78±1.66	5.08±1.33	9.23±1.48	7.39±1.25	5.98±1.69
	CWS	18.94±2.25	15.25±3.59	10.51±1.98	8.76±1.59	4.62±0.94	13.13±1.56	7.59±0.56	1.87±1.20
NO ₂ ⁻	GWS	0.55±0.05	0.81±0.08	0.42±0.04	1.46±0.15	0.24±0.02	0.24±0.16	0.27±0.13	0.42±0.07
	CWS	0.48±0.03	0.95±0.07	0.44±0.12	2.62±1.15	0.27±0.10	0.50±0.33	0.34±0.03	0.14±0.02
NH ₄ ⁺	GWS	NA	3.62±0.69	0.02±0.00	2.47±0.57	0.30±0.25	4.13±0.94	2.25±0.07	4.63±0.94
	CWS	NA	3.83±0.88	0.03±0.01	2.62±1.15	1.75±1.25	4.46±0.68	3.72±0.34	3.81±0.39
PO ₄ ³⁻	GWS	0.36±0.26	1.45±0.14	0.93±0.16	0.54±0.11	0.25±0.14	9.20±0.58	0.30±0.11	0.23±0.12
	CWS	0.35±0.12	1.64±0.06	1.39±0.15	0.69±0.30	0.34±0.16	7.94±0.45	0.44±0.01	0.31±0.11
Chl.a	GWS	1.29±0.79	0.51±0.09	2.02±0.31	1.02±0.90	1.45±0.69	0.54±0.13	5.74±1.85	12.53±3.03
	CWS	1.05±0.79	0.36±0.08	1.31±0.85	0.70±0.08	0.72±0.08	0.33±0.25	5.39±1.17	14.87±2.16

Table 3-2 Environmental factors of 1-m water column at the two study zones. $p < 0.05$ was in bold.

Abbreviation: GW1m, 1-m depth water in GZ; CW1m, 1m depth water in CZ.

表 3-2 龙须菜栽培区和对照区 1 米深水体主要环境因子的比较 ($p < 0.05$ 的环境因子加粗显示; GW1m, 龙须菜栽培区 1 米深水体; CW1m, 对照区 1 米深水体)

		Dec. 2014	Jan. 2015	Feb. 2015	Mar. 2015	Apr. 2015	May 2015	Jun. 2016	Jul. 2015
Temperature	GW1m	18.40±0.00	14.50±0.00	14.30±0.10	15.30±0.10	18.63±0.06	23.10±0.30	26.33±0.15	27.90±0.40
	CW1m	18.30±0.00	13.93±0.06	14.17±0.12	15.30±0.00	18.60±0.00	23.20±0.17	26.83±0.32	27.80±0.50
Salinity	GW1m	30.73±0.00	29.68±0.04	31.01±0.02	32.09±0.03	33.67±0.02	34.14±0.02	30.72±0.22	31.93±0.19
	CW1m	30.76±0.00	29.53±0.09	31.01±0.05	32.12±0.01	33.68±0.05	34.12±0.00	31.16±0.38	32.16±0.15
DO	GW1m	8.08±0.12	10.02±0.55	8.98±0.09	8.59±0.10	5.45±0.44	7.98±0.84	7.88±0.38	10.38±1.07
	CW1m	7.81±0.07	8.64±0.11	8.99±0.14	8.33±0.28	3.35±0.49	6.50±0.33	7.41±0.35	10.52±0.94
pH	GW1m	8.05±0.12	8.00±0.04	8.08±0.03	8.06±0.03	8.17±0.04	8.30±0.06	7.79±0.07	8.05±0.06
	CW1m	7.98±0.03	8.04±0.04	8.11±0.05	8.04±0.08	8.13±0.04	8.21±0.01	7.93±0.02	8.06±0.02
TN	GW1m	44.78±4.43	22.53±2.21	36.59±3.21	17.72±0.16	14.76±4.04	30.28±4.57	36.34±9.29	37.33±16.30
	CW1m	45.40±0.76	22.79±3.83	39.03±3.35	16.22±1.09	14.60±7.92	36.40±1.88	28.19±1.14	12.10±1.69
TP	GW1m	3.90±1.59	2.15±0.04	4.45±0.50	2.93±0.30	0.80±0.12	4.13±0.72	2.12±0.16	0.85±0.04
	CW1m	3.39±1.12	3.32±0.57	5.14±0.79	2.74±0.28	0.75±0.19	3.69±0.54	1.27±0.23	1.04±0.25
NO ₃ ⁻	GW1m	17.29±3.79	15.50±3.39	10.57±0.82	7.99±1.42	3.57±0.73	12.62±3.21	7.50±0.70	3.18±1.66
	CW1m	19.07±4.11	14.04±1.34	10.22±2.12	10.28±0.72	3.31±2.92	12.60±3.68	6.97±1.03	2.30±0.74
NO ₂ ⁻	GW1m	0.46±0.07	0.81±0.04	0.36±0.08	1.47±0.16	0.29±0.23	0.25±0.04	0.33±0.09	0.22±0.17
	CW1m	0.66±0.14	1.01±0.16	0.47±0.10	1.56±0.12	0.20±0.04	0.31±0.11	0.35±0.06	0.10±0.02
NH ₄ ⁺	GW1m	NA	3.72±0.85	0.03±0.00	1.05±2.23	0.48±0.13	3.85±1.21	2.56±0.42	3.99±1.07
	CW1m	NA	4.64±1.18	0.03±0.00	2.61±0.49	1.34±1.19	4.35±0.18	3.85±0.28	3.14±0.64
PO ₄ ³⁻	GW1m	0.63±0.32	1.50±0.22	1.07±0.03	0.50±0.03	0.34±0.13	9.35±0.47	0.28±0.11	0.12±0.18
	CW1m	0.35±0.13	1.45±0.17	1.21±0.30	0.64±0.01	0.31±0.18	8.58±0.95	0.28±0.08	0.25±0.06
Chl.a	GW1m	1.40±0.88	0.62±0.25	0.86±0.07	0.72±0.07	1.45±0.69	0.55±0.13	7.47±1.04	15.58±2.01
	CW1m	1.76±0.41	0.54±0.13	1.30±0.60	1.10±0.64	0.72±0.08	0.50±0.15	5.76±0.74	14.71±1.81

Table 3-3 Environmental factors of bottom water column at the two study zones. $p < 0.05$ was in bold. Abbreviation: GWB, bottom water in GZ; CWB, bottom water in CZ.

表 3-3 龙须菜栽培区和对照区底层水体主要环境因子的比较 ($p < 0.05$ 的环境因子加粗显示; GWB, 龙须菜栽培区底层水体; CWB, 对照区底层水体)

		Dec. 2014	Jan. 2015	Feb. 2015	Mar. 2015	Apr. 2015	May 2015	Jun. 2016	Jul. 2015
Temperature	GWB	18.23±0.21	14.23±0.06	14.23±0.06	15.37±0.06	18.60±0.00	22.87±0.21	25.17±0.59	24.97±0.45
	CWB	18.20±0.17	14.07±0.12	14.17±0.12	15.27±0.06	18.60±0.00	22.80±0.00	25.83±0.21	25.53±0.42
Salinity	GWB	30.76±0.02	30.06±0.06	31.00±0.05	32.10±0.02	33.69±0.01	34.15±0.03	31.78±2.01	33.47±0.31
	CWB	29.47±1.94	29.80±0.16	30.95±0.09	32.12±0.01	33.69±0.03	34.16±0.03	30.76±1.84	33.06±0.24
DO	GWB	7.77±0.26	9.44±0.14	8.59±0.21	8.20±0.13	5.15±0.16	5.66±0.08	4.53±1.14	8.25±0.51
	CWB	7.70±0.06	8.49±0.01	8.64±0.12	8.19±0.11	3.20±0.19	5.71±0.36	5.44±0.08	8.37±0.20
pH	GWB	8.12±0.26	7.90±0.05	8.06±0.03	8.00±0.03	8.12±0.02	8.24±0.04	7.79±0.07	7.92±0.01
	CWB	7.94±0.05	8.00±0.03	8.05±0.02	7.94±0.06	8.10±0.01	8.20±0.01	7.94±0.01	7.93±0.03
TN	GWB	44.37±2.90	19.80±4.19	39.50±0.50	17.56±0.76	13.88±2.49	35.04±2.11	30.00±4.29	29.90±5.53
	CWB	44.97±1.57	22.45±2.96	39.12±0.89	17.30±2.24	15.28±1.19	32.25±2.86	31.78±10.44	17.54±2.39
TP	GWB	3.90±1.24	2.04±0.53	5.00±0.78	4.55±1.34	0.84±0.12	3.63±0.29	3.10±1.70	1.62±0.69
	CWB	3.45±1.06	3.12±0.31	5.69±0.54	1.88±0.56	0.83±0.40	3.81±0.55	2.33±1.85	1.57±0.78
NO ₃ ⁻	GWB	16.83±2.42	16.56±5.08	10.36±0.45	8.88±1.65	4.55±0.79	14.34±3.22	7.59±0.75	4.44±1.76
	CWB	16.42±3.29	10.06±1.08	10.66±0.64	9.94±2.72	3.15±1.15	10.60±3.16	7.27±0.30	3.73±0.96
NO ₂ ⁻	GWB	0.52±0.07	0.88±0.09	0.36±0.04	1.44±0.09	0.35±0.06	0.39±0.11	0.32±0.11	0.45±0.03
	CWB	0.58±0.13	0.97±0.02	0.42±0.05	1.48±0.11	0.28±0.07	0.49±0.35	0.34±0.03	0.24±0.13
NH ₄ ⁺	GWB	NA	2.14±0.20	0.02±0.00	1.88±0.27	0.31±0.08	4.71±1.05	2.93±0.35	4.56±0.40
	CWB	NA	3.09±1.21	0.04±0.01	2.50±0.12	0.99±0.83	4.11±0.31	3.77±0.31	3.40±0.48
PO ₄ ³⁻	GWB	0.54±0.24	1.35±0.03	0.97±0.09	0.45±0.12	0.32±0.07	8.37±1.34	0.26±0.02	0.23±0.31
	CWB	0.48±0.10	1.48±0.11	1.33±0.27	0.47±0.18	0.45±0.17	8.25±0.75	0.23±0.10	0.28±0.04
Chl.a	GWB	1.01±0.88	0.64±0.32	0.74±0.21	1.20±0.79	1.44±0.68	0.62±0.08	6.44±0.72	11.74±0.91
	CWB	1.00±0.93	0.48±0.20	1.96±1.00	1.06±0.67	0.67±0.13	0.50±0.15	5.82±0.85	15.24±2.93

At 1-m depth water samples, positive relationships were observed between PO_4^{3-} , temperature, salinity, or pH and cultivation time, while there were negative relationships between DO and cultivation time ($p < 0.05$). The increase in rates of PO_4^{3-} , temperature, and pH were greater in GW1m than in CW1m. However, the slopes of salinity and DO of CW1m were steeper than that of GW1m (Fig. 3-2). In bottom water samples, PO_4^{3-} , temperature, salinity and pH showed a positive relationship with cultivation time, while the DO had a negative correlation with cultivation time ($p < 0.05$). Among these regressions between environmental factors and cultivation time, the slopes of PO_4^{3-} and pH in GWB were steeper compared to CWB. The slopes of temperature and salinity in GWB were flatter than those of CWB ($p < 0.05$) (Fig. 3-3).

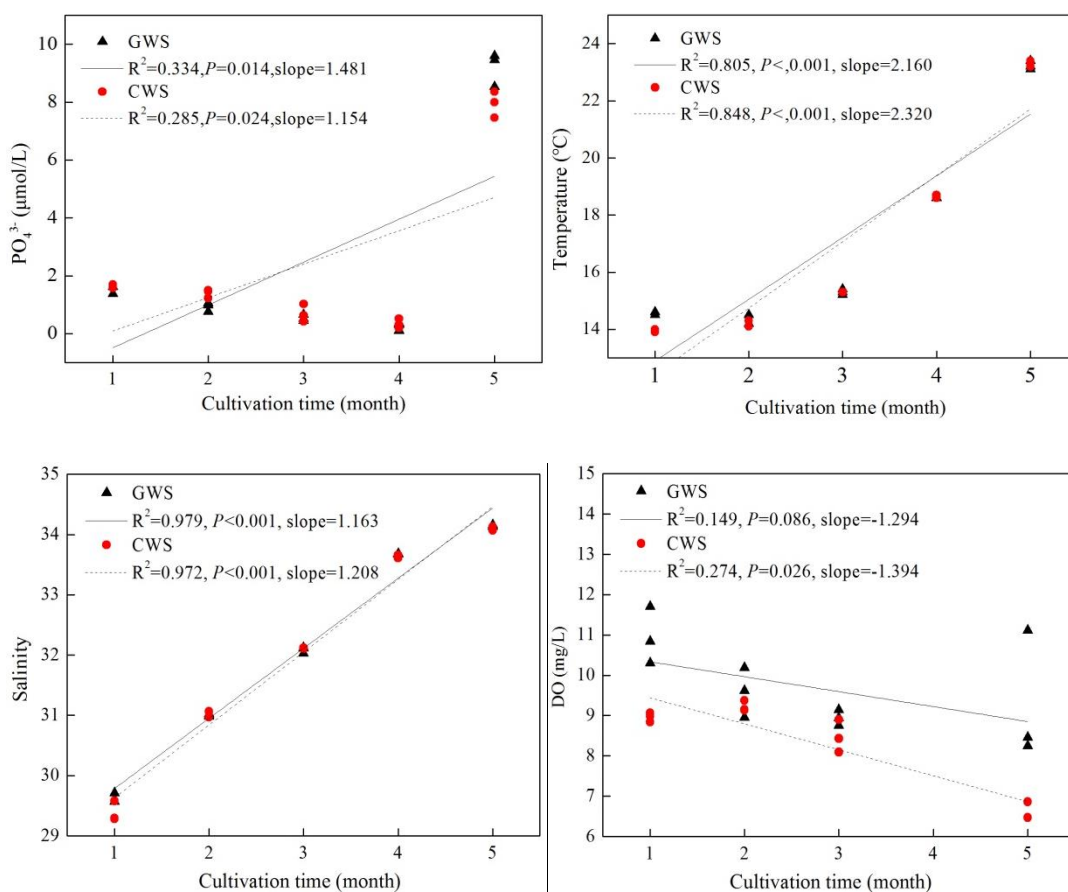


Fig. 3-1 Linear regressions between environmental factors and cultivation time at the surface water samples.

图 3-1 表层水体环境因子与龙须菜栽培的关系

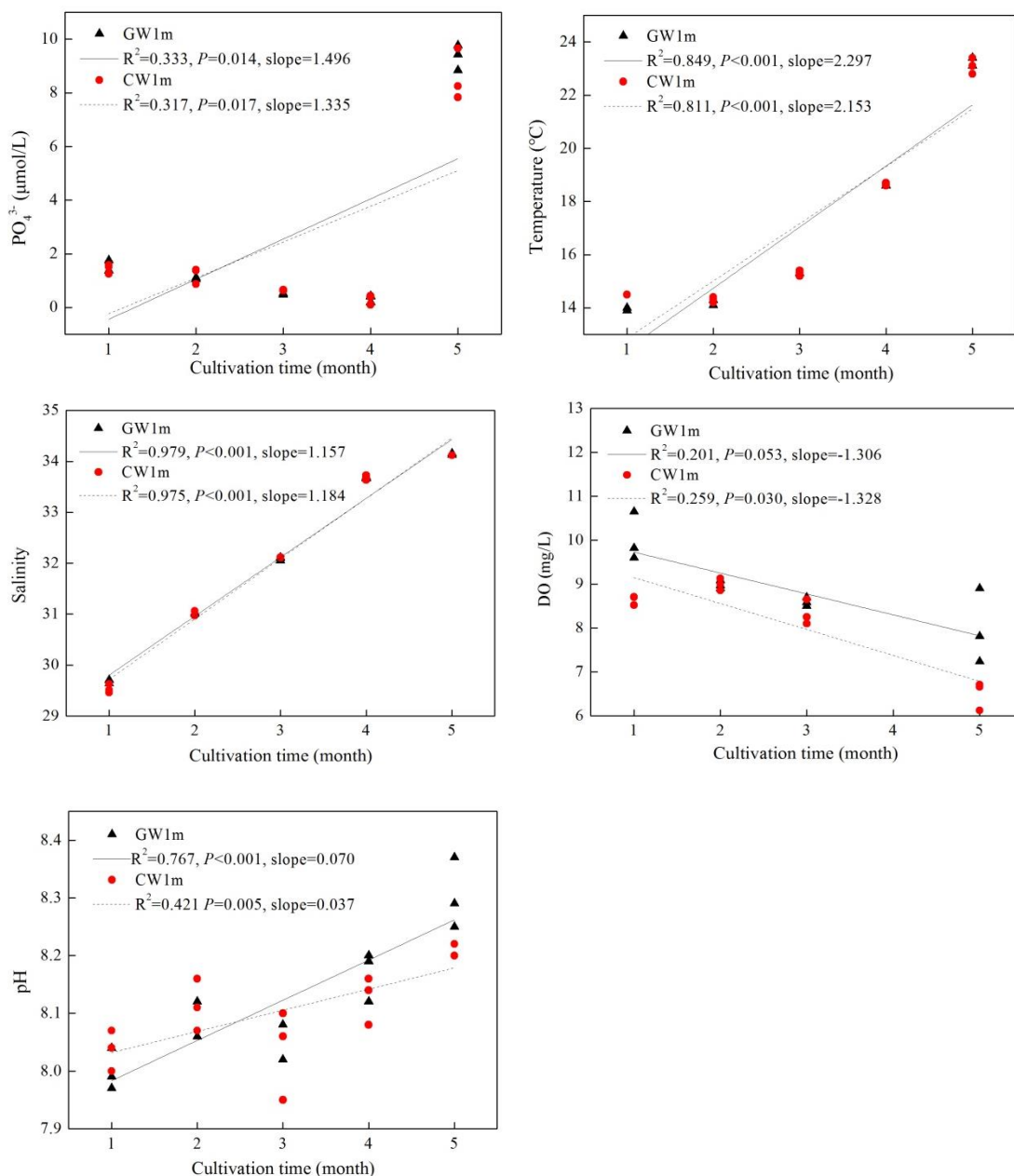


Fig. 3-2 Linear regressions between environmental factors and cultivation time at 1-m depth water samples.

图 3-2 1 米深水体环境因子与龙须菜栽培的关系

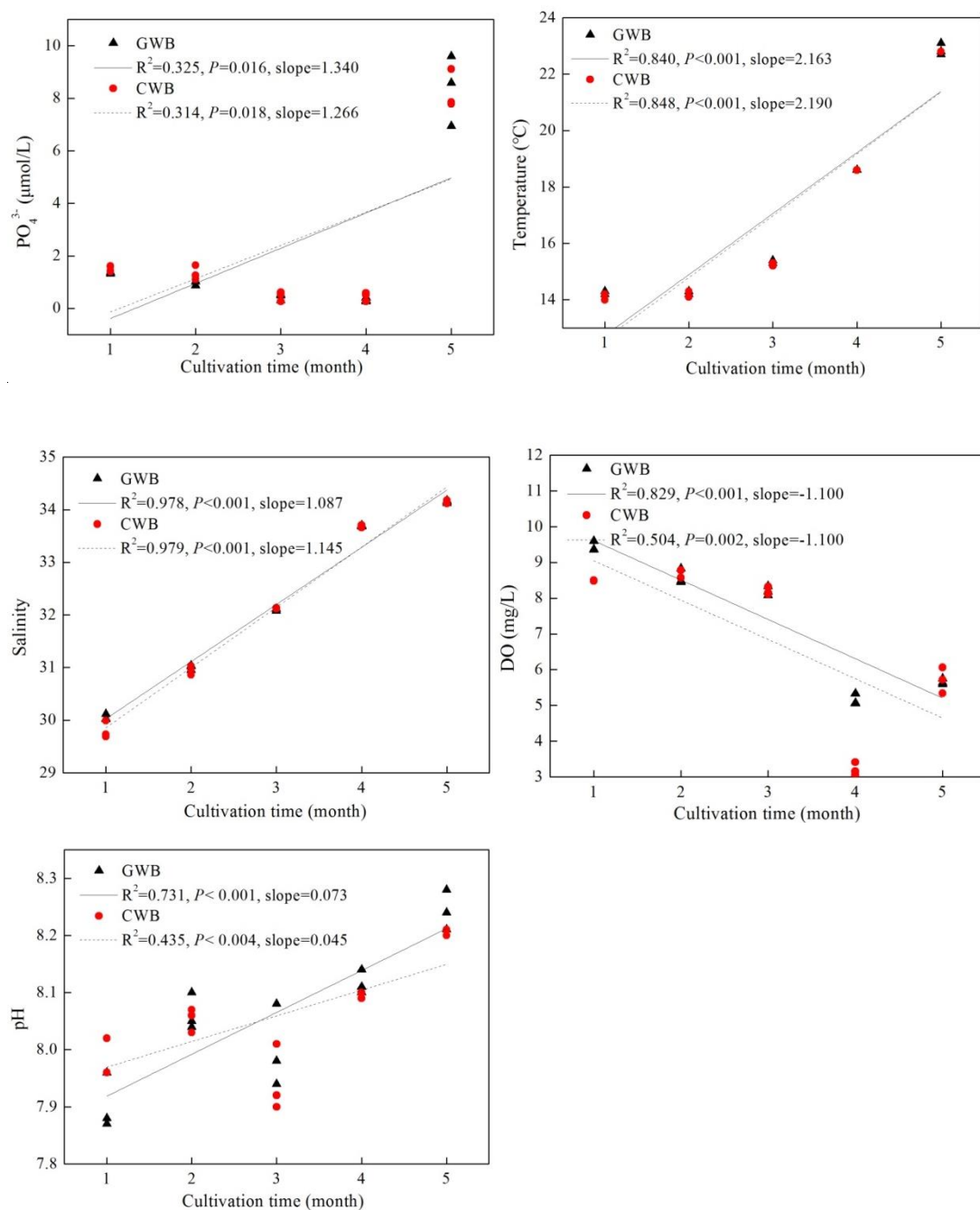


Fig. 3-3 Linear regressions between environmental factors and cultivation time for the bottom water samples.

图 3-3 底层水体环境因子与龙须菜栽培的关系

3.3.2 The dynamics of the microbial community structure in response to *Gracilaria* cultivation

Microbial communities in GZ and CZ were analyzed by sequencing 16S rRNA gene amplicons with Illumina MiSeq technology. The microbial community at GZ showed

a decreased richness over time as evidenced by OTU numbers (a total of 210, 231 OTUs with an average of 2920 ± 48 OTUs) compared to CZ (a total of 200, 681 OTUs with an average of 2787 ± 201 OTUs). The results of linear regression showed that α -diversity of microbial communities in the surface water was negatively correlated with cultivation time ($R^2=0.31$, $P=0.02$) and no correlation was observed between other water samples and cultivation time. Also, α -diversity of the microbial communities was negatively correlated with the biomass of seaweed *G. lemaneiformis* (Fig. 3-4).

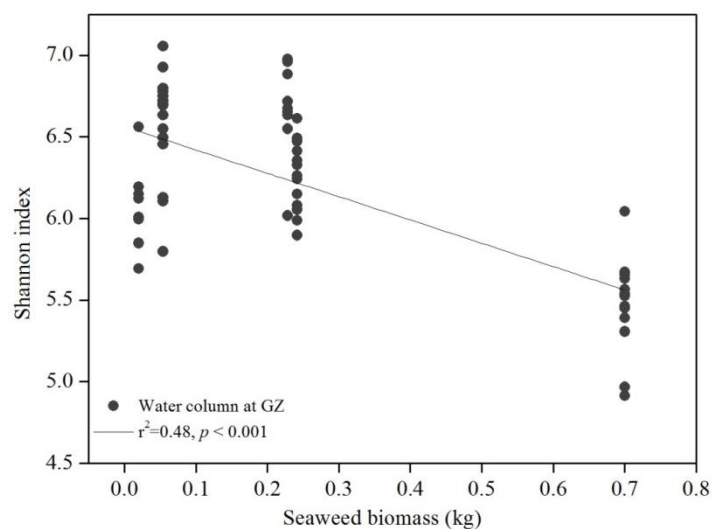


Fig. 3-4 Linear relationship between α -diversity and seaweed biomass.

图 3-4 微生物群落 α -多样性与龙须菜生物量的关系

Dissimilarity tests were conducted for all depths to delineate how *G. lemaneiformis* cultivation, sampling time and water depth affected the microbial community structure based on the permutational multivariate analysis of variance (PERMANOVA) with the Adonis function. Time, interactions between time and seaweed cultivation, and interactions between time and depth were the prevailing factors causing the divergences in microbial communities based on R^2 values (Table 3-4). The Adonis test was also performed to reveal the influences of seaweed cultivation, time and their interactions on the microbial community structure at all depths. The results revealed that time, and interactions between time and seaweed cultivation were the major driving factors in shaping the microbial community structure (Table 3-5).

Table 3-4 Significance tests of seaweed cultivation, month and depth on the overall microbial community structure by Adonis

表 3-4 龙须菜栽培、时间及水深对微生物群落结构影响的 Adonis 分析

	F	R ²	p
Cultivation	3.761	0.006	0.001
Month	28.715	0.309	0.001
Depth	24.049	0.185	0.001
Cultivation×Time	3.820	0.041	0.001
Cultivation×Depth	1.399	0.011	0.025
Time×Depth	4.171	0.160	0.001
Cultivation×Time×Depth	1.418	0.054	0.001

Table 3-5 Significance tests of cultivation and month on the microbial community structure of each depth by Adonis

表 3-5 龙须菜栽培和时间对不同水深微生物群落结构影响的比较

	Surface water		1-m depth water		Bottom water	
	R ²	p	R ²	p	R ²	p
Cultivation	0.022	0.006	0.015	0.029	0.017	0.020
Time	0.592	0.001	0.622	0.001	0.601	0.001
Cultivation×Time	0.121	0.001	0.116	0.001	0.137	0.001

NMDS visualized the overall changes in microbial succession at GZ and CZ across seaweed cultivation time and sampling water depths. The close clustering of samples at all depths was observed at GZ from January to May 2015, demonstrating a similarity of microbial community assembly during the cultivation period (Fig. 3-5). As expected, similar clusters were found among samples before cultivation, during cultivation time and after harvest, meaning that the microbial community assembly was shaped by the cultivation time of *G. lemaneiformis*. An obvious different distribution pattern of microbial communities was observed before-cultivation, cultivation period and after harvest.

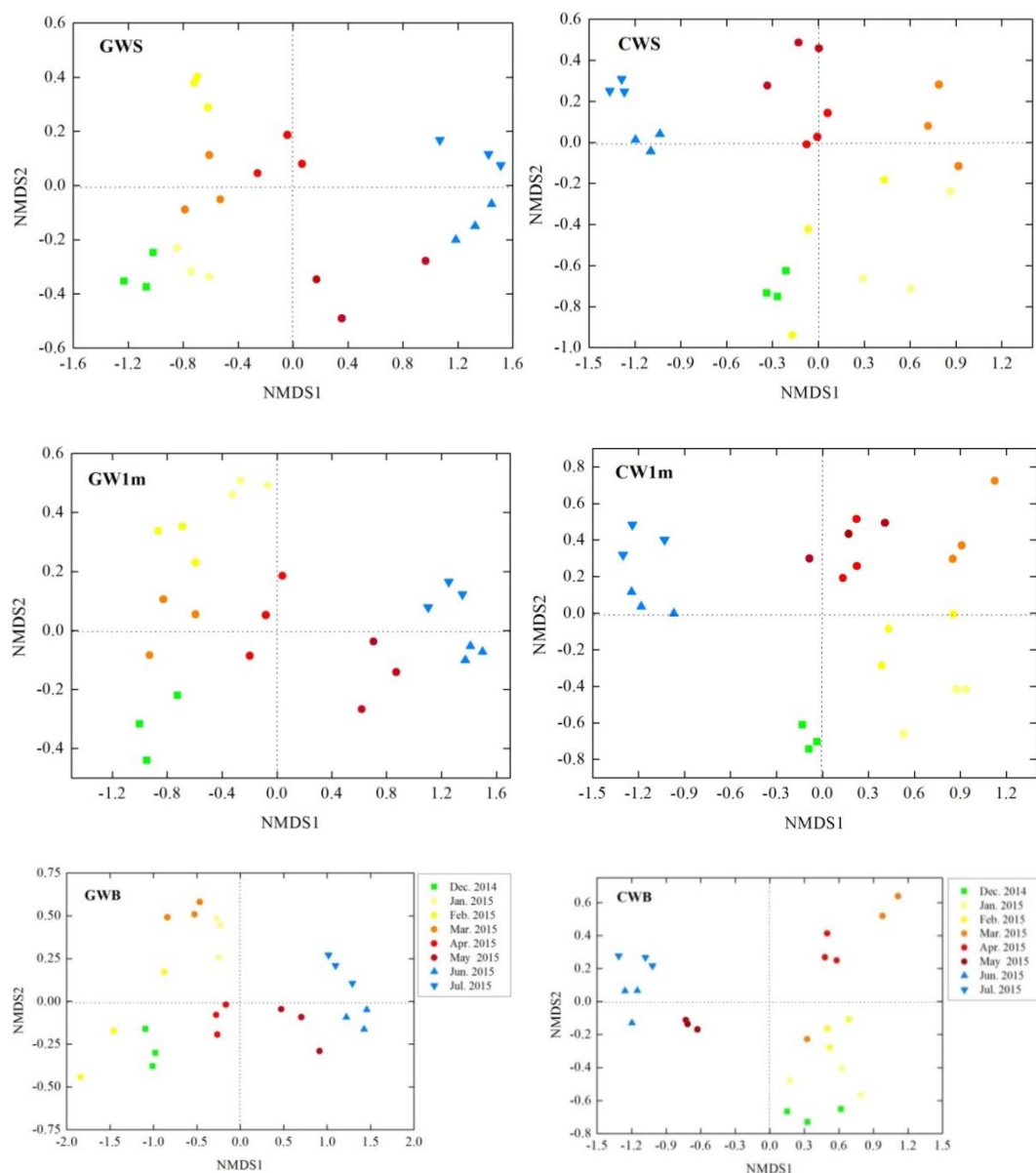


Fig. 3-5 NMDS based on Bray–Curtis distances showing the changes of microbial community patterns at GZ and CZ before cultivation (green square), during cultivation period (circle), and postharvest (blue triangle). Colors of circle from deep to shallow represent community succession with the increases of seaweed biomass from January to May 2015.

图 3-5 龙须菜栽培区和对照区在不同栽培时期微生物群落特征的 NMDS 分析 (绿色正方形表示栽培前, 圆圈表示栽培期间, 蓝色三角形表示收获后; 栽培期间圆圈颜色由浅至深代表龙须菜生物量从 2015 年 1-5 月有所增加)

The result of NMDS was confirmed by the overall dissimilarity test of the microbial communities between GZ and CZ using Adonis (non-parametric multivariate), which illustrated that the structure of microbial communities was different during the seaweed cultivation period (January to May 2015) at 1-m depth and the bottom. *Gracilaria* cultivation did shift the assembly of the microbial communities (Table 3-6). To visualize the dissimilarity in microbial communities between GZ and CZ, linear regressions were performed to detect any difference at each depth based on the Bray–Curtis distance (Fig. 3-6). The results showed that the distances between samples in the surface water ($r^2=98$, $p<0.001$), 1-m depth water ($r^2=0.95$, $p<0.001$), and near the bottom water ($r^2=0.93$, $p<0.001$) all increased with time. In addition, no linear regressions were detected between α -/ β -diversity and cultivation time, suggesting that water depth had no effect on the microbial community structure.

Table 3-6 Dissimilarity in structure of the microbial community along different water depths between GZ and CZ. $p < 0.05$ was in bold.

表 3-6 龙须菜栽培区和对照区不同月份不同水层微生物群落结构差异 ($p < 0.05$ 加粗显示)

	Januray	February	March	April	May	June	July
GWS vs. CWS	0.126	0.085	0.487	0.124	0.309	0.082	0.073
GW1m vs. CW1m	0.189	0.059	0.007	0.001	0.041	0.202	0.417
GWB vs. CWB	0.001	0.028	0.095	0.001	0.001	0.02	0.195

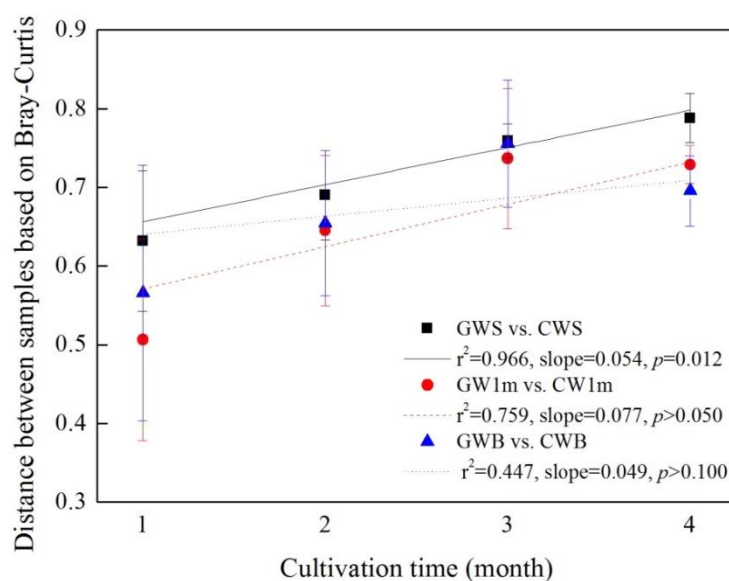


Fig. 3-6 Distances between samples of GZ and CZ (based on Bray–Curtis dissimilarity) at surface (square), 1-m depth (circle) and bottom water (triangle).

图 3-6 龙须菜栽培区和对照区在表层水体(正方形表示)、1 米深水体(圆圈表示)及底层水体(三角形表示)的 Bray–Curtis 距离

Comparisons of the microbial communities at the phylum level showed that Proteobacteria was the most dominant phylum in the two study zones during before-cultivation and cultivation period ($p < 0.01$). In contrast, the relative abundances of Bacteroidetes increased to become the dominant phylum ($p < 0.05$) in all water depths during the postharvest time (Fig. 3-7, Fig. 3-8, Fig. 3-9). The significances of each phylum affected by seaweed cultivation were determined based on ANOVA tests for surface, 1m depth and near the bottom water (Table 3-7). The results indicated that *Actinobacteria*, *Bacteroidetes*, *Proteobacteria* and *Synergistetes* were significantly influenced by *G. lemaneiformis* cultivation at all depths ($p < 0.05$).

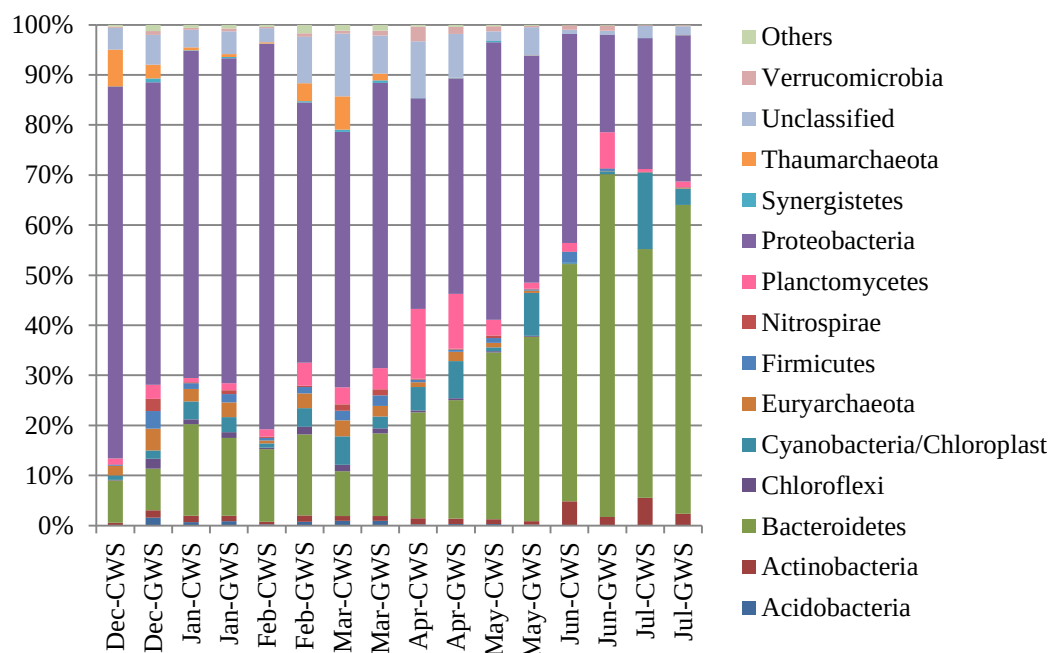


Fig. 3-7 The microbial community structure of surface water column based on 16S rRNA gene sequencing.

图 3-7 龙须菜栽培区和对照区表层水体微生物群落结构

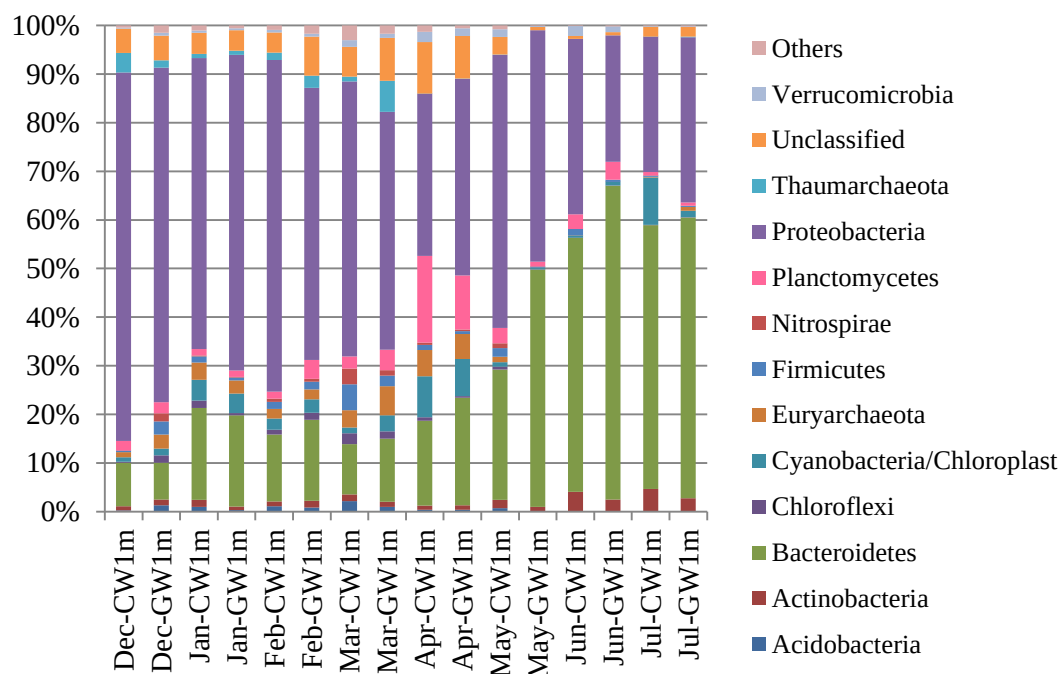


Fig. 3-8 The microbial community structure of 1m depth water column based on 16S rRNA gene sequencing.

图 3-8 龙须菜栽培区和对照区 1 米深水体微生物群落结构

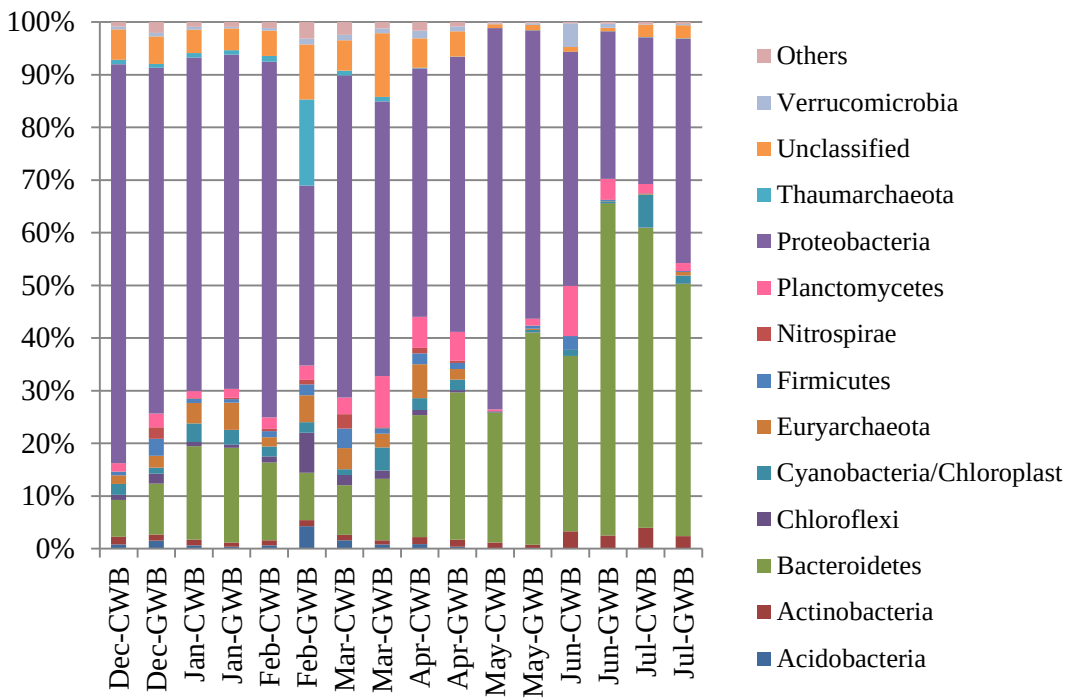


Fig. 3-9 The microbial community structure of near the bottom water column based on 16S rRNA gene sequencing.

图 3-9 龙须菜栽培区和对照区底层水体微生物群落结构

Table 3-7 Significances of each phylum affected by seaweed cultivation based on ANOVA.

表 3-7 龙须菜栽培对门水平微生物类群的影响

	Surface		1-m depth		Bottom	
	F	<i>p</i>	F	<i>p</i>	F	<i>p</i>
<i>Acidobacteria</i>	6.222	0.017	2.692	0.109	1.070	0.307
<i>Actinobacteria</i>	5.283	0.027	4.502	0.040	7.780	0.008
<i>Bacteroidetes</i>	6.163	0.017	7.560	0.009	4.515	0.040
<i>Chloroflexi</i>	7.520	0.009	0.684	0.413	1.716	0.198
<i>Cyanobacteria/Chloroplast</i>	0.017	0.898	2.026	0.163	1.037	0.315
<i>Euryarchaeota</i>	2.850	0.099	0.859	0.360	0.001	0.969
<i>Firmicutes</i>	1.206	0.279	2.888	0.097	0.535	0.469
<i>Nitrospirae</i>	3.525	0.068	1.246	0.271	0.077	0.783
<i>Planctomycetes</i>	1.469	0.233	0.567	0.456	0.389	0.537
<i>Proteobacteria</i>	6.185	0.017	2.334	0.135	7.389	0.010
<i>Synergistetes</i>	4.750	0.035	0.973	0.330	4.312	0.044
<i>Thaumarchaeota</i>	0.660	0.422	0.264	0.610	4.956	0.032
<i>Unclassified</i>	0.308	0.582	2.969	0.093	2.431	0.127
<i>Verrucomicrobia</i>	0.268	0.608	2.692	0.109	1.070	0.307
Others	-	-	-	-	-	-

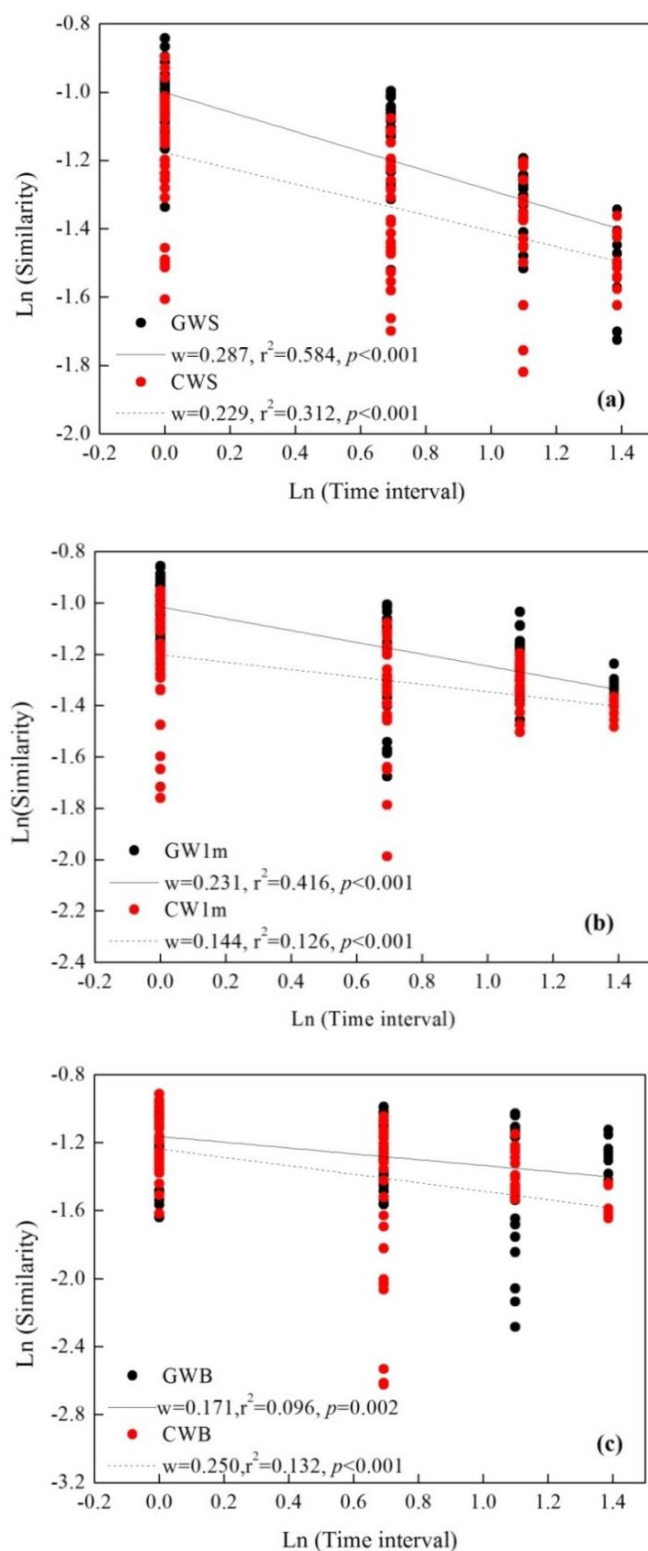


Fig. 3-10 Temporal succession of microbial communities in surface (a), 1-m depth (b) and near the bottom water (c).

图 3-10 龙须菜栽培区和对照区表层水体(a)、1 米深水(b)和底层水体(c)微生物群落演替

To further understand the microbial community structure altered by seaweed cultivation and/or water depth, linear regressions were conducted between time/depth and β -diversity at the two study zones. Microbial community succession in surface, 1-m depth water and near the bottom water was negatively correlated with cultivation time at GZ ($w=0.287$, $p<0.001$; $w=0.231$, $p<0.001$; $w=0.171$, $p<0.01$, respectively) and CZ ($w=0.229$, $p<0.001$; $w=0.144$, $p<0.001$; $w=0.250$, $p<0.001$, respectively). The slopes of surface (Fig. 3-10a) and 1-m depth water (Fig. 3-10b) at GZ were steeper than CZ ($t=18.363$, $p<0.001$; $t=25.506$, $p<0.001$). However, the slopes of bottom water at CZ was steeper than GZ ($t=-18.414$, $p<0.001$) (Fig. 3-10c). No linear relationships were observed for depth and β -diversity at the two study zones (Fig. 3-11).

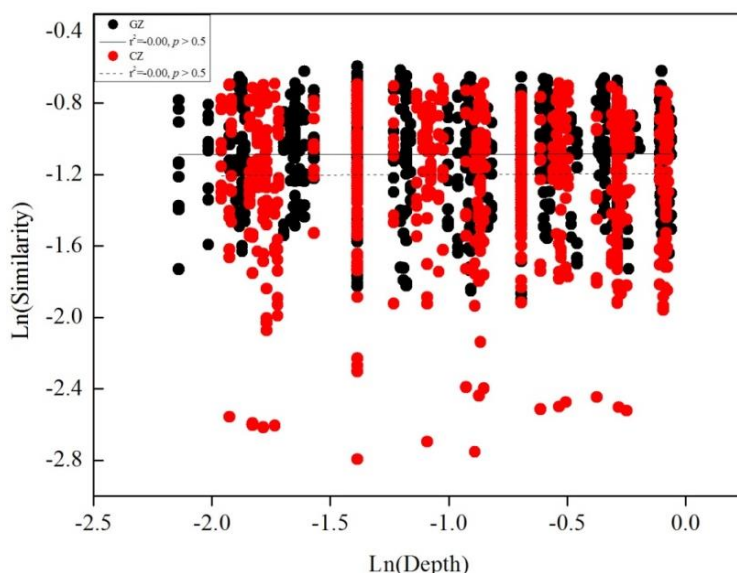


Fig. 3-11 Relationship between depth and β -diversity in water samples.

图 3-11 微生物群落 β -多样性与水深之间的关系

3.3.3 Relationships between environmental factors and the microbial community structure

CCA was performed to identify the relationship between microbial turnover and environmental factors. All CCA models were significant ($p < 0.01$) for explaining the relationships between microbial communities and environmental variables. Temperature, salinity and PO_4^{3-} were excluded from CCA because of variance inflation factors larger than 20.

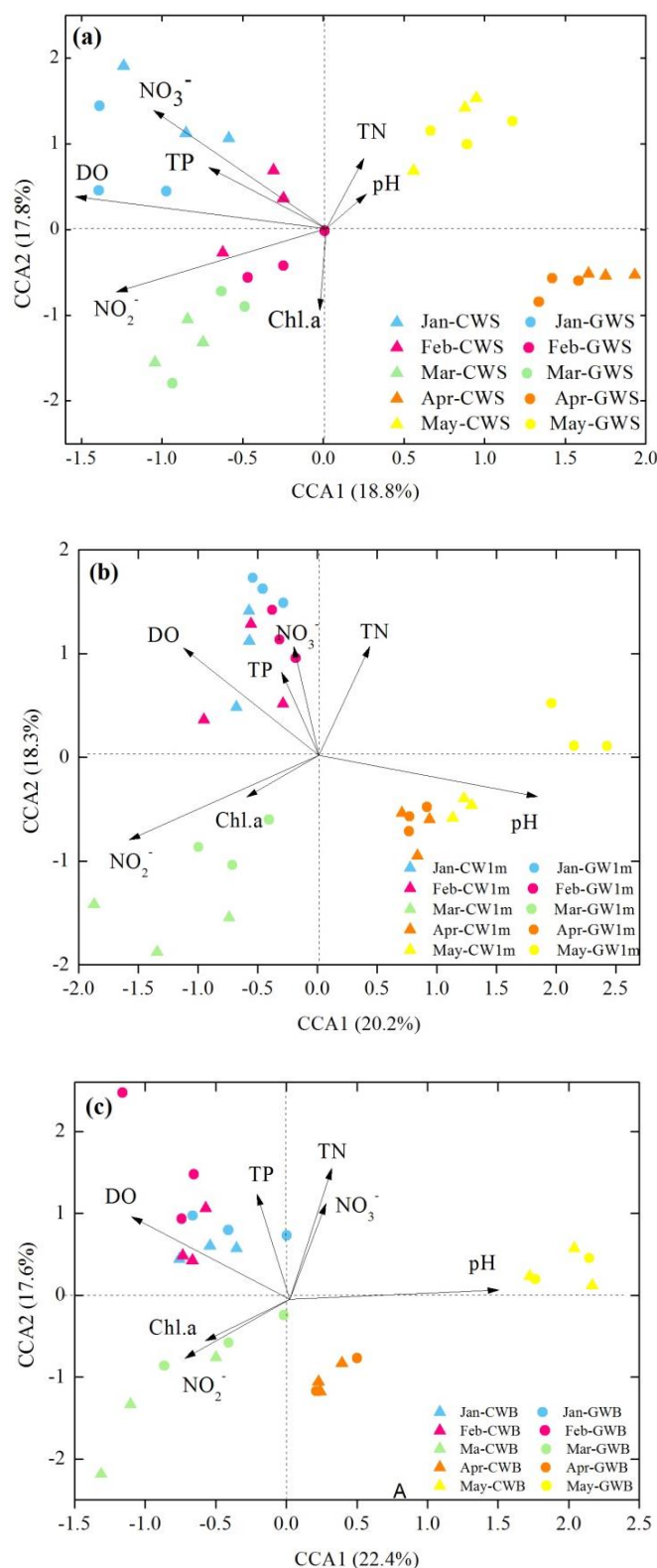


Fig.3-12 Canonical correspondence analysis (CCA) of microbial community composition and environmental variables in surface water (a), 1-m depth water (b) and near the bottom water (c).

图 3-12 表层水体(a)、1 米深水体(b)和底层水体(c)微生物群落组成与环境因子的 CCA 分析

CCA plots visualized the linkage between environmental properties and the microbial community. According to CCA, the microbial community assembly was driven by seaweed cultivation time (Fig. 3-12). Physical (DO, pH) and chemical factors (NO_3^- , NO_2^-) were principal variables ($p < 0.05$) in shaping the microbial community structure (Table 3-8). In the surface and 1-m depth water, pH and TN were positively correlated with samples of April and May when the biomass of *G. lemaneiformis* reached a relatively high level. However, bottom samples show a positive relationship with pH, TN and NO_3^- . GZ samples were generally separated from CZ samples, and samples of each month formed a close clustering, suggesting that seaweed cultivation time had a significant effect on the microbial community.

Table 3-8 Significance of each environmental factor by ANOVA of CCA

表 3-8 环境因子对不同水层微生物群落结构影响的显著性检验

Environmental attributes	Surface water		1-m water		Bottom water	
	r^2	P	r^2	P	r^2	P
DO	0.737	0.001	0.523	0.001	0.599	0.001
pH	0.060	0.428	0.801	0.001	0.660	0.001
TN	0.190	0.081	0.267	0.015	0.595	0.001
TP	0.287	0.014	0.157	0.111	0.381	0.004
NO_3^-	0.839	0.001	0.240	0.021	0.324	0.009
NO_2^-	0.627	0.001	0.704	0.001	0.284	0.015
Chl.a	0.239	0.023	0.116	0.175	0.166	0.098

3.4 Discussion

Since *G. lemaneiformis* cultivation is environmentally beneficial, increased numbers of fishermen conduct large-scale cultivation of the red algae in the coastal area of China. Previous studies indicate that *G. lemaneiformis* can be widely used for human consumption food, bait and a premium agar sources as well as in pharmaceutical and other industries. Due to the indispensable role of microbes in marine ecosystems, one of the most important objectives was to reveal how the microbial community response

to large-scale seaweed cultivation across the cultivation cycle at different water depths. However, few studies have been reported on the diversity, structure and succession of microbial communities in seaweed *G. lemaneiformis* cultivation ecosystem. With the development of culture-independent, the dynamics and turnover of microbial communities can be better understood. In this study, we demonstrated that *G. lemaneiformis* cultivation decreased α -diversity, increased the succession rate across the seaweed cultivation time and altered the microbial community assembly. These findings greatly enhance our understanding of microbial community responses to large-scale seaweed cultivation.

3.4.1 Divergences in microbial communities in response to *Gracilaria* cultivation

The results in this study support the hypothesis that *G. lemaneiformis* cultivation alters the microbial community assembly. The overall structure of microbial communities was significantly different between GZ and CZ at the surface water, at 1-m depth and near the bottom water. The results showed that seaweed cultivation times affected the structure of microbial communities with higher R^2 even more than seaweed cultivation. This suggests that the microbial community responds sensitively to *G. lemaneiformis* growth and the concomitant increase of seaweed biomass. According to the dissimilarity test, the structure of microbial communities differed significantly between the two study zones at 1-m depth and near the bottom during the *Gracilaria* growth cycle and afterharvest in GZ and CZ at all water depths. Since the microbial community structure was similar after the harvest, *Gracilaria* cultivation did have an effect. These results were confirmed by NMDS analysis, showing that cultivation time of *G. lemaneiformis* shaped the distribution pattern of microbial communities. Furthermore, linear regressions based on Bray-Curtis dissimilarity and cultivation time comparing the pairwise distances between microbial communities at GZ and CZ in each depth, indicated that the distance between samples increased along with the seaweed cultivation time only in surface water. That is, no linear relationship was observed between 1-m depth water or near the bottom water samples and seaweed cultivation time. The thalli of *G. lemaneiformis* generally grows in the

surface water, creating a micro-environment. The dynamics of microbial communities at the phylum level showed that the *Bacteroidetes* replaced *Proteobacteria* as the most dominant phylum in the cultivation area of *G. lemaneiformis*, further indicating that large-scale seaweed cultivation altered the structure of microbial communities.

The species-sorting paradigms built on theories of community change over environmental gradients (Whittaker, 1962) and assume that spatial niche separation is above and beyond spatial dynamics. Thus, variations in community composition can be tracked by changes in the local environment due to environmental changes or perturbations (Leibold et al., 2004), that is to say, community adaptation to a new and changing environments by altering the species composition. Previous studies on the meta-community have defined the ability of a community to change its properties in response to changing local environmental conditions as the “adaptive capacity” of the system (Leibold, 2004). Large-scale cultivation of *G. lemaneiformis* has increased DO and pH, and decreased nutrients in GZ especially in the surface water. Also, *G. lemaneiformis* secretes many organic compounds, such as algal polysaccharides, and releases detritus into the surrounding environment. On the other hand, the α -diversity has a negative correlation with seaweed biomass in this study, which may indicate that *Gracilaria* selects for a certain specific microbial community composition. Our previous study showed consistent results, indicating that *Gracilaria* cultivation enriched the water column and sediment with specific taxa, such as *Arenibacter*, *Croceitalea*, *Leucothrix* and *Maribacter* (Xie, 2017).

3.4.2 *Gracilaria* cultivation alters the microbial succession rate

Our second hypothesis was that seaweed growth leads to increased rates of succession in the microbial community. The temporal turnover (w values) of the time-decay relationship is a crucial indicator of succession in the microbial community. Significantly linear decrease in the similarity of microbial communities with time (Log transformed) was indicated for all samples. The w values of surface and 1-m depth waters were significantly higher at GZ compared to CZ, whereas the w values of near the bottom water were higher at CZ than that of GZ. The w values in this

study (0.144-0.289) confirmed that the range of temporal turnover has the same range of w values as in other studies (0-0.3) (Hatosy et al., 2013). Microbial similarity decreased across seaweed cultivation strongly supported by time-decay (decay of community similarity over time) in aquatic communities (Shade et al., 2013). Recently, Hatosy et al (2013) addressed whether the mechanisms that underlie microbial temporal β -diversity vary across different scales, referring to three different marine microbial communities. Temporal turnover of microbial communities varied intra-seasonal, seasonal and inter-annual scales. Moreover, temporal turnover across different scales was correlated to various environmental factors, easily driven by episodic events, and ultimately caused divergent turnover rates in the microbial community (Hatosy et al., 2013). In our study, the seaweed cultivation period across winter, spring and early summer, composed of a seasonal temporal scale. The temporal turnover of microbial communities differed between GZ and CZ across seasonal scale, showing that *G. lemaneiformis* cultivation shifted the succession rate of the microbial communities.

To further uncover the reason why large-scale seaweed cultivation leads to increases in turnover in the surface and 1-m depth water, and a decrease in near the bottom water, we analyzed the results based on ecological theories. Generally, a community with higher diversity appears to have greater stability and dampen the disturbance of episodic (e.g. mariculture, typhoon, oil spill) events more powerfully (Gonzalez et al., 2012). The steeper slope in surface and 1m depth water samples at GZ suggested higher succession rate due to *G. lemaneiformis* cultivation. Since α -diversity of microbial communities at GZ was negatively correlated to seaweed biomass, seaweed may be viewed as the episodic event that disturbs the environment of GZ and increases the temporal turnover. In addition, seaweed selects taxa and the α -diversity decreased and β -diversity increased across seaweed cultivation time. The lower slopes in near the bottom water at GZ may be caused by a longer distance between seaweed thalli and the bottom water column, assuming it is a function of interaction with the sediment. These results are supported by “complex adaptive system”, which emphasizes that a community has adaptive capability to change its

properties in response to changing environmental conditions (Leibold, 2004). Similarly, higher temporal turnover at GZ indicated that *G. lemanleiformis* cultivation altered succession rate, which supported our second hypothesis.

3.5 Chapter Summary

Large-scale cultivation of *G. lemanleiformis* caused improvements of water quality and altered the α -diversity and succession microbial communities throughout the water columns. A faster temporal turnover occurred in the surface and 1-m depth water samples at GZ than at CZ. This study provides new insight into the environmental benefits of seaweed cultivation on the structure, dynamics and succession of the marine microbial communities throughout the *G. lemanleiformis* growth life cycle.

Chapter 4: The dynamics of functional microbial communities in Shen'ao Bay

4.1 Introduction

Marine plants such as mangroves, seaweeds and sea grasses have the potential as a C sink for anthropogenic carbon emission (Nellemann et al., 2009). Using marine plants is one of the important pathways for C fixation and C sequestration in order to remove elevated CO₂ (Muraoka, 2004). For examples, China and Korea were the top two countries cultivating seaweed and contribute significantly to C sequestration (Chung et al., 2010). Since microbes mediate many key processes in marine ecosystem, such as biogeochemical cycling of C, N, P and S (Falkowski et al., 2008), knowing the changes of functional genes due to seaweed cultivation is fundamental to understand the functional structure of microbial community responses to seaweed cultivation. However, to date, insufficient information is available on the functional diversity, structure and dynamic of microbial communities in seaweed cultivation ecosystem. Previous studies mainly focused on the taxonomic structure and composition of microbial communities, little is known about the effect of seaweed cultivation on microbial functional profiles. Considering its fast growth rate and high harvested biomass, *G. lemaneiformis* was chosen as our study seaweed and was cultivated near Nan'ao Island, Shantou City, Guangdong Province.

With the development of high-throughput technologies such as high-throughput sequencing and GeoChip, the functional structure and composition of microbial communities could be determined in different ecosystems (Caporaso et al., 2012; Tu et al., 2014). For example, GeoChip 5.0 is one of the most used functional gene microarray-based technologies and has become a powerful tool to analyze the functional structure and composition of microbial communities in various habitats (He et al., 2012a). GeoCip 5.0 (60K) contains 57,604 probes involved in gene categories including C, N, P and S cycling, organic remediation, metal homeostasis and virulence. Thus, GeoCip 5.0 can be used to analyze the functional structure, composition and dynamic of microbial communities in *Gracilaria* cultivation ecosystem rapidly and comprehensively.

In order to understand the temporal and spatial dynamics of microbial functional structure in response to seaweed cultivation, water samples at 1-m depth throughout the growth cycle of *Gracilaria* (December 2014 to July 2015) and at different depths (i.e. the surface water, 1-m depth, half of water depth, 75% of water depth and the bottom) were collected at the *Gracilaria* cultivation zones (GZ) and the control zones (CZ). Those samples were analyzed using GeoCip 5.0 to test the following hypotheses that (i) *Gracilaria* cultivation would altered the microbial functional structure with the increasing biomass of seaweed; and (ii) the overall structure or functional gene composition in various categories would be shifted at different water depth due to seaweed cultivation. Our results indicated that *Gracilaria* cultivation indeed altered the overall functional profiles of microbial communities with the growth of seaweed and at different depths. Most of functional genes, such as C degradation, nitrification, N₂ fixation and organic remediation were inhibited at GZ to adapt to changes in micro-environment caused by seaweed cultivation. Whereas, genes involved in dissimilatory N reduction and assimilatory N reduction increased at 1-m depth of GZ in April and May. The changes in community functional structure were closely correlated with environmental variables, even though some of them did not show significant differences between GZ and CZ. In summary, this study provides original data of the functional structure and dynamic of microbial communities in response to seaweed cultivation and provides new insights into the interactions in the seaweed-microbe-environment ecosystem.

4.2 Materials and methods

4.2.1 Sample collections

Two study zones were chosen the same as in Chapter 3 and each zone has three sampling sites: *G. lemaneiformis* cultivation zones (GZ, G1, G2, G3) and control zones (CZ, C1, C2, C3) near Shen'ao Bay, Nan'ao Island coastal area (23° 28' N, 117° 06' E). To analyze the temporal dynamic of functional profiles in microbial communities, 48 samples at 1-m depth were collected in eight months (December 2014 to July 2015). Also, 30 water samples in five different depths (surface water,

1-m depth, half of the water depth, 75% of the water depth and bottom water) were collected to test the dynamics of spatial dynamics in microbial communities. Water samples for DNA extraction were filtered immediately serially filtered through 10 and 0.2 μm pore-size filters, and all filters were preserved in liquid nitrogen during transportation and then were transferred to -80°C freezer. Also, water samples for environmental factor determination were stored in car refrigerator at 4°C . The physical properties were examined *in-situ* using YSI 556 systems (YSI Incorporated, USA), including temperature, salinity, pH and DO. Chemical variables (total nitrogen, total phosphorus, NO_3^- , NO_2^- , NH_4^+ , PO_4^{3-} , Chl. a) of seawater were determined at lab immediately according to Standard Method of seawater analysis (AQSIQ, 2007).

4.2.2 DNA extraction, purification and labeling

DNA extraction and purification were performed the same as in Chapter 3. The criterion of DNA for labeling is as follows: $A_{260}/A_{280} \sim 1.8$, $A_{260}/A_{230} \sim 2.0$. For each sample, 29.5 μL (800 μg) of DNA was mixed thoroughly with 5.5 μL random primer (3 $\mu\text{g}/\mu\text{L}$) and incubate at 99°C for 5 min using Thermocycler. The DNA/random primer mixture was immediately chilled on ice. For each reaction, 5 μL $10\times$ buffer, 2.5 μL dNTP mix, 1 μL Klenow, 0.5 μL CyDye and 6 μL H_2O was combined in a separate tube to prepare the 15 μL labeling premix. Since CyDye is light sensitive, it should be added at last in dark room. Fifteen μL labeling premix was transfer to DNA/random primer mixture and mixed well. Then mixture was incubated at 37°C for 6 hrs, heat-inactivated at 95°C for 3 min and then cooled at 4°C . The labeled DNA was purified using Qiagen QIAquick Kit according to the manufacturer's introduction and eluted using 100 μL H_2O . After purification, CyDye incorporation was determined by NanoDrop. As the last step of DNA labeling, all labeled DNA were dried using a Speed Vacuum at 45°C for 2 hrs.

4.2.3 GeoChip hybridization, scanning and GeoChip data processing

In this study, samples were hybridized with 60 K GeoChip containing 57406 probes. Before use, $10\times$ Blocking Agent was prepared, left at room temperature for at least 6

hrs and vortex well. The labeled DNA was re-hydrate with 11.9 μL of ddH₂O, spin down and kept in dark. Hybridization Solution Master Mix was mixed thoroughly with labeled DNA and spined down. Labeled DNA was denatured at 95 $^{\circ}\text{C}$ for 3 minutes and then immediately transferred to a 37 $^{\circ}\text{C}$ incubator for 30 minutes. Then labeled DNA was spined for 1-minute at 6000 $\times g$ and kept in 37 $^{\circ}\text{C}$ incubator. For hybridization, a gasket slide was loaded into the Agilent SureHyb chamber base with label “Agilent” facing up and take 50 μL of the mixture, and about 48 μL was loaded onto a gasket well. A microarray slide with label “Agilent” down was put onto the gasket slide with the numeric barcode side facing up. Samples were hybridized at 67 $^{\circ}\text{C}$ for 22-24 hours (Tu et al., 2014).

Before scanning the hybridized slides, these arrays were washed by Wash buffer 1 and Wash buffer 2. Each slide was inserted with label “Agilent” side up and barcode end first into the slide holder. Hybridized slides were scanned on a Microarray Scanner (NimbleGen). The raw signal was uploaded to the Microarray Data Manager on the website (<http://ieg.ou.edu/microarray>). Scanned images were grid, transferred and qualified using ImaGene 6.0 (Biodiscovery Inc., El Segundo, CA, USA) and poor quality spots were removed. Signal intensity is determined based on scanned images and normalized by mean. Spots with signal-to-noise ratio (SNR) > 2.0 were positive spots and signal intensity was log transformed, and then used in the following analyses (He et al., 2007; Xue et al., 2016).

4.2.4 Statistical analysis

Environmental properties were compared between GZ and CZ using the Student *t*-test. Detrended correspondence analysis (DCA) was performed to analyze the microbial functional structure. To compare the functional profiles of microbial communities in different months/depths, three complementary non-parametric multivariate analyses, non-parametric multivariate analysis of variance (adonis), analysis of similarity (ANOSIM), and the multi-response permutation procedure (MRPP), were used to examine the dissimilarity in microbial communities between GZ and CZ. The significance of differences in signal intensities of genes between control and treatment

was measured by *t*-test. Canonical correspondence analysis (CCA), Mantel test and partial-Mantel test were performed to linking the relationship between the function structure of microbial communities and environmental properties. All statistical analyses were conducted by the *vegan* package using Rstudio when not specified.

4.3 Results

4.3.1 Temporal dynamics of functional microbial communities

The environmental factors of 1-m depth water across 8 months were the same as Table 3-2. Significance of environmental properties between seaweed cultivation and the control samples was examined by the Student *t*-test. During the seaweed cultivation, DO were significantly higher at GZ than at CZ ($p < 0.05$), while TP and PO_4^{3-} were lower at GZ than at CZ, indicating that higher water quality was observed at GZ.

A total of 33,275 functional gene probes were detected for all 48 samples after data normalization and transformation, and samples at GZ and CZ covering 92.56% and 98.16% of the total detected probes, respectively. Among all the detected probes, 30,189 (90.73%) were shared by GZ and CZ, 611 probes (1.84%) and 2,475 probes (7.44%) were detected uniquely at GZ and CZ, suggesting a different distribution of probes between samples of the two study zones. For those unique probes at GZ, the largest proportion was related to C cycling (271 probes, 44.4%), followed by organic remediation (121 probes, 19.8%) and N cycling (79 probes, 12.9%). More unique probes were detected at CZ than at GZ with 1,159 probes (46.8%) involved in C cycling, 401 probes (16.2%) for organic remediation and 313 probes (12.6%) for N cycling. The α -diversity of functional microbial communities was higher at GZ than at CZ based on the Simpson evenness and Pielou's evenness ($p < 0.05$) during the seaweed cultivation period (Table 4-1). The above results indicated that microbial functional diversity especially evenness increased in response to *Gracilaria* cultivation at GZ.

Table 4-1 α -diversity indices of microbial functional genes detected by GeoChip in 1-m depth water samples at GZ and CZ across 8 months. Data are presented as mean $\pm$ sd. The difference between the two study zones was tested using the Student's *t*-test.

表 4-1 龙须菜栽培区和对照区 1 米深水体微生物群落功能多样性的比较 (功能基因芯片数据以平均值 $\pm$ 标准差表示; 学生 *t*-检验分析龙须菜栽培区和对照区 α -多样性指数的差异)

		December	January	February	March	April	May	June	July
Shannon (H')	GW1m	5.27 $\pm$ 0.24	5.27 $\pm$ 0.19	5.30 $\pm$ 0.32	5.60 $\pm$ 0.04	5.45 $\pm$ 0.08	5.44 $\pm$ 0.08	5.39 $\pm$ 0.17	5.36 $\pm$ 0.11
	CW1m	5.47 $\pm$ 0.07	5.56 $\pm$ 0.08	5.62 $\pm$ 0.03	5.54 $\pm$ 0.05	5.41 $\pm$ 0.17	5.44 $\pm$ 0.01	5.40 $\pm$ 0.38	5.30 $\pm$ 0.42
	<i>P</i> -values	0.30	0.11	0.23	0.17	0.78	0.94	0.96	0.82
Simpson evenness	GW1m	0.99 $\pm$ 0.00	0.98 $\pm$ 0.00	0.98 $\pm$ 0.00	0.98 $\pm$ 0.00	0.99 $\pm$ 0.00	0.99 $\pm$ 0.00	0.99 $\pm$ 0.00	0.99 $\pm$ 0.00
	CW1m	0.98 $\pm$ 0.00	0.98 $\pm$ 0.00	0.98 $\pm$ 0.00	0.98 $\pm$ 0.00	0.98 $\pm$ 0.00	0.98 $\pm$ 0.00	0.98 $\pm$ 0.00	0.98 $\pm$ 0.01
	<i>P</i> -values	0.38	0.29	0.04	0.12	0.12	0.00	0.39	0.38
Pielou evenness	GW1m	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
	CW1m	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
	<i>P</i> -values	0.41	0.48	0.04	0.12	0.05	0.00	0.32	0.28

Table 4-2 Significance tests of cultivation and month on the microbial community function by adonis

表 4-2 龙须菜栽培对微生物群落功能影响的差异显著性分析

	F	R ²	<i>p</i>
Cultivation	3.209	0.051	0.025
Month	2.095	0.235	0.011
Cultivation $\times$ Month	1.778	0.200	0.029

In order to compare if seaweed cultivation and cultivation time affect the functional microbial communities, adonis test was performed based on GeoChip 5.0 data. The results showed that cultivation time, the interaction between cultivation and cultivation time, and seaweed cultivation had significant influence on the functional

microbial community structure ($p < 0.05$) (Table 4-2). Also, three different dissimilarity tests (MRPP, Anosim and adonis) were conducted to analyze if *Gracilaria* affected divergences of functional microbial community structure between GZ and CZ. As Table 4-3 indicated, overall shifts in functional potential of microbial communities were observed by all dissimilarity tests ($p < 0.05$). DCA implied that 1-m depth water samples of April and May formed a similar cluster compared to other samples (Fig. 4-1). In order to further understand the effects of seaweed cultivation on functional microbial communities, normalized signal intensities of different gene categories (e.g. C, N, P, S cycling) were compared between GZ and CZ.

Table 4-3 Non-parametric multivariate dissimilarity tests of functional gene profiles between GZ and CZ in 1-m depth water. Results presented are based on distance matrices calculated with Bray-Curtis index. p values < 0.05 are in bold.

表 4-3 龙须菜栽培对 1 米深水体微生物群落功能影响的非参数多变量差异性检验 ($p < 0.05$ 加粗显示)

	<i>Statistic</i>	<i>p</i> -value
MRPP	0.271	0.015
Anosim	0.07	0.011
adonis	2.497	0.048

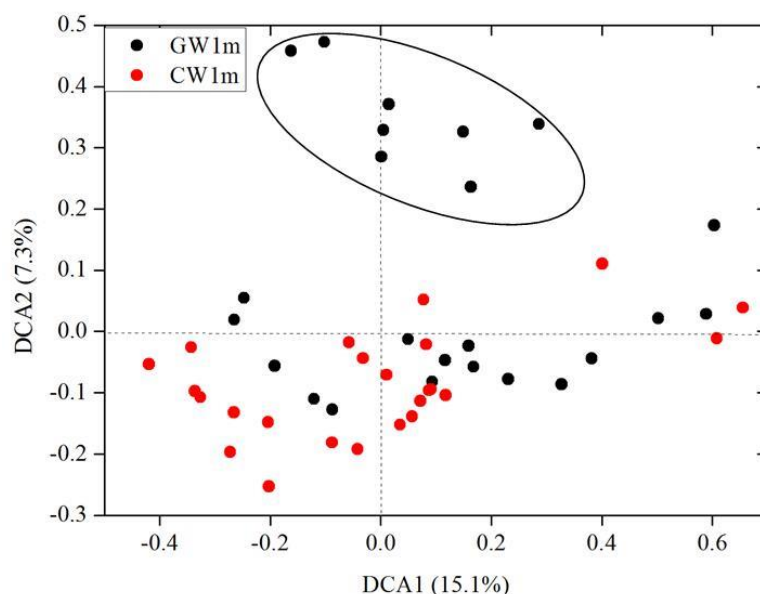


Fig. 4-1 DCA revealed the microbial functional structure at different months. Black circles and red circles represent 1-m depth water samples at GZ and CZ, respectively.

图 4-1 不同时间微生物群落功能差异的 DCA 分析 (黑色圆点代表龙须菜栽培区 1 米深水体, 红色圆点代表对照区 1 米深水体)

C degradation genes

A total of 10,447 probes representing 120 functional gene families that involved in C degradation were detected for 48 samples. Normalized signal intensity of each gene during the cultivation of *Gracilaria* was analyzed in order from labile C to recalcitrant C. Before the seaweed was cultivated (December 2014), only gene *rgl* and *axe* showed significant differences between GZ and CZ. With the growth of *Gracilaria* (May 2015), six gene families exhibited higher intensities in 1-m depth water samples at CZ ($P < 0.05$), and most of them involved in recalcitrant C degradation. On the contrary, two genes including *pulA* for starch degradation and *rgl* for pectin showed higher signal intensity at GZ ($p < 0.05$) (Fig. 4-2). After harvest of seaweed (July 2015), intensities of genes involved in C degradation almost had no significantly difference between GZ and CZ. These results indicated that *Gracilaria* cultivation shifted the C degradation potential of microbial communities with positive impacts on labile C degradation and negative effect on recalcitrant C degradation.

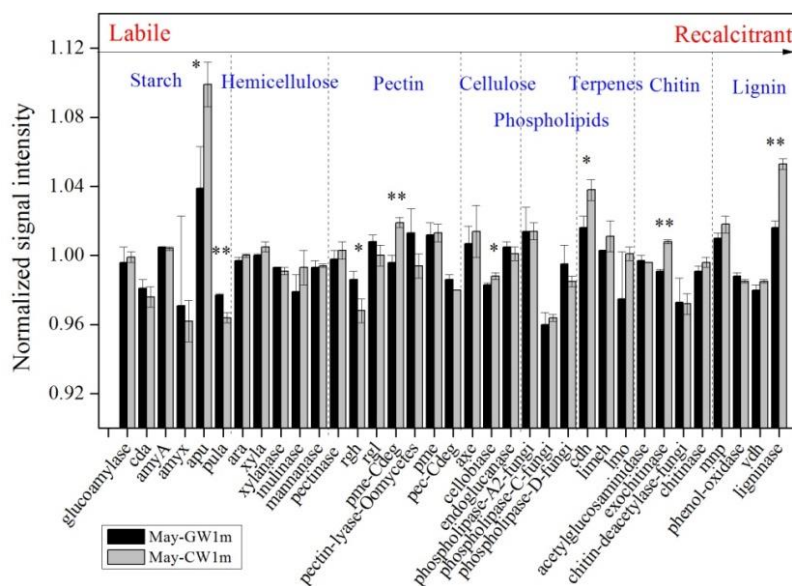


Fig. 4-2 Normalized signal intensity of genes involved in C degradation. The target genes were arranged in order from labile to recalcitrant C. All data are presented as mean $\pm$ sd. ** $p < 0.01$, * $p < 0.05$ by t -test.

图 4-2 龙须菜栽培区和对照区参与碳降解的基因丰度比较 (目标基因从活性有机碳到不易降解碳进行排列, 所有数据均为平均值 $\pm$ 标准差, t -检验: ** $p < 0.01$, * $p < 0.05$)

N cycling genes

A total of 4,287 probes representing 20 functional gene families involved in N cycling were detected. But few genes showed significantly different intensities before cultivation or after-harvest of the seaweed. When the biomass of *Gracilaria* was high during the growth cycle of seaweed (April-May 2015), the relative abundance of *amoA* gene involved in nitrification decreased significantly at GZ compared to CZ ($p < 0.01$), while the gene *nifH* involved in N fixation also decreased at GZ, suggesting a suppressed microbial functions of N fixation and subsequent nitrification. The genes *nasA* and *nir* for assimilatory N reduction ($p < 0.05$) as well as the gene *nrfA* ($p < 0.01$) for dissimilatory N reduction exhibited higher signal intensity at GZ than at CZ, suggesting an increased potential of N reduction affected by seaweed cultivation (Fig. 4-3). These results implied that cultivation of *Gracilaria* increase the synthesis of NH_4^+ by higher N reduction and lower nitrification since NH_4^+ is the most favorable nutrient of *Gracilaria*.

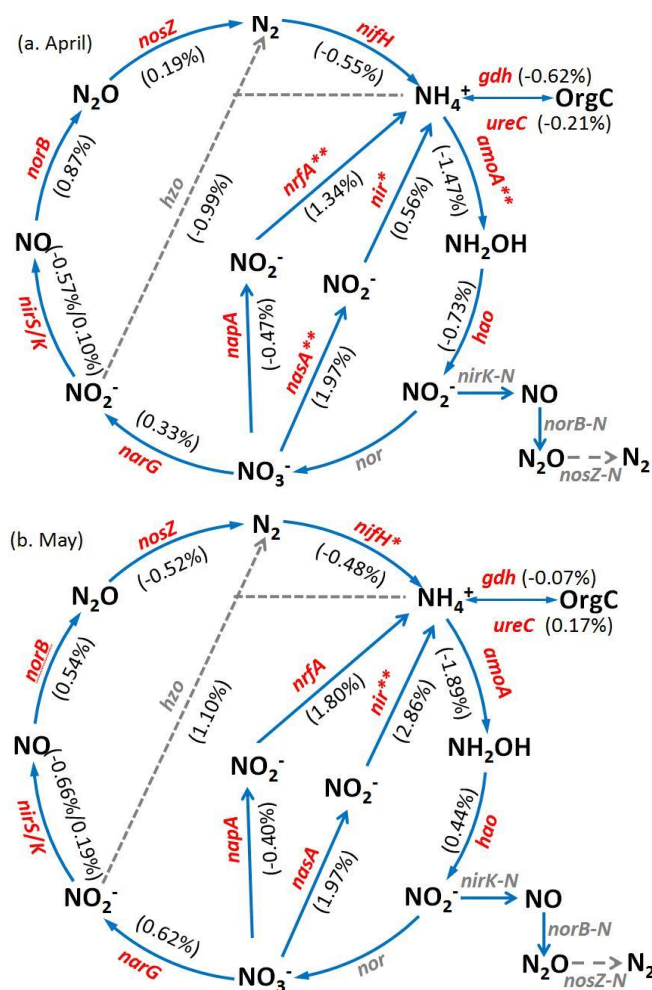


Fig. 4-3 The average changes in normalized signal intensity of genes involved in N cycling when the biomass of *Gracilaria* is relatively high (April, May). a. N cycling in April; b. N cycling in May. All data are presented as mean $\pm$ sd. ** $p < 0.01$, * $p < 0.05$ by *t*-test.

图 4-3 龙须菜生物量较大时(2015 年 4-5 月)参与氮循环的基因丰度变化情况 (a. 2015 年 4 月氮循环变化, b. 2015 年 5 月氮循环变化, 所有数据均为平均值 $\pm$ 标准差, *t*-检验: ** $p < 0.01$, * $p < 0.05$)

***P* cycling genes**

For P cycling, a total of 1,444 probes representing three functional gene families were determined for 48 samples. Similar to N cycling genes, signal intensities of genes involved in P cycling did not have difference between GZ and CZ before cultivation or after seaweed harvest. When seaweed biomass reached a relatively high level (April-May 2015), only one of the three genes, *ppk* encoding polyphosphate kinase protein, showed a significant higher intensity at GZ than that of CZ ($P < 0.05$) (Fig.

4-4). These results indicated that *Gracilaria* cultivation accelerated polyphosphate degradation and might increase the P cycling rate during the seaweed cultivation period.

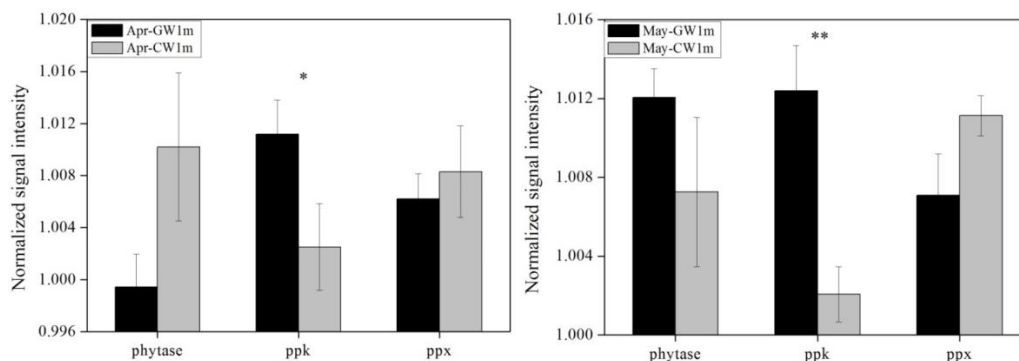


Fig. 4-4 Normalized signal intensity of genes involved in P cycling when the biomass of *Gracilaria* is relatively high (April, May). All data are presented as mean $\pm$ sd. ** $p < 0.01$, * $p < 0.05$ by t-test.

图 4-4 龙须菜栽培区和对照区参与磷循环的基因丰度比较 (所有数据均为平均值 $\pm$ 标准差, t -检验: ** $p < 0.01$, * $p < 0.05$)

Other functional genes

GeoChip 5.0 also targets genes in other functional categories such as organic remediation and metal homeostasis. For organic remediation category, a total of 6,988 probes representing 148 genes families were detected. Before cultivation of *Gracilaria* (December 2015), only few genes showed different intensity between GZ and CZ. With the growth of seaweed, more and more genes exhibited different signal intensities between GZ and CZ. When biomass of seaweed was relatively high (May 2015), 12 genes involved in aromatics remediation had significant differences between the two study zones, and 5 of them showed higher intensity at GZ while 7 of them showed higher intensity at CZ. The gene *pcpB* involved in herbicides related compounds, gene *linC* involved in pesticides related compounds, and the gene *chnC* and *chnD* involved in remediation of other hydrocarbons all had higher signal intensities at GZ ($p < 0.05$). The gene *mauab* involved in herbicides related compound, *msmabcd*, *scnabc* and *thmAB* involved in other remediation showed higher intensity at CZ than at GZ (Fig. 4-5). The above results suggested that the potential of organic

remediation at the seaweed cultivation zone was weaker than that at the control zone.

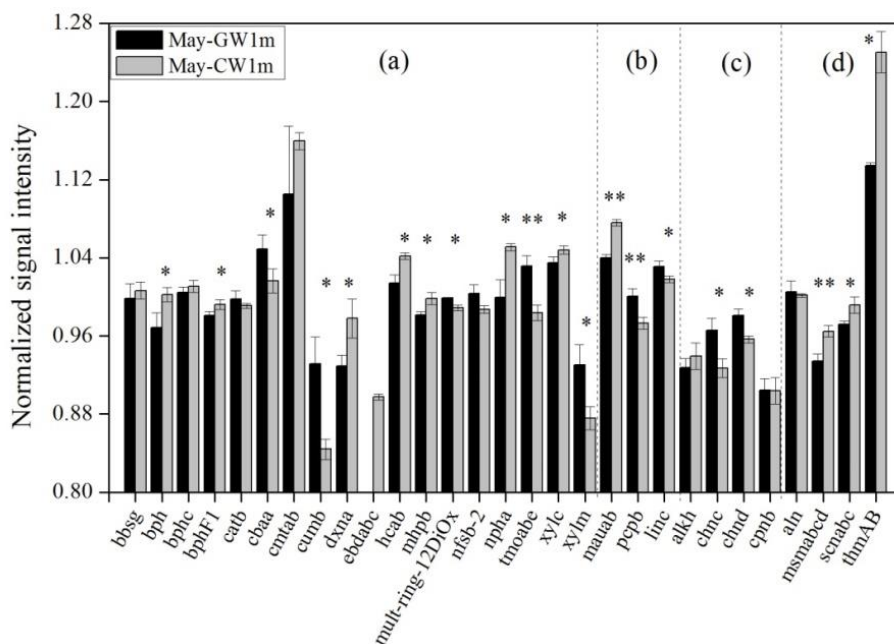


Fig. 4-5 Normalized signal intensity of genes involved in organic remediation. (a) Aromatics, (b) herbicides/pesticides related compound, (c) other hydrocarbons, (d) others. All data are presented as mean $\pm$ sd. ** $p < 0.01$, * $p < 0.05$ by t-test.

图 4-5 龙须菜栽培区和对照区参与有机污染物修复的基因丰度比较

(a) 芳香族, (b) 除草剂/杀虫剂相关物质, (c) 其他有机物, (d) 其它; 所有数据均为平均值 $\pm$ 标准差, t-检验: ** $p < 0.01$, * $p < 0.05$

For metal homeostasis category, there were 2,406 probes representing 15 genes families were detected and many of them showed increased intensities at GZ during the seaweed cultivation period. Before seaweed cultivation (December 2015), no significantly different gene intensity was observed between GZ and CZ. When the biomass of *Gracilaria* was relatively high in 2015 (April-May), gene encoding reductase for arsenic (*arsC*) and mercury (*merB*), and gene *merG* involved in mercury transport were abundant in samples at GZ ($p < 0.05$) (Fig. 4-7). These results indicated that seaweed cultivation can promote the potential of heavy metal detoxification by microbial communities.

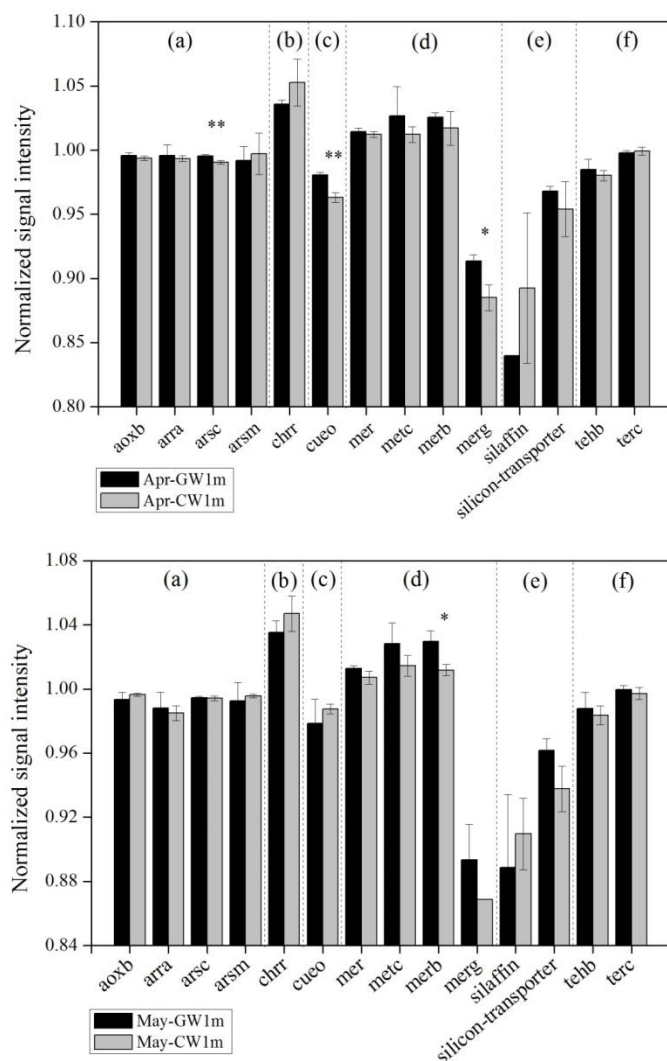


Fig. 4-6 Normalized signal intensity of genes involved in metal homeostasis. (a) Arsenic detoxification, (b) Chromium detoxification, (c) Copper detoxification, (d) Mercury detoxification/transport, (e) Silicon biosynthesis/transport, (f) Tellurium detoxification. All data are presented as mean $\pm$ sd. ** $p < 0.01$, * $p < 0.05$ by t-test.

图 4-6 龙须菜栽培区和对照区参与金属动态平衡的基因丰度比较

(a) 砷解毒, (b) 铬解毒, (c) 铜解毒, (d) 汞解毒/运输, (e) 硅生物合成/运输, (f) 碲解毒; 所有数据均为平均值 $\pm$ 标准差, t -检验: ** $p < 0.01$, * $p < 0.05$

Canonical correspondence analysis (CCA), Mantel test and partial Mantel test were performed to link the environmental factors and microbial functional potential in different months at the two study zones. The CCA model was significant ($F = 1.59$, $P < 0.01$) with the first two axes explained 54.27% of the total variation in functional microbial communities. Physical factors including temperature and pH, and chemical

factors including TP, PO_4^{3-} , NO_2^- and Chl.a were important environmental factors affecting the microbial functional gene composition ($p < 0.05$). Different environmental attributes affected samples at different months. Temperature, TN and Chl.a were positively related with samples before cultivation (December 2014) or after harvest (June-July 2015) of *Gracilaria*; NO_3^- , NO_2^- and TP affected samples at the beginning of seaweed cultivation (January to March 2015); and pH and PO_4^{3-} influenced samples when seaweed biomass was high (April to May 2015).

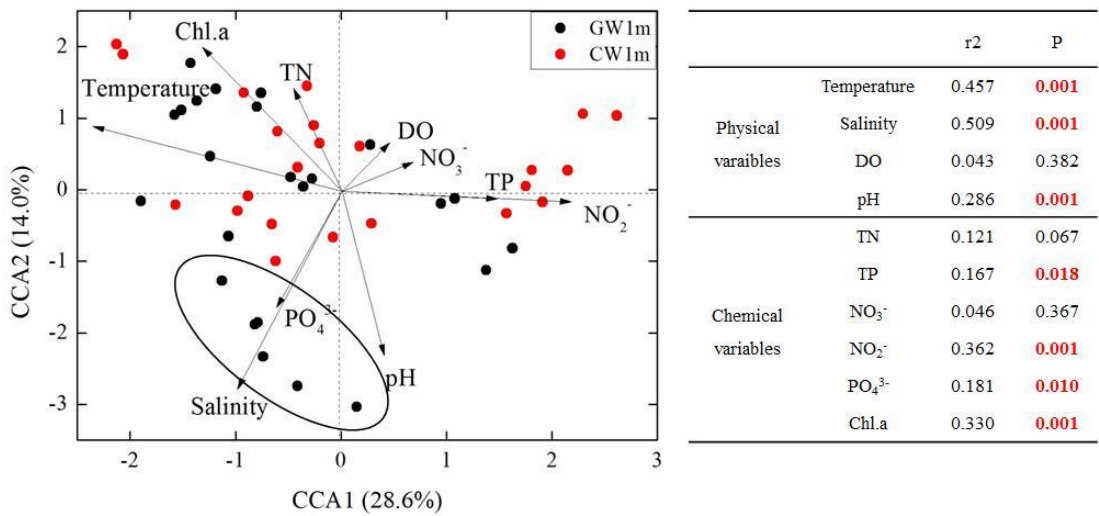


Fig. 4-7 Canonical correspondence analysis (CCA) of microbial functional gene profiles and selected environmental factors revealed the key variables shaping the microbial functional structure. Samples in close cluster were 1-m depth water samples in April and May at GZ. Black circles and red circles represent 1-m depth water samples at GZ and CZ, respectively. The insert table showed the significances of each environmental variable in explaining the variations of functional profile of microbial communities based on permutation test.

图 4-7 微生物群落功能与环境因子的 CCA 分析

4-5 月样品在图中形成明显聚类(椭圆标注); 图中黑色圆形代表龙须菜栽培区 1 米深水样, 红色圆形代表对照区 1 米深水样; 表格中的 p 值表示各环境因子是否对微生物群落功能差别具有显著解释作用

In addition, only samples of 1-m depth in April and May at GZ formed a similar cluster, indicating that seaweed cultivation shaped the functional microbial community structure when the biomass of *Gracilaria* was high (Fig. 4-7). Partial

Mantel test revealed that temperature ($p < 0.01$) and Chl.a ($p < 0.05$) were the most influential factors causing the variation of functional microbial structure between GZ and CZ. These results indicated that physical and chemical factors significantly affected the functional microbial community structure during the *Gracilaria* growth cycle. Some specific groups of functional genes were further analyzed below.

4.3.2 Spatial dynamics of functional microbial communities

Environmental factors of all depths of water samples were shown in Table 4-4 and the Student *t*-test was performed to reveal the effects of *Gracilaria* cultivation on the environment. According to the results of *t*-test, environmental properties of the surface water were responsive to seaweed cultivation with higher DO, pH and PO_4^{3-} and lower TN and NO_3^- at GZ than that at CZ. At 1-m depth water, DO was higher at GZ than at CZ. At 75% of water depth, significantly higher TN was observed at GZ than at CZ ($p < 0.05$). Environmental factors at half of water depth and the bottom water have no difference between GZ and CZ. Generally, *Gracilaria* cultivation improved the water quality in the surface water by increasing DO and pH and decreasing nutrients especially nitric salts.

A total of 29,808 probes were detected by GeoChip 5.0 for 30 samples after data filtration. Samples of GZ and CZ covered 96.0% (28,616 probes) and 85.8% (25,587 probes) of the total probes detected, respectively. 24,395 probes (81.8%) were shared by GZ and CZ, while more unique probes (4,221 probes, 14.2%) were observed at GZ compared to CZ (1192 probes, 4.0%). With the respect of the composition of these unique probes, large proportions of probes involved in C cycling (46.9% and 57.0%) and remediation of organic contaminants (20.9% and 21.1%) were detected at both GZ and CZ, followed by N cycling (14.4% and 14.3%) and S cycling (10.3% and 11.2%). The α -diversity did not have difference between GZ and CZ, except that higher Pielou's evenness was observed at 1-m depth water at GZ than that at CZ ($p < 0.05$).

Table 4-4 Environmental factors of water samples at all depth collected in May 2015. All data are presented as mean $\pm$ sd. $p < 0.05$ was bold.

表 4-4 深澳湾 2015 年 5 月不同水层环境因子的比较 (所有数据均为平均值 $\pm$ 标准差, $p < 0.05$ 加粗显示)

	May	Surface water	1-m depth water	Half of the water depth	75% of the water depth	Bottom water
Temperature	GZ	23.23 $\pm$ 0.15	23.10 $\pm$ 0.30	22.97 $\pm$ 0.15	22.93 $\pm$ 0.15	22.87 $\pm$ 0.21
	CZ	23.30 $\pm$ 0.10	23.20 $\pm$ 0.17	23.3 $\pm$ 0.10	22.80 $\pm$ 0.00	22.80 $\pm$ 0.00
Salinity	GZ	34.14 $\pm$ 0.02	34.14 $\pm$ 0.02	34.12 $\pm$ 0.01	34.12 $\pm$ 0.02	34.15 $\pm$ 0.03
	CZ	34.10 $\pm$ 0.03	34.12 $\pm$ 0.00	34.14 $\pm$ 0.02	34.15 $\pm$ 0.02	34.16 $\pm$ 0.03
DO	GZ	9.27$\pm$1.60	7.98$\pm$0.84	6.24 $\pm$ 0.05	5.74 $\pm$ 0.19	5.66 $\pm$ 0.08
	CZ	6.60 $\pm$ 0.23	6.50 $\pm$ 0.33	6.03 $\pm$ 0.22	6.00 $\pm$ 0.23	5.71 $\pm$ 0.36
pH	GZ	8.25$\pm$0.02	8.30 $\pm$ 0.06	8.29 $\pm$ 0.05	8.27 $\pm$ 0.04	8.24 $\pm$ 0.04
	CZ	8.19 $\pm$ 0.02	8.21 $\pm$ 0.01	8.22 $\pm$ 0.02	8.20 $\pm$ 0.00	8.20 $\pm$ 0.01
TN	GZ	24.58$\pm$0.33	30.28 $\pm$ 4.57	29.15 $\pm$ 4.98	24.85$\pm$1.22	35.04 $\pm$ 2.11
	CZ	38.50 $\pm$ 5.33	36.40 $\pm$ 1.88	31.25 $\pm$ 4.81	32.29 $\pm$ 2.71	32.25 $\pm$ 2.86
TP	GZ	3.83 $\pm$ 0.43	4.13 $\pm$ 0.72	4.19 $\pm$ 0.95	4.46 $\pm$ 0.70	3.63 $\pm$ 0.29
	CZ	3.72 $\pm$ 0.65	3.69 $\pm$ 0.54	4.43 $\pm$ 0.64	4.14 $\pm$ 0.91	3.81 $\pm$ 0.55
NO ₃ ⁻	GZ	9.23$\pm$1.48	12.62 $\pm$ 3.21	11.64 $\pm$ 1.24	7.07 $\pm$ 0.62	14.34 $\pm$ 3.22
	CZ	13.13 $\pm$ 1.56	12.60 $\pm$ 3.68	8.34 $\pm$ 4.39	9.14 $\pm$ 3.56	10.60 $\pm$ 3.16
NO ₂ ⁻	GZ	0.24 $\pm$ 0.16	0.25 $\pm$ 0.04	0.28 $\pm$ 0.05	0.26 $\pm$ 0.05	0.39 $\pm$ 0.11
	CZ	0.50 $\pm$ 0.33	0.31 $\pm$ 0.11	0.44 $\pm$ 0.27	0.32 $\pm$ 0.09	0.49 $\pm$ 0.35
NH ₄ ⁺	GZ	4.13 $\pm$ 0.94	3.85 $\pm$ 1.21	4.08 $\pm$ 0.95	4.41 $\pm$ 0.79	4.71 $\pm$ 1.05
	CZ	4.46 $\pm$ 0.68	4.35 $\pm$ 0.18	4.40 $\pm$ 0.22	4.09 $\pm$ 0.33	4.11 $\pm$ 0.31
PO ₄ ³⁻	GZ	9.20$\pm$0.58	9.35 $\pm$ 0.47	8.44 $\pm$ 0.22	8.36 $\pm$ 1.33	8.37 $\pm$ 1.34
	CZ	7.94 $\pm$ 0.45	8.58 $\pm$ 0.95	8.07 $\pm$ 0.33	8.23 $\pm$ 0.55	8.25 $\pm$ 0.75
Chl.a	GZ	0.54 $\pm$ 0.13	0.55 $\pm$ 0.13	0.54 $\pm$ 0.12	0.63 $\pm$ 0.08	0.62 $\pm$ 0.08
	CZ	0.33 $\pm$ 0.25	0.50 $\pm$ 0.15	0.49 $\pm$ 0.16	0.53 $\pm$ 0.15	0.50 $\pm$ 0.15

Adonis tests were performed to discern the impacts of seaweed cultivation on microbial functional profiles, and the results indicated that the interaction of seaweed cultivation and water depth, seaweed cultivation and water depth all have significant effects on functional microbial communities ($p < 0.01$) (Table 4-5). The functional potential of microbial communities differed significantly between GZ and CZ by

three different dissimilarity tests (MRPP, Anosim, and adonis) ($p < 0.05$) (Table 4-6). DCA was also conducted to reveal the overall effects of *Gracilaria* cultivation on functional profiles of microbial communities at different water depths. Fig. 4-8 indicated that samples of 1-m depth water at GZ (GW1m) appeared to be closely clustered even though samples of other depths did not separate completely. These results suggested that seaweed cultivation shifted the overall functional profile of microbial communities. Further analysis of different functional gene categories was performed to understand the effect of seaweed cultivation on specific functional genes/processes.

Table 4-5 Significance tests of seaweed cultivation and depth on the microbial community function by adonis

表 4-5 龙须菜栽培及水深对微生物群落功能影响的 adonis 分析

	F	R ²	p
Cultivation	10.470	0.153	0.001
Depth	2.661	0.194	0.003
Cultivation×Depth	4.362	0.318	0.001

Table 4-6 Significance of effects of seaweed cultivation on overall microbial functional profile in different depths using three dissimilarity test methods

表 4-6 龙须菜栽培对微生物群落功能特征影响的非参数多变量差异性分析

	Statistic	P-value
MRPP	0.304	0.010
Anosim	0.094	0.016
adonis	2.41	0.010

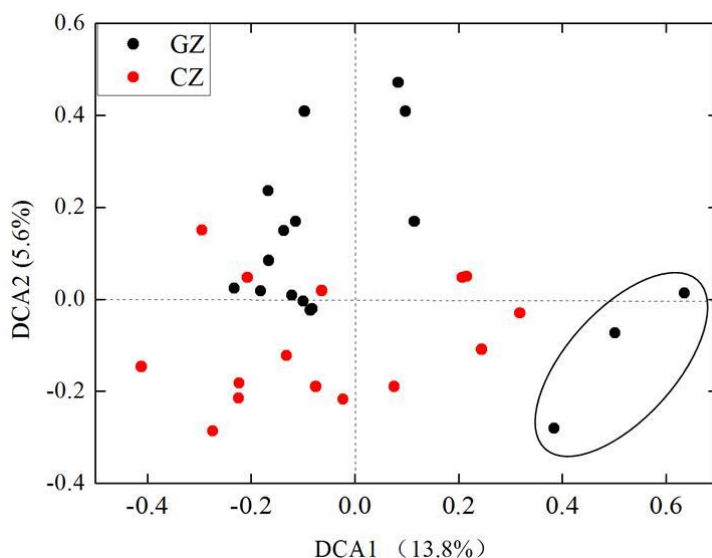


Fig. 4-8 DCA revealed the microbial functional structure of water column at different depths in May. Black circles and red circles represent water samples of different depths at GZ and CZ in May 2015.

图 4-8 不同水层微生物群落功能特征的 DCA 分析 (黑色和红色圆点分别代表 2015 年 5 月龙须菜栽培区和对照区不同水层水样)

C degradation genes

A total of 9,378 probes representing 60 functional genes families involved in C degradation were detected for 30 samples. For all water samples at five depths, only those at 1-m depth and at the bottom showed differences in signal intensities between GZ and CZ. At 1-m depth water, the signal intensities of gene *apu* for starch degradation, *cdh* for terpenes degradation and *ligninase* genes for lignin degradation were higher at CZ (CW1m) than at GZ (GW1m). At the bottom water, more genes showed higher intensities at CZ (CWB) than at GZ (GWB), such as *endopolygalacturonase-fungi* and *exopolygalacturonase-fungi* genes for pectin degradation, *phospholipase-D-fungi* genes for phospholipids degradation, and *endochitinase* and *exochitinase* genes for chitin degradation (Fig. 4-9). These results indicated that seaweed cultivation could suppress C degradation of microbial communities especially at the bottom water.

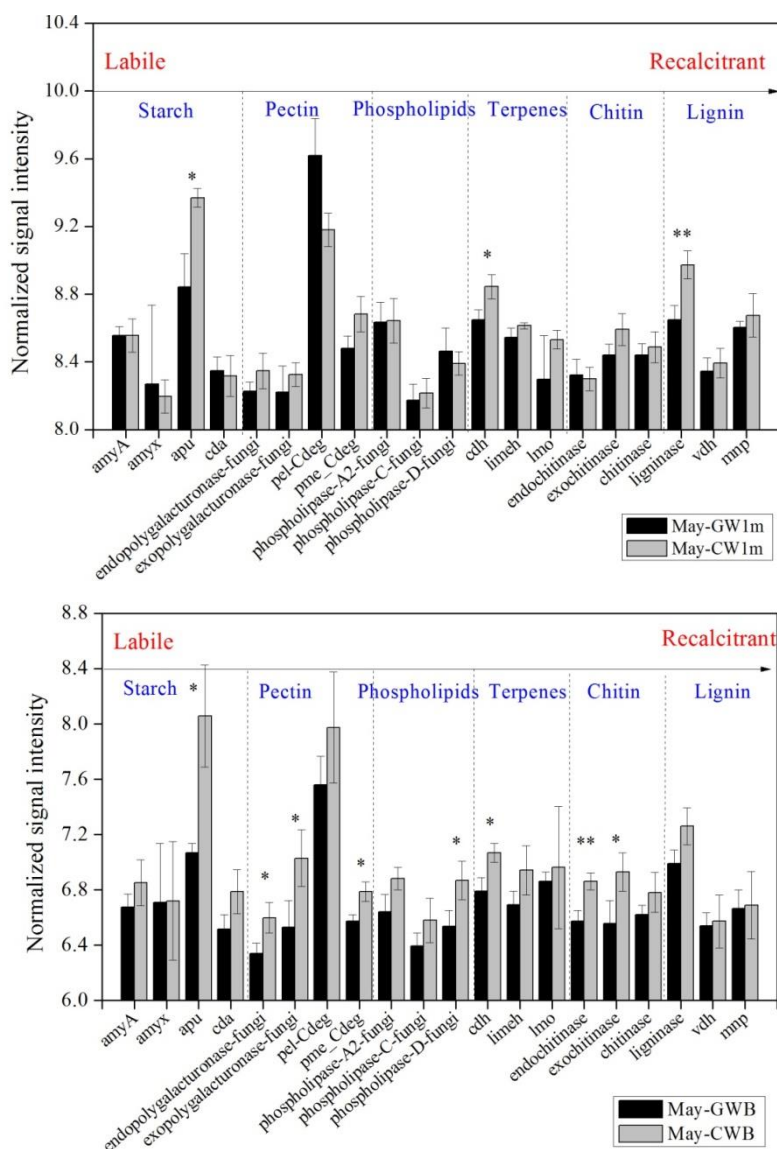


Fig. 4-9 Normalized signal intensity of genes or enzymes involved in C degradation at different water depths. The targeted genes were arranged in order from labile to recalcitrant C. GW1m: 1-m depth at GZ, CW1m: 1-m depth at CZ; GWB: the bottom at GZ, CWB: the bottom at CZ. All data are presented as mean $\pm$ sd. ** $P < 0.01$, * $P < 0.05$ by t-test.

图 4-9 龙须菜栽培区和对照区不同水层(1 米深及底层水体)参与碳降解的基因丰度比较 (目标基因从活性有机碳到不易降解碳进行排列, 所有数据均为平均值 $\pm$ 标准差, t -检验: ** $p < 0.01$, * $p < 0.05$)

Other functional categories

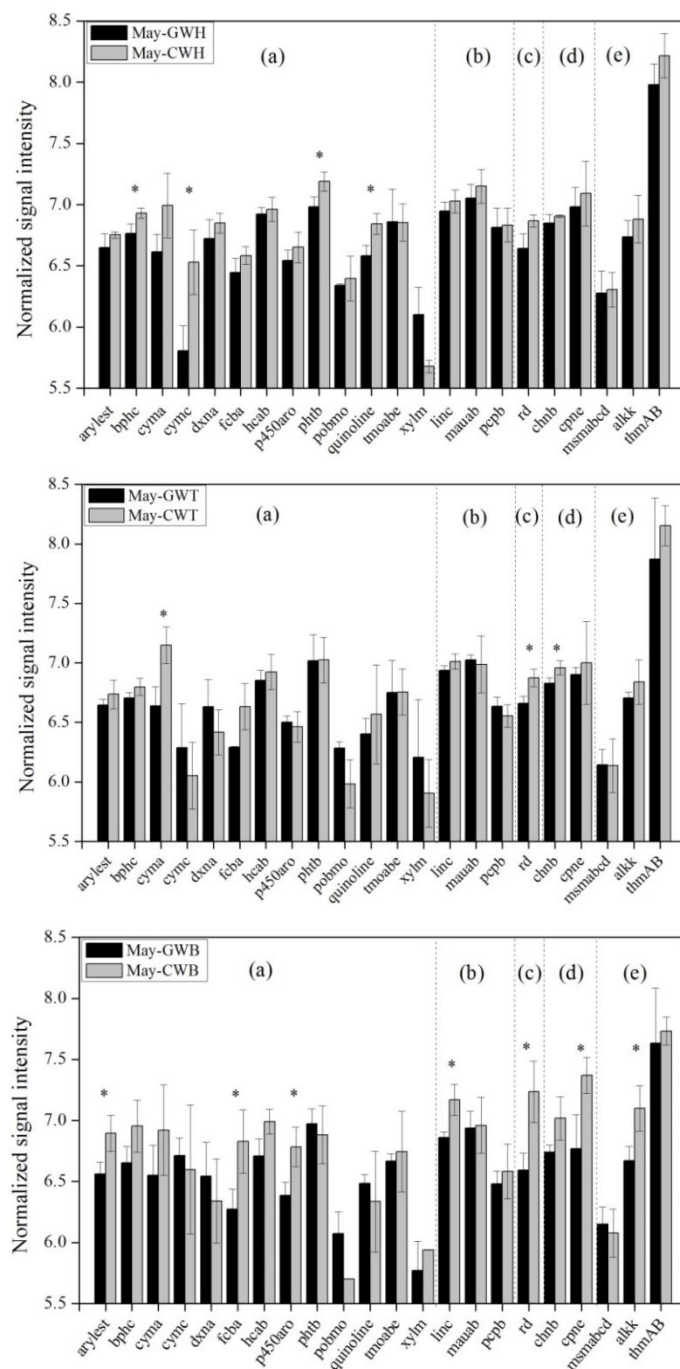


Fig. 4-10 Normalized signal intensity of genes involved in organic remediation. (a) Aromatics, (b) herbicides/pesticides related compound, (c) chlorinated solvents, (d) other hydrocarbons, (e) others. All data are presented as mean $\pm$ sd. ** $p < 0.01$, * $p < 0.05$ by t -test.

图 4-10 参与有机污染物修复的基因丰度比较

(a) 芳香族, (b) 除草剂/杀虫剂相关物质, (c) 氯化溶剂, (d) 其他有机物, (e) 其它; 所有数据均为平均值 $\pm$ 标准差, t -检验: ** $p < 0.01$, * $p < 0.05$

GeoChip 5.0 also detected genes in other functional categories such as organic remediation and metal homeostasis. A total of 6,445 probes were detected for 30 samples, representing 147 gene families involved in remediation of organic matters. For those genes with significantly different intensities between GZ and CZ, most of them involved in aromatics degradation (for examples, *bphC*, *cymA*, *cymC*, *fcBA*, *p450* and *phtB*) and had lower intensities at 1-m depth, half of water depth, 75% of water depth and the bottom water of GZ than that of CZ (Fig. 4-10). Also, lower intensities of genes, such as those involved in pesticides related compound (*linC*), chlorinated solvents (*rd*), other hydrocarbons (*chnB* and *cpne*) and others (*alkk*) were observed at GZ than at CZ. These results indicated that seaweed cultivation suppressed organic remediation of microbial communities from 1-m depth to the bottom water.

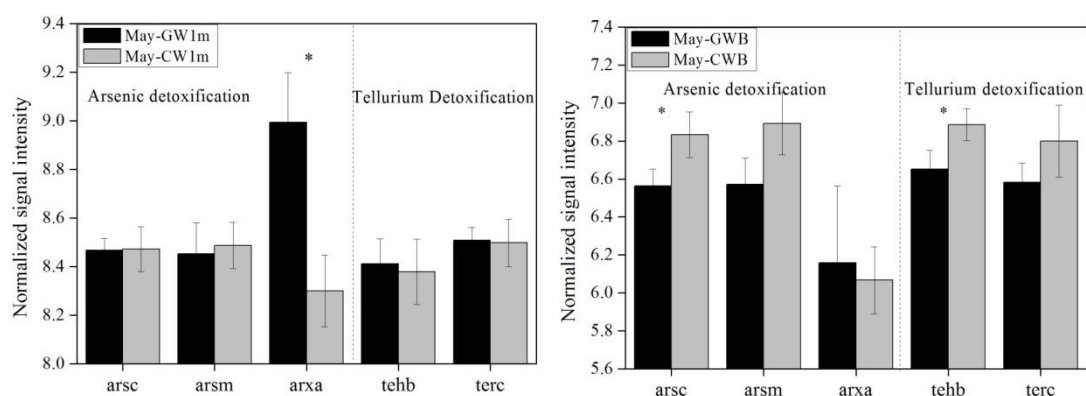


Fig. 4-11 Normalized signal intensity of genes involved in metal homeostasis. All data are presented as mean $\pm$ sd. ** $p < 0.01$, * $p < 0.05$ by t-test.

图 4-11 参与金属动态平衡的基因丰度比较

所有数据均为平均值 $\pm$ 标准差, t -检验: ** $p < 0.01$, * $p < 0.05$

For metal homeostasis, 2,171 probes representing 15 gene families were detected for 30 samples. At 1-m depth, only gene *arxA* for arsenic detoxification showed higher intensity at GZ than that of CZ (Fig. 4-11). At the bottom water, lower intensities of *arsC* involved in arsenic detoxification and *tehB* involved in tellurium detoxification were observed at GZ than at CZ ($p < 0.05$). The opposite results of heavy metal detoxification at different depths could be related with the interaction

between seaweed and microbes. Seaweed cultivation promoted heavy metal detoxification by microbial communities in the cultivation waters.

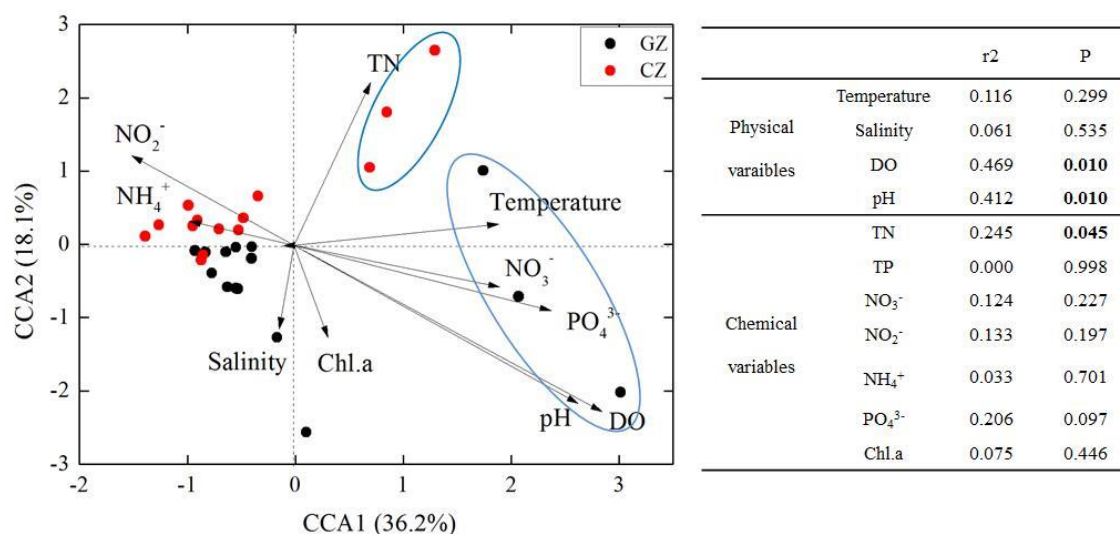


Fig. 4-12 CCA indicated the linkage between microbial functional structure and selected environmental variables. Black circles and red circles represent 1-m depth water samples at GZ and CZ, respectively. Samples of 1-m depth at CZ in Quadrant 1 formed a close cluster were 1-m, and 1-m depth water samples at GZ in Quadrant 4 and Quadrant 1 also formed a close cluster. The insert table showed the significances of each environmental factor in explaining the variations of microbial functional gene composition based on permutation test.

图 4-12 微生物群落功能与环境因子的 CCA 分析

图中黑色圆形代表龙须菜栽培区 1 米深水样，红色圆形代表对照区 1 米深水样。龙须菜栽培区 1 米深水样在第一象限和第四象限形成了明显的聚类；对照区 1 米深水样在第一象限也形成了明显聚类；表格中的 p 值表示各环境因子是否对微生物群落功能差别具有显著解释作用

The linkage between the microbial functional structure and a selected set of environmental factors were analyzed using CCA, Mantel test and partial Mantel test. The CCA results indicated a significant model ($F=1.62$, $P<0.01$) was tested with 54.17% explained by the first two axes. Permutation test of each environmental attribute showed that DO, pH and TN were important factors influencing the microbial functional gene composition ($P<0.05$). Samples at 1-m depth of two study zones were well separated from samples at other depths, suggesting that seaweed cultivation has effect on microbial functional structure along water depths. To be specific, 1-m

depth water samples at GZ (GW1m) were positively correlated with temperature, pH, DO, NO_3^- and PO_4^{3-} , while 1-m depth water samples at CZ (CW1m) were positively affected by TN (Fig. 4-12).

4.4 Discussion

Many anthropogenic activities such as fishing, entertainment and mariculture have great impacts on the diversity, structure and function of microbial communities in marine ecosystems. As marine microbial communities mediate important biogeochemical processes, such as C, N, P and S cycling (Cho, 1988), understanding their response to mariculture is essential to reveal the effects of anthropogenic activities on marine ecosystems. However, few studies were reported about the influence of mariculture, especially seaweed cultivation, on functional profiles of microbial communities (Herndl and Weinbauer, 2003). In this study, GeoChip 5.0 based high throughput technology was applied to identify the microbial functional diversity, composition and structure to reveal the impact of seaweed *G. lemaneiformis* cultivation on functional profiles of microbial communities. Our hypotheses are that (i) *Gracilaria* cultivation alters the microbial functional structure with increasing seaweed biomass; and (ii) functional microbial communities in various gene families were shifted due to seaweed cultivation. Our results support thesis hypotheses. High evenness of microbial functional profiles at GZ suggested a more consistent functional community structure due to seaweed cultivation. Significance tests indicated that functional structure of microbial communities was affected by seaweed cultivation. Also, genes involved in various functional groups showed differences in signal intensities between GZ and CZ. Analyses for linking environmental variables and microbial communities suggested that key environmental variables could shape the functional profiles of microbial communities.

4.4.1 The functional microbial community structure in 1-m depth water

Cultivation of seaweed *G. lemaneiformis* can increase the C input from plant and change the micro-environment of marine microbial communities in cultivation

ecosystems. A previous study at our lab reported that *Gracilaria* cultivation altered the structure and composition of microbial communities via changes in environmental variables (Xie, 2017). Consistent with those previous studies, higher DO and lower nutrients were observed at GZ than that at CZ. Low nutrients could lead to a lower primary production, causing a decrease in organic C in the surrounding environment. Genes involved in C degradation showed lower signal intensities, especially in recalcitrant C when seaweed biomass was high due to low organic C. However, no direct evidence was available due to a lack of total organic carbon (TOC) data. These results indicated a shifted microbial community structure with accelerated rate of labile C degradation and inhibited rate of recalcitrant C degradation when seaweed biomass was high, suggesting that *Gracilaria* cultivation altered the C degradation functional potential of microbial communities at GZ.

N cycling showed significant variance in the microbial functional structure between the two study zones when seaweed biomass was high. During April and May, functional composition of microbial communities involved in N cycling changed significantly. Microbial communities of GZ accelerated N reduction with increased intensities of *nasA*, *nir* and *nrfA*, and inhibited nitrification with decreased intensity of *amoA*. All of the above changes led to an increase in NH_4^+ . Since NH_4^+ is the most favorable nutrient of *Gracilaria* and can be taken preferentially than NO_3^- and NO_2^- by the seaweed (Li and Lin, 2005), its accumulation would benefit the growth of *Gracilaria* greatly. However, lower intensity of gene *nifH* was detected at GZ and can be explained by a general lack of nutrients in this ecosystem. The shifts of functional genes composition involved in N cycling implied that the functional microbial community structure could respond quickly to environmental changes and interact with seaweed positively. P cycling rate may be accelerated with increase in signal intensity of gene *ppk* encoding polyphosphate kinase protein, suggesting that microbial communities at GZ can accelerate the polyphosphate synthesis.

Except for genes involved in biogeochemical cycling (e.g. C, N, P and S cycling), genes involved in organic remediation and metal homeostasis also responded differently to *G. lemaneiformis* cultivation. Genes involved in remediation of

aromatics, herbicides/pesticides related compounds and other hydrocarbons all had lower signal intensities at GZ with the growth of seaweed. The phenomenon was caused by co-metabolism of microbes (Horvath, 1972). Since the organic C source was less at GZ than at CZ, the functional potential of organic degradation would decrease correspondingly. With respect to genes in metal homeostasis category, many heavy metal detoxification genes showed increased intensities at GZ during the seaweed cultivation period. Seaweed has the potential to uptake heavy metal from the surrounding water (Caliceti et al., 2002). The increase of metal detoxification/transport ability may be correlated with their uptake by seaweed. All of these together indicated that seaweed cultivation could alter the functional structure of microbial communities to adapt the seaweed cultivation ecosystem.

The microbial functional gene composition was also affected by environmental variables, such as temperature, salinity, pH, DO, TN, TP and Chl.a. In this study, salinity, pH and PO_4^{3-} were positively correlated with samples at 1-m depth of GZ in May. Other samples did not form similar cluster even though they had strong correlations with environmental variables such as temperature, Chl.a, NO_2^- and TP. Although many environmental variables did not show differences between GZ and CZ, seaweed cultivation indeed altered the micro-environment and shaped the functional community structure via their growth.

4.4.2 *Gracilaria* cultivation shifted the microbial functional gene composition at different depths

G. lemaneiformis cultivation is one of the key industries of Nan'ao Island and makes great contributions to ecological and economic development. As an important role of seaweed cultivation ecosystem, little is known about the functional profiles of microbial communities at different water depths. In this study, we examined samples at five different depths at GZ and CZ in May 2015 using a comprehensive functional gene array, GeoChip 5.0 and hypothesized that the overall structure or functional gene composition was shifted at different water depths due to seaweed cultivation. Our results supported this hypothesis. *Gracilaria* cultivation altered the

micro-environment by changing physical and chemical variables, especially at the surface water. Adonis tests indicated that seaweed cultivation, depth, and the interaction between cultivation and depth all shifted the microbial functional structure significantly. Three different dissimilarity tests, MRPP, anosim and adonis, also revealed that *Gracilaria* cultivation affected the functional profiles of microbial communities. DCA and CCA both showed that 1-m depth water samples at GZ (GW1m) were closely clustered and had positive relationship with many key environmental variables such as temperature, pH, DO, NO_3^- and PO_4^{3-} , while 1-m depth water samples at CZ (CW1m) was positively correlated with TN. These above results together suggested that seaweed cultivation altered the overall functional composition of microbial communities.

Since the micro-environment of GZ changed, the ecological processes should have various responses at different water depths. For genes involved in C degradation from labile to recalcitrant carbon, all water samples at 1-m depth and the bottom showed lower intensities at GZ than at CZ. At 1-m depth, the inhibited C degradation under seaweed cultivation may be the result of protection of *Gracilaria* thalli. Meanwhile, the suppressed potential of C degradation at the bottom water was possibly due to lower TC concentrations in benthic environments at GZ compared with CZ.

For those genes involved in other functional categories, significant differences were detected between the two study zones. Most of genes involved in organic remediation (e.g., aromatics degradation, pesticides related compound, chlorinated solvents and other hydrocarbons) had lower intensities at 1-m depth, half of water depth, 75% of water depth and the bottom water of GZ than at CZ, indicating that seaweed cultivation may improve the water quality and decrease C sources at GZ which suppress organic remediation of microbial communities at different water depths. With the respect to metal homeostasis, opposite results of signal intensities were detected at 1-m depth and the bottom. At 1-m depth, higher potential of arsenic detoxification was observed at GZ than that of CZ. However, lower intensities of gene involved in arsenic detoxification and tellurium detoxification were observed at the bottom water of GZ than at CZ. The opposite results of heavy metal detoxification at

different depths may be related with the interaction between seaweed and microbes. Seaweed can take heavy metals from water and microbes could help this process by increasing the potential of metal detoxification in the surrounding water.

The variances of functional microbial community composition at different water depths were closely related with different micro-environments even though few physical or chemical factors showed differences at all depths. From the divergent functional structure of microbial communities between GZ and CZ, we found that the most affected depths were 1-m depth and the bottom. These phenomena could be explained by their divergent surrounding environments. As the close clusters shown in CCA figure, 1-m depth water was directly influenced by seaweed since *Gracilaria* live around the surface to 1-m depth water and can secrete many organic matters, such as algal polysaccharide, to affect the functional structure of microbial communities (Decho and Lopez, 1993). Also, microbial communities of the bottom water were affected strongly by seaweed detritus, benthic organisms, environment and sediment resuspension (Anderson and S. Meadows, 1978), which cause variances in the functional microbial communities structure. Therefore, *Gracilaria* cultivation altered the functional structure of microbial communities at different water depths when the seaweed biomass was high.

4.5 Chapter Summary

In this study, we used GeoChip 5.0 technology to analyze the effects of *Gracilaria* cultivation on the functional microbial community structure throughout one growth cycle (temporal dynamic) and at different depths when seaweed biomass was high (spatial dynamic). Our results showed that significant variances of microbial community functional potentials were observed between GZ and CZ along with the growth of seaweed, especially in April and May when seaweed biomass was high. Also, seaweed cultivation shifted the functional microbial community structure between the two study zones at different depths in May, especially at 1-m depth and the bottom water. The close relationship between functional gene changes and seaweed biomass could reflect the intricate interactions in the

seaweed-microbe-environment ecosystem. These results indicated that micro-environment affected by *Gracilaria* cultivation could mediate the effects of seaweed cultivation and environmental changes on the functional community composition. The temporal and spatial study of microbial functional structure in this seaweed cultivation ecosystem can provide basic data about key ecological processes of marine microbial communities affected by mariculture and shed light on future work of seaweed-microbe interactions.

Chapter 5: Microbial communities on the *G. lemaneiformis* surface

5.1 Introduction

As ecosystem engineers, the surfaces of seaweed provide organic nutrients and protective living substrates for many epiphytic organisms including bacteria, fungi, protozoa, diatoms, spores, and larvae of marine invertebrates (Bulleri et al., 2002; Haywood et al., 1995). The interaction between epiphytic organisms and the host seaweed is critical for the completion of their live cycles. For example, epiphytic bacteria induce liberation of carpospores in the red alga, *Acrochaetium* (Weinberger et al., 2007) and settlement of zoospores in the green algae, *Enteromorpha* and *Ulva* (Joint et al., 2002; Joint et al., 2007). Also, they produce the plant growth-promoting substances (auxin, cytokinin and kinetin) (Maruyama et al., 1986; Provasoli, 1958; Singh et al., 2011), quorum sensing signaling molecules (Williams, 2007) and bioactive molecules (Goecke et al., 2010) that are responsible for the normal morphological development and growth of the host seaweeds. Epiphytic bacterial communities, therefore, are indispensable both to the host seaweed and the coastal ecosystem where they live. Previous studies have shown that the diversity, composition and structure of epiphytic microbial communities are species-specific and function-driven in different host seaweeds across seasons and habitats (Burke et al., 2011a; Burke et al., 2011b; Lachnit et al., 2009; Lachnit et al., 2011). However, the divergences in composition and function of the epiphytic and free-living microbial communities have rarely been reported (Burke et al., 2011a; Lachnit et al., 2011).

Given the importance of seaweed *G. lemaneiformis* in coastal ecosystems, it is essential to understand the interaction between epiphytic microbes and *G. lemaneiformis*, and by link the community composition with their overall ecological functions. Reports based on the 16S rRNA gene clone library (Wu, 2012) and denaturing gradient gel electrophoresis (DGGE) fingerprinting have shown that the surface of *G. lemaneiformis* hosts diverse microbial communities. However, due to the limitations in technology and the samples surveyed in previous studies, the composition and function of epiphytic microbial communities of *G. lemaneiformis*

have not been thoroughly characterized.

To understand the diversity, composition, structure and function of epiphytic microbial communities of seaweed *G. lemaneiformis*, we used multiple high-throughput technologies in this study. Amplicons of 16S rRNA and ITS gene amplicons by the Illumina Miseq platform to reveal the diversity and structure of microbial and fungal communities in epiphytes of *G. lemaneiformis* (Caporaso et al., 2012). Also, a functional gene arrays, GeoChip 5.0 was applied to examine the functional profiles of epiphytic microbial communities, which can provide us with an overall functional analysis of epiphytes such as genes affiliated with seaweed-associated life, biogeochemical cycling, stress, and pathogen (He et al., 2010; He et al., 2007). In this study, the epiphytic and free-living microbial communities were analyzed using 16S rRNA sequencing, ITS sequencing and GeoChip 5.0 to test the following hypotheses: (i) the structure and function of epiphytic communities would be significantly shifted from the free living microbial communities; (ii) certain seaweed taxa groups and functional genes would be enriched in epiphytic communities; and (iii) *G. lemaneiformis* would select epiphytic microbes to perform a stable interaction between host seaweed and their epiphytic microbes. Our results showed significant shifts in composition and function of epiphytic microbial communities between the epiphytic and free-living groups, and specific groups of microbes were enriched possibly due to seaweed's selection. This study provides useful information about the interactions between epiphytic microbes and host seaweeds in coastal ecosystems.

5.2 Materials and methods

5.2.1 Sample collection and DNA extraction

Healthy *G. lemaneiformis* thalli (100 g per sample) and the *in situ* sea water (5 L per sample) were collected at low tide (23° 28'N, 117° 06'E) in April 2015 from three sites in the large-scale cultivation zones of *G. lemaneiformis* near Nan'ao Island, China. The thalli were enclosed individually inside press-seal bags *in-situ* and stored at 4°C.

Water samples were immediately serially filtered through 10 and 0.2 μm pore-size filters (EMD Millipore GTPP04700, USA) under low temperature and stored in liquid nitrogen onsite. Temperature, salinity, pH, dissolved oxygen (DO), transparency and water depth were obtained from the *in-situ* data by using the Potable Water Quality Analyzer-YSI 556 systems (YSI Incorporated, USA), a Secchi disc and a Digital Depth Sounder (Speedtech, USA), respectively. Total nitrogen (TN), total phosphorus (TP), nitrate (NO_3^-), nitrite (NO_2^-), ammonia (NH_4^+), soluble reactive phosphate (PO_4^{3-}) and chlorophyll a (Chl-*a*) were analyzed according to standard methods (AQSIQ, 2007) in the laboratory.

Approximately two gram of seaweed thalli were rinsed with filtered sterile seawater and placed into sterile flasks in 100 mL sterile seawater, then vibrated in the Ultrasonic Processor (Auto Science, Tianjin, China) in 50 W for 1.5 min (Buesing and Gessner, 2002; Mao, 2012). 2.5 L seawater per sample containing epiphytic microbes was serially filtered through 10 and 0.2 μm filters (EMD Millipore GTPP04700, USA). All samples, including filters enriched with microbes from seawater, were transferred to an -80°C freezer awaiting processing. Microbial genomic DNA was extracted from each filter using Omega E.Z.N.A.® Water DNA Kit (Omega Bio-Tek, USA) as per the manufacturer's instructions. The DNA quality and quantity were measured using PicoGreen through a FLUOstar Optima fluorescence plate reader (BMG LABTECH, Jena, Germany) and purified using the MicroElute® Cycle-Pure Kit (Omega Bio-Tek, USA) to meet the criteria as follows: $A_{260\text{nm}}/A_{280\text{nm}} > 1.8$, $A_{260\text{nm}}/A_{230\text{nm}} > 2.0$.

5.2.2 Amplification of 16S rRNA and ITS genes, and MiSeq sequencing

The diversity and composition of microbial and fungal communities were analyzed using MiSeq sequencing of the 16S rRNA gene and ITS. The V4 of 16S rRNA genes were amplified with the primer pair 515F (5'-GTGCCAGCMGCCGCGGTAA-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3') (Caporaso et al., 2012), and ITS was amplified with the primer pair ITS7F (5'-GTGARTCATCGARTCTTTG-3') and ITS4R (5'-TCCTCCGCTTATTGATATGC-3'). PCR was performed in triplicate for

each sample as follows: 94 °C for 3 min, then 94 °C for 25 s, 53 °C (16S rRNA gene) or 51 °C (ITS) for 20 s, and 68 °C for 45 s, followed by a final extension at 68 °C for 10 min. The triplicate PCR products of each sample were pooled together and quantified by PicoGreen using a FLUOstar Optima. Then PCR products of each sample were combined equally and purified by QIAquick Gel Extraction Kit (Qiagen, Valencia, USA). The purified DNA was loaded on a 500 cycles (2×250 bp paired ends) kit and run on a Miseq machine (Illumina, San Diego, CA, USA) at the Institute for Environmental Genomics, University of Oklahoma, USA.

Raw sequences were split with barcodes and were trimmed by Btrim (Kong, 2011) with a minimum length of 100 bp. The paired reads were joined using FLASH (Magoc and Salzberg, 2011) with at least 10 bp overlap and lower than 5% mismatches. After removing sequences containing N, sequences in the length of 245~258 bp (for 16S rRNA genes) and 210~450 (for ITS) were subjected to chimeras detecting using U-Chime (Edgar et al., 2011). Operational Taxonomic Units (OTUs) clustering was performed using UCLUST (Edgar, 2010) and then assigned to microbial or fungal taxonomy by RDP Classifier (Wang et al., 2007a). Singletons were removed and the resample was rarefied at 10,000 sequences per sample for downstream data processing. The above steps were performed through the Galaxy pipeline (<http://zhoulab5.rccc.ou.edu/>) developed by Y. Qin *et al.* (unpublished).

5.2.3 GeoChip hybridization, scanning and GeoChip data processing

Functional gene array (FGA), GeoChip 5.0, containing 161,961 probes (180K) designed for covering 1447 genes in 12 functional categories for biogeochemical elemental cycling and different microbial functional processes (He et al., 2012a; He et al., 2007), was used to detect the functional gene abundance of epiphytic and free-living microbial communities in the *G. lemaneiformis* cultivation ecosystem. 0.8 µg DNA from each sample was mixed with random primers and denatured before dNTP, fluorescent dye Cy3 Dntp and DNA polymerase were added for labeling at 37°C for 6 h, followed by heating at 95°C for 3 min. Labeled DNA was purified and dried for hybridization. DNA was re-hydrated with ddH₂O, suspended in hybridization solution

and denatured at 95°C for 3 min. The mixtures were then loaded onto the glass microarray and hybridized at 67 °C for 24 h. Finally, the arrays were washed with Wash Buffer and scanned by an MS 200Microarray Scanner (NimbleGen) at 532nm and 635 nm (Xue et al., 2016). To grid and process the scanned images, NimbleScan software version 2.5 was used to transform them into signal intensities (He et al., 2010; He et al., 2012b). The raw signals files were uploaded to the Microarray Data Manager on the website (<http://ieg.ou.edu/microarray>). A cutoff of Signal to Noise (SNR) > 2 was set as the signal threshold. The signal intensities data were then normalized and transferred using the logarithm method.

5.2.4 Statistical analysis

The alpha- and beta-diversity of microbial communities were analyzed using multiple indices (Shannon Index, Simpson Index, Pielou Evenness and Simpson Evenness) and the Bray-Curtis distance. Detrended correspondence analysis (DCA) was applied to visualize the overall difference in the distribution patterns of microbial communities among samples. Non-parametric multivariate (Adonis) was performed to test the variances of the microbial community structure and function between GW (free-living communities of the *G. lemaneiformis* cultivation ecosystem) and GE (the epiphytes on the *G. lemaneiformis* surface). Analysis of variance (ANOVA) was performed to test the differences in signal intensities of each functional gene for GeoChip or relative abundances of OTUs. Response ratio was used to compute the variances in the composition of microbial communities between GW and GE from 16S rRNA and ITS gene amplicon sequencing data at 95% confidence interval. The hierarchical clustering heatmap of functional gene abundance was used to demonstrate the overall differences in microbial community function. All the above analyses were carried out in R (version 3.2.2) using vegan, gplots and ggplot2 packages when not specified.

5.3 Results

5.3.1 Overall profiles of the epiphytic and the free-living microbial community of *G. lemaneiformis*

A comparison of α -diversity of microbial communities between GW and GE implied a special core microbial community on the surface of seaweed *G. lemaneiformis* due to its particular habitat, with significantly lower Shannon and Simpson index and higher Simpson evenness of GE samples (Table 5-1). Three different non-parametric multivariate tests (adonis) were performed to test the dissimilarity of the microbial communities. The results of adonis demonstrated distinct divergences in composition, structure and functional profiles of the microbial communities between GW and GE (Table 5-2), which was confirmed by the DCA plot (Fig. 5-1).

Table 5-1. α -diversity indices exhibited significant divergences in structure and function of the microbial community between GW and GE. The α -diversity indices were compared between the free-living community and the epiphytic community using one-tailed paired *t*-tests, indicated by * when $p < 0.1$, ** when $p < 0.05$ or *** when $p < 0.01$. GW: free-living community, GE: epiphytic community.

表 5-1 龙须菜体表附着 (GE)及其周围水体微生物群落(GW)结构和功能的多样性指数比较 (单尾*t*-检验比较 α -多样性指数的差异, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; GW: 龙须菜周围水体微生物群落, GE: 龙须菜体表附生微生物群落)

	16S rRNA		ITS		GeoChip 5.0	
	GW	GE	GW	GE	GW	GE
Shannon Index	4.95 ±0.22	4.09 ±0.14***	3.58 ±0.34	2.78 ±0.65	11.12 ±0.03	11.01 ±0.04**
Simpson Index	48.73 ±12.26	29.52 ±3.42*	12.47 ±7.70	7.64 ±4.97	66479.68 ±1721.62	59727.74 ±2386.82**
Pielou evenness	0.72 ±0.02	0.70 ±0.02	0.60 ±0.05	0.51 ±0.11	1.00 ±0.00	0.99 ±0.00*
Simpson evenness	0.05 ±0.01	0.08 ±0.01***	0.03 ±0.02	0.03 ±0.02	0.93 ±0.00	0.97 ±0.00*

Table 5-2. Adonis test of microbial community structure and function between GW and GE. *P* < 0.05 was in bold.

表5-2 龙须菜体表附着及其周围水体微生物群落结构及功能差异的adonis检验

	R^2	p
16S	0.840	0.015
ITS	0.634	0.001
GeoChip	0.870	0.001

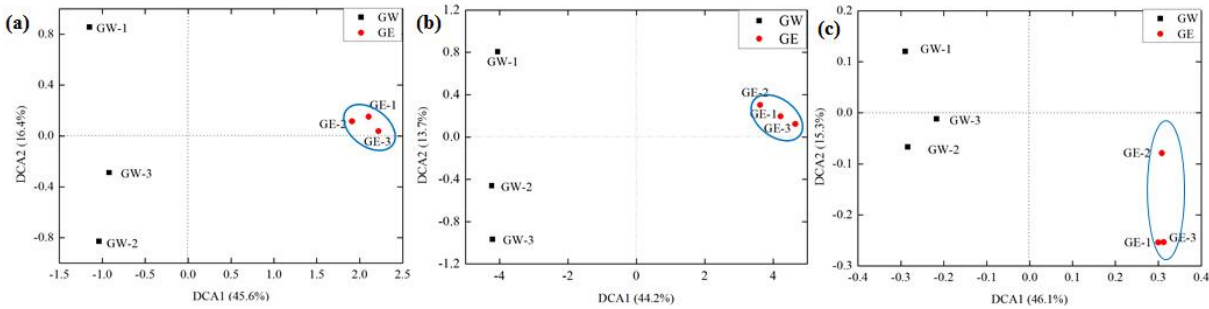


Fig. 5-1 DCA based on the Bray-Curtis distance showed the differences in microbial community (a), fungal community (b) and functional profiles (c) between GW and GE. GW-1, GW-2, GW-3: free living community; GE-1, GE-2, GE-3: epiphytic community.

图5-1龙须菜体表附着(GE)及其周围水体(GW)细菌群落结构(a)、真菌群落结构(b)及群落功能(c)差异的DCA分析 (GW-1, GW-2, GW-3: 龙须菜周围水体微生物群落; GE-1, GE-2, GE-3: 龙须菜体表附生微生物群落)

5.3.2 Seaweed and surface-associated microbial groups were enriched in the epiphytic microbial community

To discern the differences in microbial community composition between GE and GW, a comparison was made from high to low taxonomic levels. In GE, the prevailing phyla were *Bacteroidetes* (40.5%), *Alphaproteobacteria* (32.3%) (*p* < 0.05) and *Gammaproteobacteria* (22.1%) (*p* < 0.05), which held relatively higher diversity at the phylum level (Fig. 5-2a). At the family level, *Flavobacteriaceae* (38.8%) (*p* <

0.05), *Rhodobacteraceae* (29.5%) ($p < 0.01$), *Alteromonadaceae* (2.4%) and *Pseudoalteromonadaceae* (1.96%) were prevalent in GE along with higher abundance than in GW (Fig. 5-2b). The dominant genera in GE were *Formosa* (8.57%), *Methylophaga* (6.73%), *Meridianimaribacter* (4.84%), *Croceitalea* (3.86%), *Granulosicoccus* (3.73%) and unclassified-*Rhodobacteraceae* (19.9%), and many of them exhibited higher abundances than those in GW samples (Fig. 5-2c). In GW, the dominant phyla were *Bacteroidetes* (24.4%), *Deltaproteobacteria* (22.2%) and *Betaproteobacteria* (10.57%). The dominant family groups in GW belonged to *Flavobacteriaceae* (18.8%), *Rhodobacteraceae* (13.1%), *Unclassified-Gammaproteobacteria* (6.8%) and *Unclassified-Alphaproteobacteria* (6.5%). The genera, *Rhodopirellula* (7.49%), *Cellulophaga* (5.57%), *Bacillariophyta* (4.67%), *Loktanella* (2.96%) and *Blastopirellula* (2.44%) were prevalent in GW. Further analyses were performed at the OTU level to demonstrate the divergence of microbial community composition between GW and GE. Among 2166 OTUs detected (Fig. 5-3a), 294 (13.6%) were shared by GW and GE, and 1872 unique OTUs were only detected at GW (1596, 73.7%) or GE (276, 12.7%). For those unique OTUs in GE (Fig. 5-3c), most of them were isolated from the seaweed surface and exhibit algal polysaccharides-degrading ability, such as OTU-1726, OTU-13147 affiliated with *Maribacter*, OTU-2046 associated with *Phaeobacter* and OTU-1779 associated with *Krokinobacter*. The results at different taxonomic levels showed obvious differences in the microbial community composition between GW and GE.

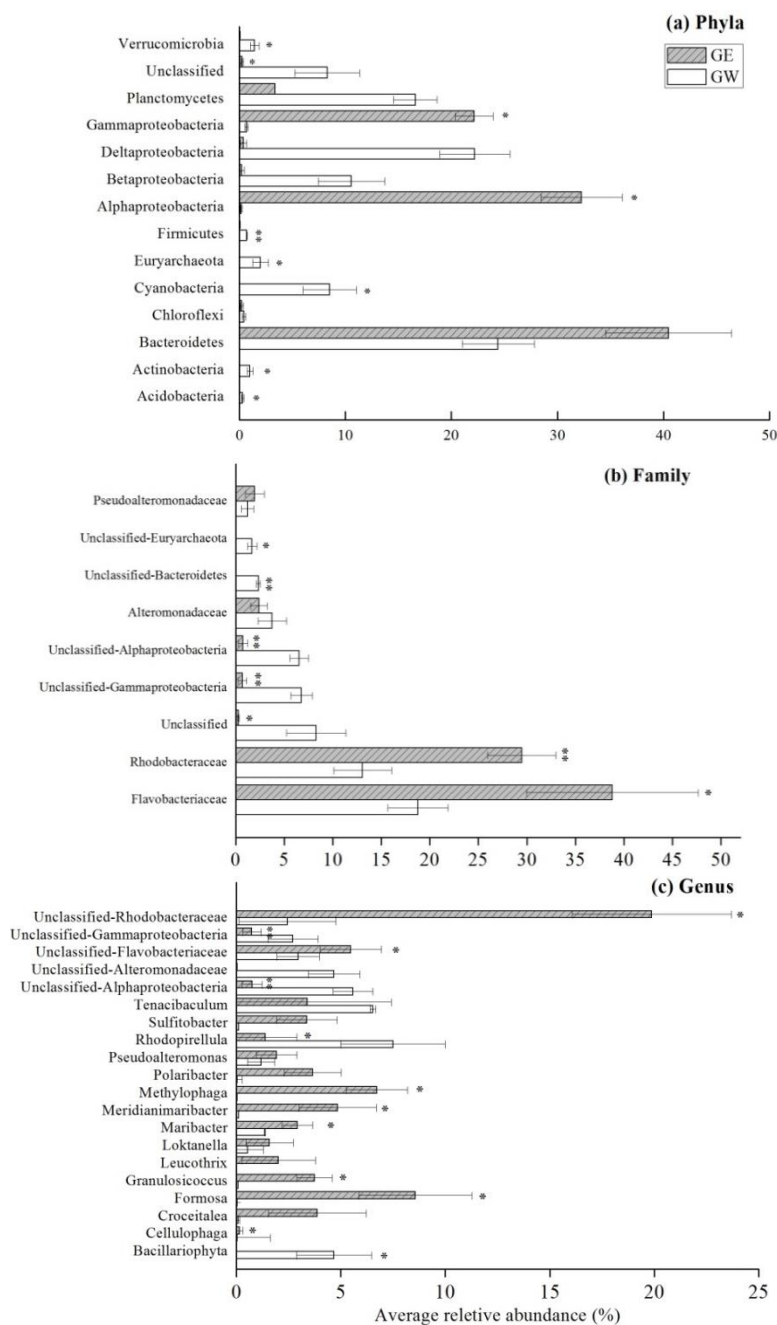


Fig. 5-2 Structure of the epiphytic and free-living microbial communities classified into (a) phyla (*Proteobacteria* at class level), (b) family and (c) genus taxonomic groups. Significances of each taxa were determined by two-tailed paired *t*-tests, indicated by * when $p < 0.05$ or ** when $p < 0.01$. GW: free living community; GE: epiphytic community.

图5-2 龙须菜体表附着细菌群落(GE)及其周围水体细菌群落(GW)在(a)门、(b)科及(c)属水平组成的比较 (双尾成对 t -检验比较每个细菌类群的相对丰度差异, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; GW: 龙须菜周围水体微生物群落, GE: 龙须菜体表附生微生物群落)

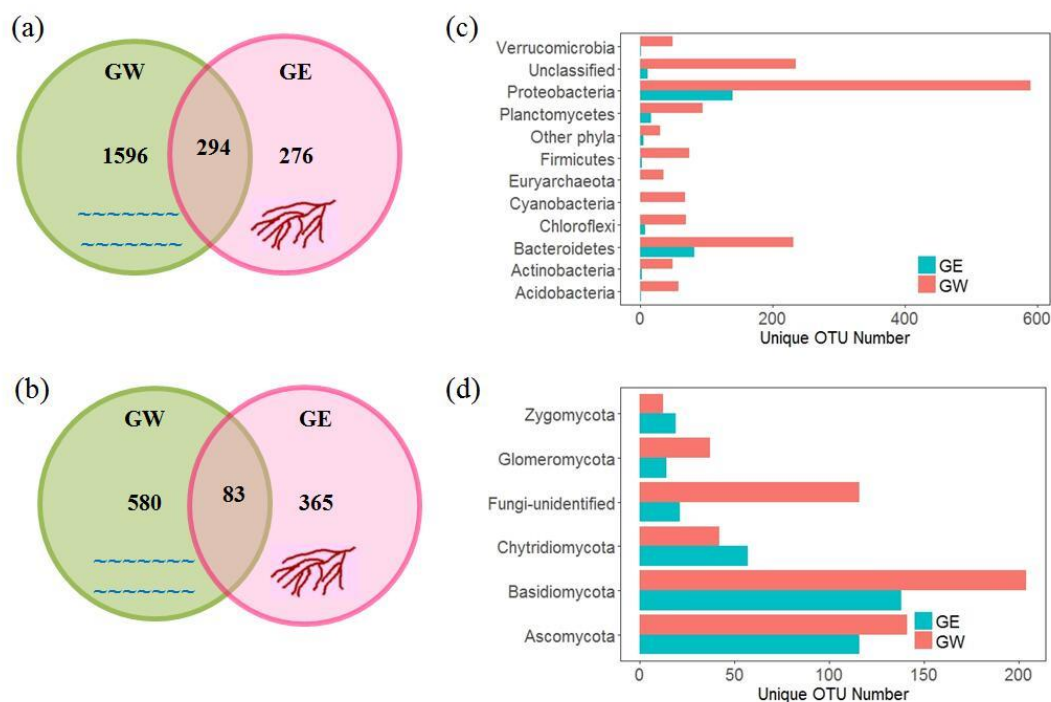


Fig. 5-3 Shared and unique OTU numbers of epiphytic and free-living microbial (a) and fungal (b) community. The phyla those unique microbial (c) and fungal (d) OTUs belonging to. GW: free-living community; GE: epiphytic community.

图 5-3 龙须菜体表附着 (GE)及其周围水体 (GW) (a)细菌群落、(b)真菌群落中共有及独有的 OTU 数目；(c)细菌群落、(d)真菌群落独有 OTU 所属的门

A comparison of the fungal community between GW and GE was also made at various taxonomic levels. At the phylum level (Fig. 5-4a), the dominant phyla of GE samples were Ascomycota (13.92%), Basidiomycota (18.37%), Chytridiomycota (23.48%) and Zygomycota (41.88%), while the most dominant phylum in GW samples was Fungi-unidentified, with a higher average relative percentage (55.78%) compared to GE ($p < 0.01$). At the family level (Fig. 5-4b), higher abundances of Mucoraceae (16.2%) and Tremellales-unidentified (1.8%) were observed in GE samples compared to GW ($p < 0.01$). The dominant families in GW samples were Ophiocordycipitaceae (2.21%), Pezizaceae (2.47%) and Russulaceae (1.87%) with higher average percentage than that of GE samples ($p < 0.05$).

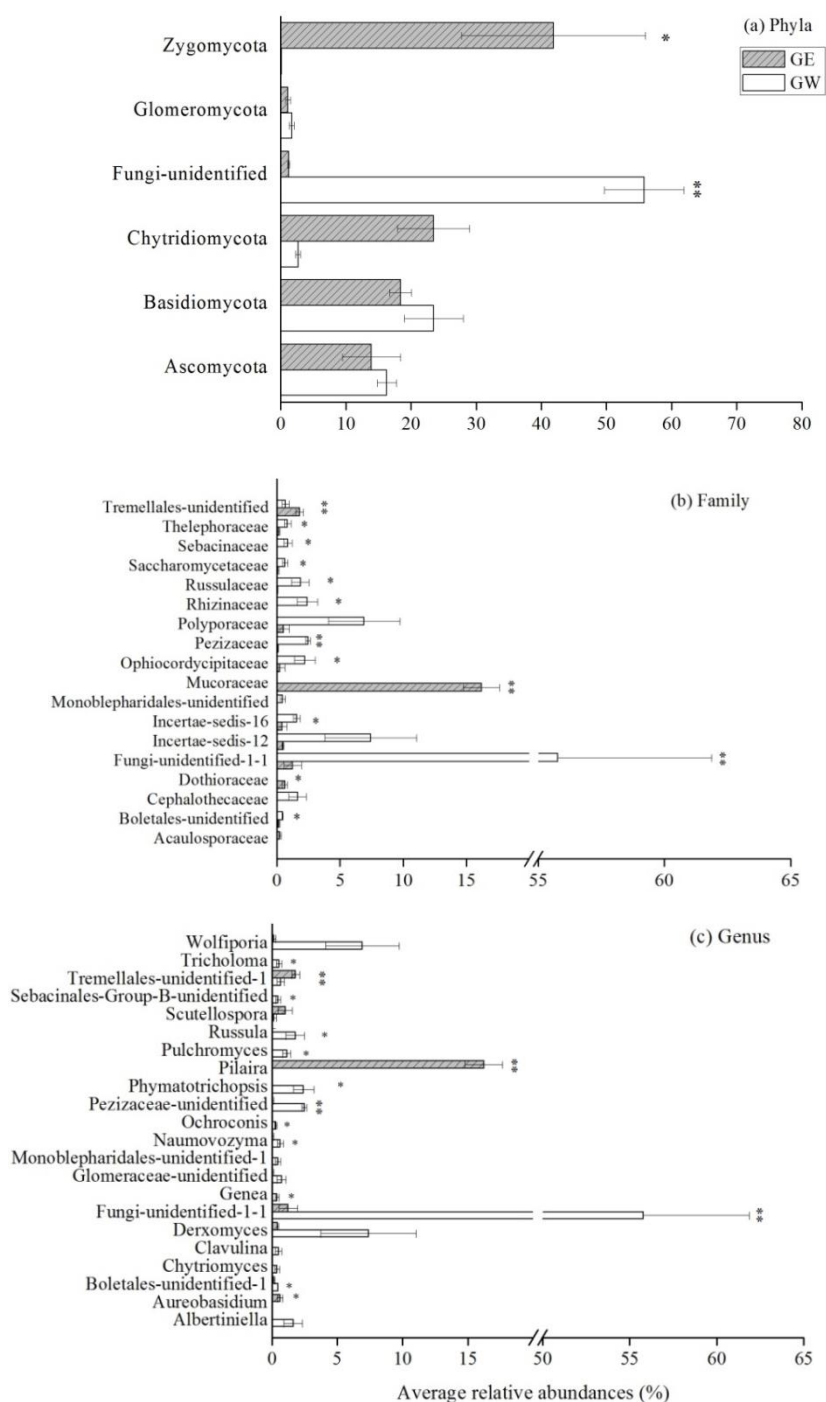


Fig. 5-4 Fungal community composition of epiphytic *G. lemaneiformis* and planktonic seawater classified into (a) phyla, (b) family and (c) genus taxonomic levels. Significances of each taxon were determined by two-tailed paired t-tests, indicated by * when $p < 0.05$ or ** when $p < 0.01$. GW: free living community; GE: epiphytic community.

图 5-4 龙须菜体表附着及其周围水体真菌群落(GW)在(a)门、(b)科及(c)属水平的组成比较 (双尾成对t-检验比较每个细菌类群的相对丰度差异, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; GW: 龙须菜周围水体微生物群落, GE: 龙须菜体表附生微生物群落)

For the prevailing genera in GE samples (Fig. 5-4c), *Pilaira* (16.2%) and *Tremellales-undefined-1* (1.8%) showed higher abundances in GE compared to GW ($p < 0.01$). In GW samples, the dominant genera were *Phymatotrichopsis* (2.42%), *Pulchromyces* (1.13%) and *Russula* (1.79%) with higher abundances compared to GE ($p < 0.05$). For 1028 OTUs generated from all of the six samples (Fig. 5-3 b), only 83 (8.1%) OTUs were shared by GE and GW samples, 365 (35.5%) and 580 (56.4%) OTUs were unique to GE and GW (Fig. 5-3d), respectively.

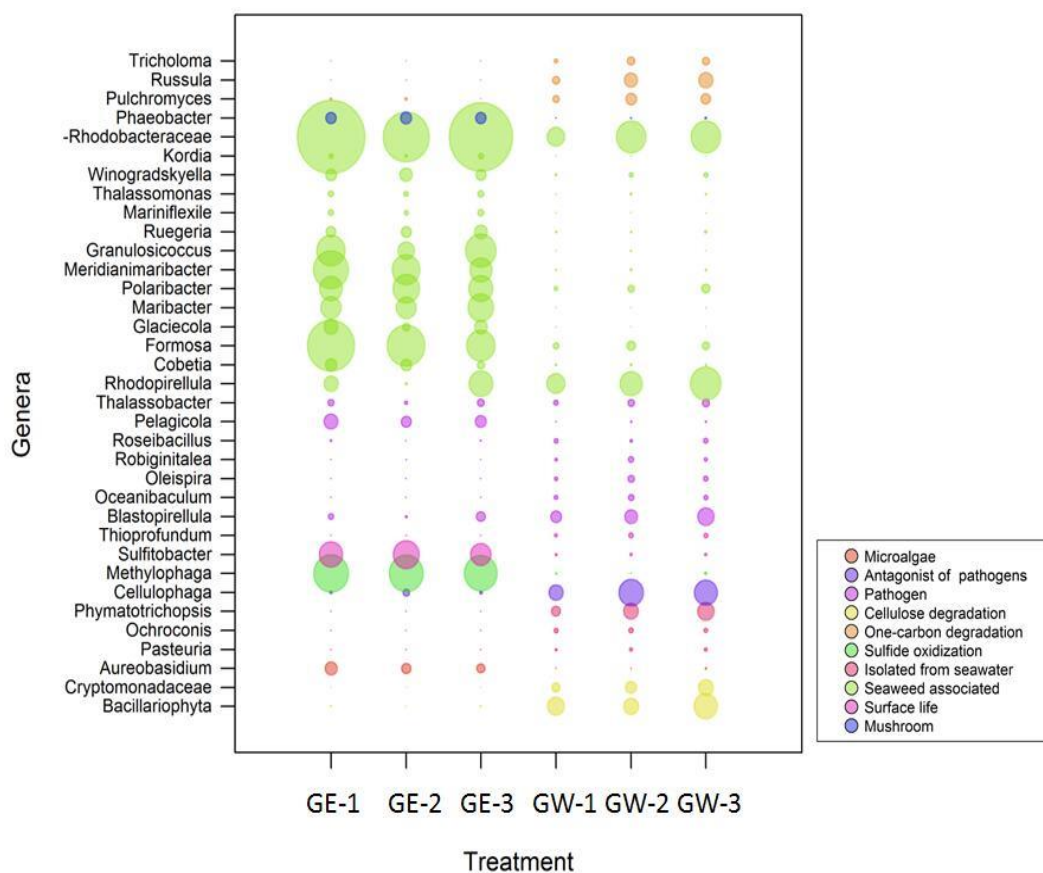


Fig. 5-5 Bubble plot of genera grouped by lifestyle or ecological functions in GE and GW samples. The size of the bubble represents the relative abundance (%) of each genus. All genera have significant divergences between GE and GW determined by two-tailed paired t-tests ($p < 0.05$).

图 5-5 龙须菜体表及其周围水体微生物群落在属水平按生活方式或生态功能群的比较 (气泡大小代表各属的相对丰度(%), 并由双尾成对 t -检验对龙须菜栽培区和对照区所有属相对丰度进行差异显著性分析)

In addition, genera of microbial communities were grouped by lifestyle or ecological functions (Fig.5-5), which revealed divergent structural and functional profiles between GE and GW ($p < 0.05$). The results indicated that most seaweed-associated and surface life genera exhibited higher abundances in the epiphytic community. Free living communities contained more genera isolated from seawater compared with the epiphytes. Also, more microalgae and pathogens were detected in GW.

5.3.3 Microbial community functional protential of *G. lemaneiformis* epiphytes

Venn diagram and donut plots based on probe numbers implied significant different functional profiles between GE and GW samples (Fig. 5-6). Analyses and comparisons of microbial community functional profiles in various gene categories were performed to determine whether a functional core exists in the epiphytic microbial community of *G. lemaneiformis*.

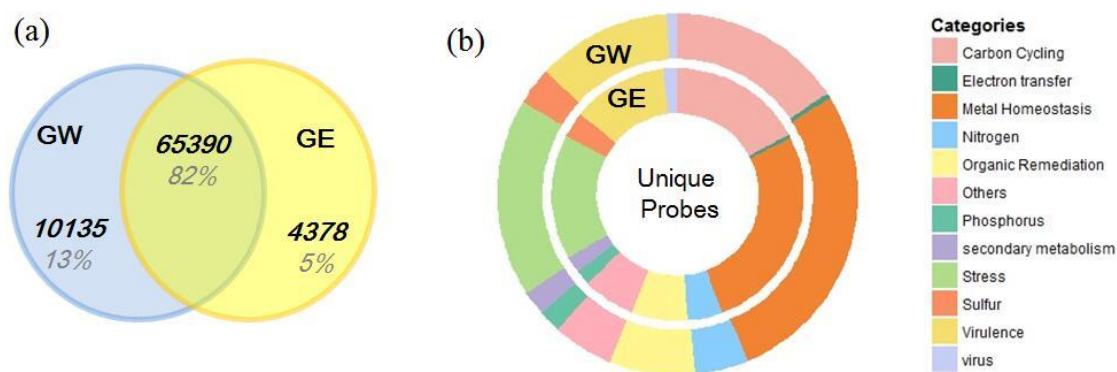


Fig. 5-6 Unique and shared probe number detected in the epiphytes and free living samples **(a)** and the gene categories those unique probes belonged to **(b)**.

图 5-6 龙须菜体表及其周围水体微生物群落共有和独有的基因探针数目(a)及独有探针的基因类群(b)

5.3.3.1 Microbial responses to surface-associated life and seaweed environments

Higher abundances of genes involved in seaweed-associated functions were observed in the epiphytic microbial community. Genes involved in adherence, colonization and biofilm formation were detected in both GE and GW samples.

However, the *amsG* gene showed higher abundances in GE samples, which is known to promote biofilm formation. Also, the genes, *aae*, *aap* and *oap*, which encode proteins functionally assigned to adhesion, virulence and opacity of surface were only observed in the epiphytes.

A substantial number of genes involved in microbial responses to the seaweed environment were detected, and many of them showed significantly higher abundances ($P < 0.05$) in the epiphytic microbial community. Genes encoding polysaccharide lyase protein (*pec-Cdeg*) were more abundant in GE samples. Presumably the epiphytic microbes use polysaccharide secreted by *G. lemaneiformis* to support their survival on the surface of the host seaweed. The signal intensities for many genes involved in heat shock (*dnaK* and *grpE*), drought tolerance (*tre-arc*) and osmotic stress (*mtrA*) were higher in the epiphytic community, which may be related to the desiccation of *G. lemaneiformis* thallus at the low tide. Also, higher signal intensities of genes involved in heavy metal detoxification were observed in GE samples than those in GW samples. Genes encoding reductase, respiration and transferase for arsenic (*arsC*, *arrA* and *arsm*), mercury-resistance protein (*metC*) and chromate reductase (*chrR*) were more abundant in the epiphytic community (Fig. 5-7). Genes associated with defense of *G. lemaneiformis* were also abundant in GE samples, including those involved in oxidative stress (*ahpC*, *oxyR*, *soda* and *soxR-OR-marC*), ABC transporters (ABC-multidrug-fungi) and antiphagocytosis (*algE* and *cap8B*). Gene *cqsS* was only observed in GE samples, which is involved in quorum sensing and may be important for the interaction between epiphytic microbes and the seaweed. In addition, many genes encoding regulation of transcription (*ctsR*, *fnr*, *oxyR*, *pspF*, *pspB* and *baeR*) and histidine kinase (*desK* and *ntrB*) showed higher abundance in the epiphytic community (Fig. 5-7b).

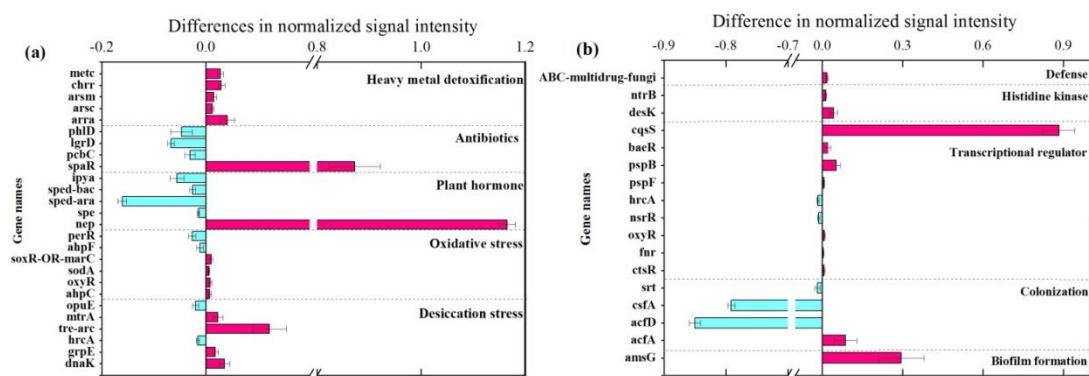


Fig. 5-7 Differences in normalized signal intensities of genes involved in seaweed surface-associated life. Pink and blue bars represent the significantly increased gene abundances and decreased gene abundances ($p < 0.05$), respectively, in epiphytic community.

图5-7 龙须菜体表附生及其周围水体微生物群落中与海藻或表面生活相关的基因丰度差异 (粉色和蓝色柱子分别代表龙须菜体表附生微生物群落中丰度显著增加和显著减少的基因)

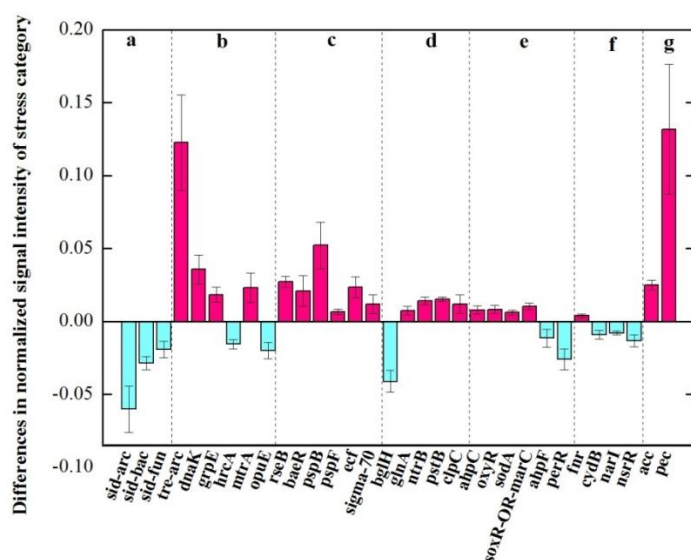


Fig. 5-8 Normalized signal intensities of genes in stress category involved in anti-pathogen (a), desiccation (drought tolerance, heat shock and osmotic stress) (b), envelope stress (c), nutrients limitation (d), oxidative stress (e), oxygen limitation (f), and stress other (g).

图5-8 胁迫基因类群中参与(a)抗致病菌、(b)干出胁迫、(c)应力胁迫、(d)营养限制、(e)氧化压力、(f)氧气限制及(g)其他胁迫的基因丰度

Many stress-related genes including those involved in heat shock, cold shock, osmotic stress, envelope stress, nutrient limitation and oxidative stress exhibited higher abundances in the epiphytes, suggesting that such microbes up-regulate their functions in response to seaweed environmental stresses (Fig. 5-8). Among the 90

genes in viruses, many were significantly fewer in the epiphytic community. Such genes are associated with infection, lysis, replication, virus structure and toxicity, and may be related to the epiphytic community protection of the host seaweed (Fig. 5-9). The results suggested that virus reproduction could be inhibited in the epiphytic community to protect the host seaweed.

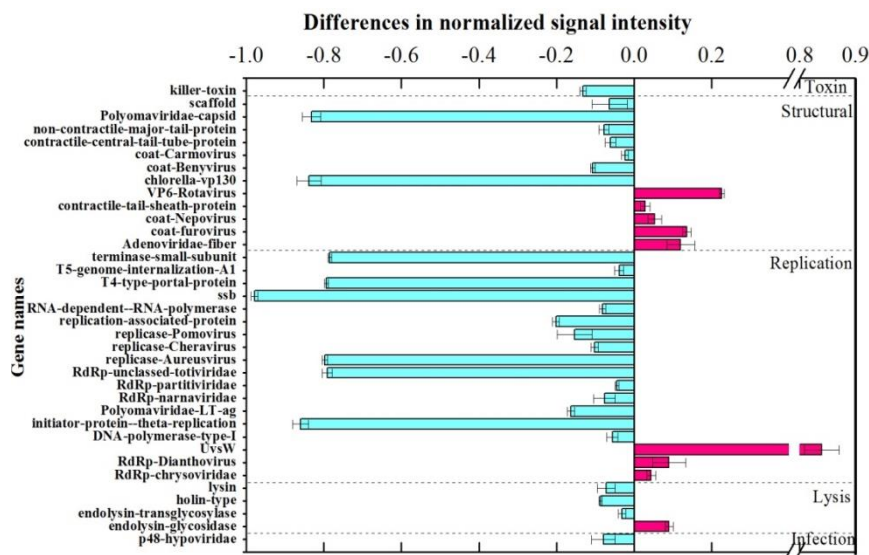


Fig. 5-9 Normalized signal intensities of genes in virus category involved in toxin, structural, replication, lysis and infection.

图5-9 病毒类群中参与毒素、结构、复制、裂解及感染的基因丰度

5.3.3.2 Biogeochemical cycling and other functional genes

All gene families involved in biogeochemical cycling were present in all samples. 7,024 probes involved in C degradation had positive signals in GE samples, such as those involved in the degradation of starch, hemicellulose, cellulose, chitin, pectin and lignin. But many of these genes showed lower abundance in GE samples than in GW samples, except for *axe* and pectate-lyase-Oomycetes (Fig. 5-10). Among N cycling genes, only those involved in N_2O reductase (*nosZ*) were significantly higher in GE samples ($p < 0.05$). Denitrification (*nirS*) and ammonification (*ureC*) genes showed significantly lower ($p < 0.05$) signal intensities in GE than in GW. The epiphytic microbial communities might stimulate N cycling by general increases in N_2O reduction and decreases in nitrite reduction and ammonification (Fig. 5-11).

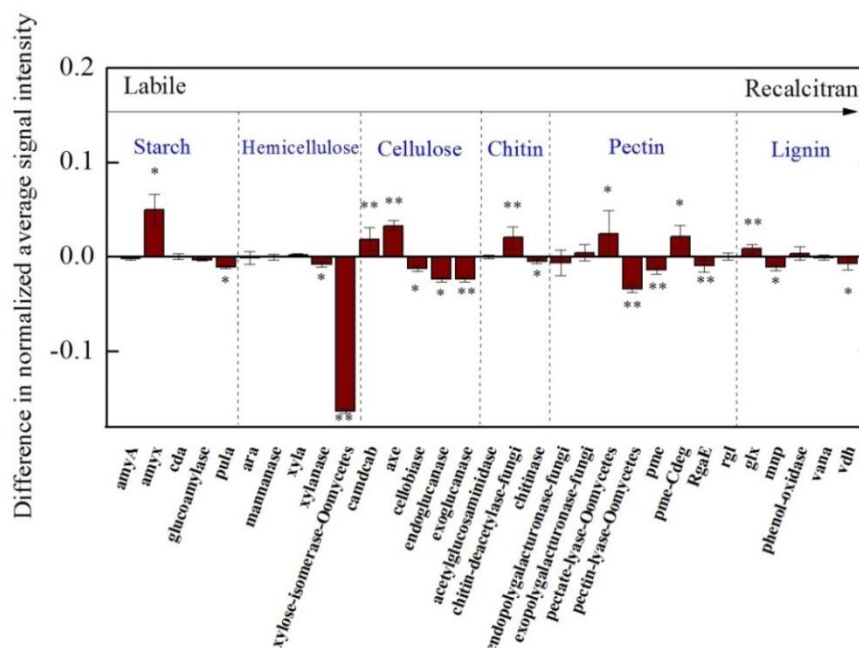


Fig. 5-10 The changes of genes involved in C degradation based on GeoChip data. The targeted genes were arranged in order from labile C to recalcitrant C. Significance was analyzed by ANOVA and indicated by * and ** when $p < 0.05$ or $p < 0.01$, respectively.

图5-10 龙须菜体表附生微生物群落参与碳循环的基因丰度变化 (目标基因从活性碳到不易降解碳排列.龙须菜体表附生及周围水体微生物群落基因丰度显著性差异由ANOVA分析, * $p < 0.05$, ** $p < 0.01$)

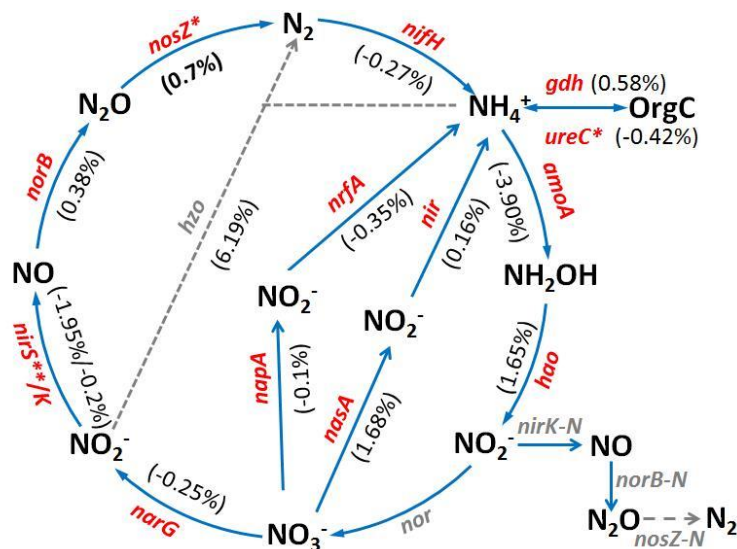


Fig. 5-11 The changes of genes involved in N cycling based on GeoChip data. Significance was analyzed by ANOVA and indicated by * and ** when $p < 0.05$ or $p < 0.01$, respectively.

图5-11 龙须菜体表附生微生物群落参与氮循环的基因变化情况 (ANOVA分析龙须菜体表附生及周围水体微生物群落基因丰度显著性差异, * $p < 0.05$, ** $p < 0.01$)

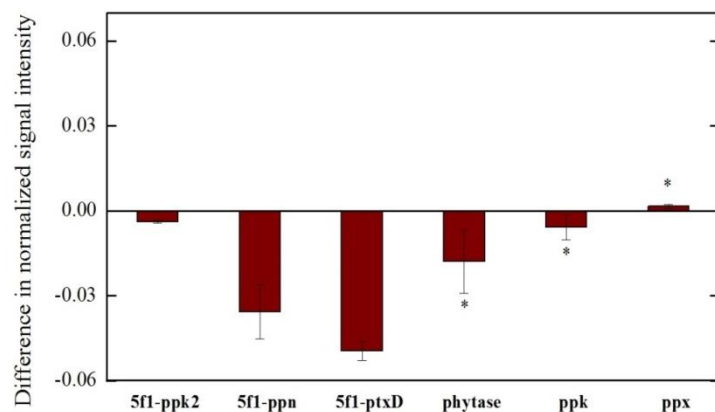


Fig. 5-12 The changes of genes involved in P cycling based on GeoChip data. Significance was analyzed by ANOVA and indicated by * and ** when $p < 0.05$ or $p < 0.01$, respectively.

图5-12 龙须菜体表附生微生物群落参与磷循环的基因变化情况 (龙须菜体表附生及周围水体微生物群落基因丰度差异的ANOVA分析: * $p < 0.05$, ** $p < 0.01$)

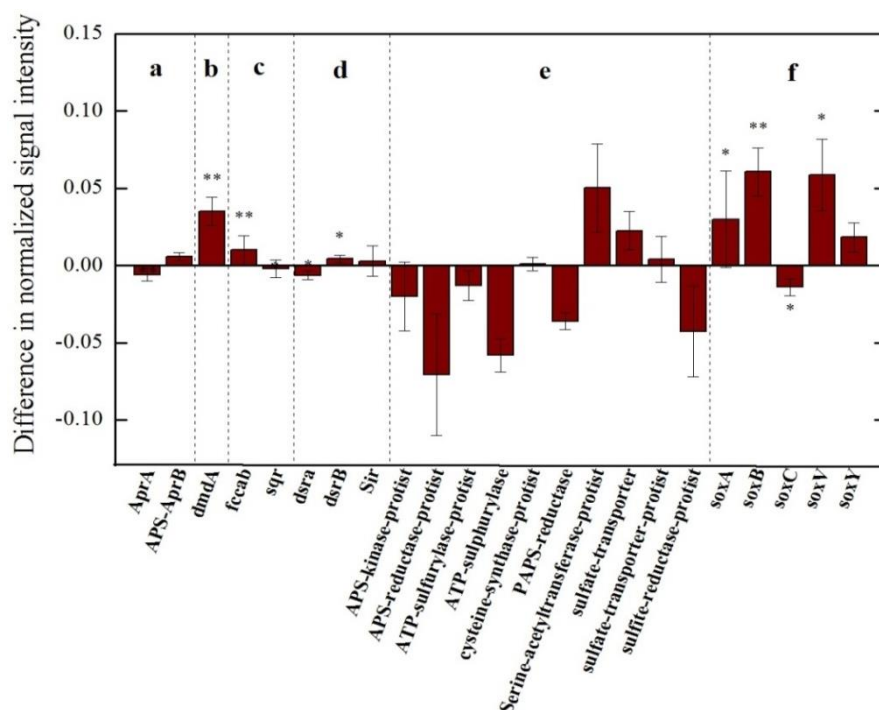


Fig. 5-13 The changes of genes involved in S cycling based on GeoChip data. The targeted genes were arranged in the following subcategories: adenylylsulfate reductase (a), DMSP degradation (b), sulfide oxidation (c), sulfite reduction (d), sulfur assimilation (e), and sulfur oxidation (f). Significance was analyzed by ANOVA and indicated by * and ** when $p < 0.05$ or $p < 0.01$, respectively.

图5-13 龙须菜体表附生微生物群落参与硫循环的基因变化情况: 腺苷酰硫酸还原酶(a), 二甲基硫丙酸降解(b), 硫化物氧化(c), 亚硫酸盐还原(d), 硫同化(e)及硫氧化(f) (龙须菜体表附生及周围水体微生物群落基因丰度差异的ANOVA分析, * $p < 0.05$, ** $p < 0.01$)

Significant differences ($P < 0.05$) in signal intensities related to P cycling were observed with higher abundance of *ppx* and lower abundance of phytase and *ppk* in GE samples (Fig. 5-12). Inorganic and organic P utilization may be stimulated according to the differences to the source of phosphorus between GW and GE. Genes involved in S cycling, DMSP degradation, sulfide oxidation and sulfur oxidation were relatively abundant in GE samples with higher signal intensities in *dmdA*, *fccAB*, *soxA*, *soxB* and *soxV*. 60% of the significantly changed genes related to S cycling were higher in GE samples, especially those involved in sulfur/sulfide oxidation and sulfite reduction (Fig. 5-13). Many secondary metabolism genes were less abundant in GE samples than in GW. Those genes involved in antioxidant enzymes and plant hormones, such as *spaR* and *nep*, had significantly higher signal intensities in GE samples (Fig. 5-14). The production of antioxidant enzymes and plant hormones were enhanced in the epiphytic communities. A total of 5,440 probes of organic contaminant remediation showed positive signals in GE samples. Many genes had significantly higher signal intensities in GE including those genes involved in reduction of chlorinated solvents (*dehH* and *rd*), herbicides related compound (*pcpB*, *tomA* and *trzN*), other hydrocarbons (*alkB* and *xamO*) and aromatics (*bco*, *cdo*, *hcaacd*, *hdno*, *nfsa-2*, *phta*, *tmoabe* and *tutfdg*) (Fig. 5-15). The results indicate that the remediation potential of organic contaminants may be higher in the epiphytic microbial community of *Gracilaria*.

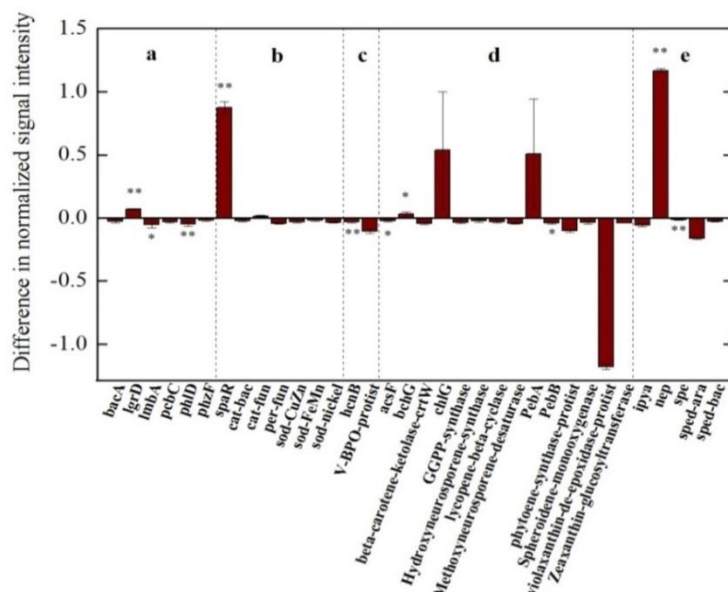


Fig. 5-14 Normalized signal intensities of genes associated with secondary metabolism. Genes were involved in antibiotic (a), antioxidant enzyme (b), other (c), pigments (d), and plant hormone (e).

图5-14 次级代谢基因类群中参与(a)抗生素、(b)抗氧化酶、(c)其他、(d)色素及(e)植物激素的基因丰度变化

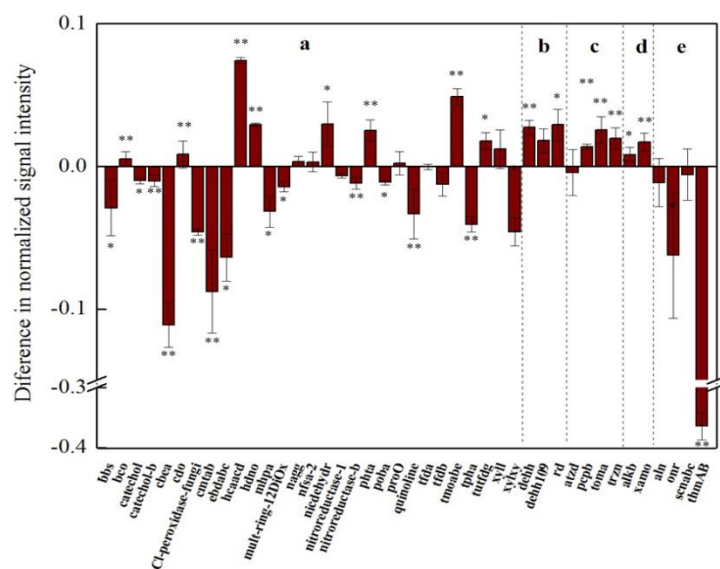


Fig. 5-15 Normalized signal intensities of genes in remediation of organic contaminants. Genes were involved in remediation of aromatics (a), chlorinated solvents (b), herbicides related compound (c), other hydrocarbons (d), and others (e).

图5-15 龙须菜体表附生微生物群落参与有机污染物修复的基因丰度变化：芳香族(a)，氯化溶剂(b)，除草剂相关化合物(c)，其他碳水化合物(d)及其他(e)

5.4 Discussion

Seaweed surfaces provide a favorable substrate and habitat for microbes to live on since they secrete multifarious organics that serve as nutrients for microbes (Beleneva and Zhukova, 2006; Wen et al., 2006; Wiese et al., 2009). The host seaweed and associated microbes form a stable mutualism as a defense against further colonization (Steinberg et al., 2002). Furthermore, those epiphytic microbes could also produce various compounds that promote growth and photosynthesis, as well as act as antifouling and antifungal agents (Singh and Reddy, 2014). Given the critical interaction between host seaweed and their epiphytes, knowing the composition and function of the community provides us with a clearer understanding of this intricate interaction in coastal ecosystems.

However, the epiphytic microbial community on the surface of *G. lemaneiformis* is not well studied. Previous studies reported the diversity of epiphytic bacteria using DGGE (Tujula et al., 2010; Zuccaro et al., 2008), 16S rRNA clone library (Lachnit et al., 2011; Staufenberg et al., 2008) and T-RFLP (Campbell et al., 2015; Hengst et al., 2010; Nylund et al., 2010), which only partially revealed the community structure on the most dominant groups. In this study, we applied 16S rRNA gene amplicon sequencing, ITS sequencing and GeoChip 5.0 to understand the epiphytes of *G. lemaneiformis* and the free-living community in the same cultivation ecosystem. Our results showed that the composition and function of the epiphytic community significantly shifted with lower diversity and higher abundances of seaweed-associated groups. The results generally supported our hypotheses.

5.4.1 Epiphytic and free-living microbial communities were phylogenetically distinct

DCA analysis and dissimilarity tests showed that the overall compositions of epiphytic and free-living microbial communities were strikingly distinct. GE samples had lower diversity and higher evenness compared to GW samples, implying *G. lemaneiformis* selects microbes on the seaweed surface leading to a lower α -diversity in the epiphytic community, which supports our first hypothesis.

A comparative analysis from high to low taxonomic levels shows that the most dominant microbial phyla of epiphytes were Bacteroidetes, Alphaproteobacteria and Gammaproteobacteria, the same as previous studies (Burke et al., 2011b). While, the free-living community at the phylum level was dominated by Bacteroidetes, Deltaproteobacteria, Planctomycetes and Betaproteobacteria, as reported previously (Rusch et al., 2007). The dominant families of epiphytes were Rhodobacteraceae and Flavobacteriaceae in agreement with other red alga research (Longford et al., 2007). Alga-associated genera were in higher abundance in the epiphytic community compared with free-living samples. *Croceitalea* (Lee et al., 2008), *Granulosicoccus* (Kurilenko et al., 2010; Park et al., 2014), *Leucothrix* (Brock, 2006b), *Flavobacterium* (Bernardet and Bowman, 2006), *Polaribacter* (Fukui et al., 2013) and *Sulfitobacter* (Ivanova et al., 2004) were isolated from the surface or rhizosphere of seaweeds. The genera *Aquimarina* (Chen et al., 2012; Lin et al., 2012) and *Formosa* (Mann et al., 2013), which have agarolytic potential to degrade many algal polysaccharides from a range of seaweeds. In addition, *Paracoccus* (Kelly et al., 2006) and *Arenicella* (Rua and Thompson, 2014) were first cultured on marine agar. Most of these genera are specialized for an alga-associated lifestyle, due to the special habitats of the seaweed surface including rhizosphere. Only 13.6% of OTUs were shared by both GE and GW. A large number of unique OTUs in the GE samples were able to degrade algal-polysaccharides, or have surface-associated life styles. For example OTU-1726 and OTU-13147 were associated with *Maribacter* (Nedashkovskaya et al., 2007), OTU-2866 with *Persicivirga* (Barbeyron et al., 2011); more than 10 OTUs were affiliated with *Lewinella* (Lee, 2007) and OTU-2046 was linked with *Phaeobacter* (Thole et al., 2012). Also, these unique OTUs were able to utilize organic compounds, especially those with linkage to *Methylophaga* (OTU-195, OTU-1670, OTU-1068) (Neufeld et al., 2007) and *Colwellia* (OTU-77, OTU-1259, OTU-1895) (Huston et al., 2004).

As for the fungal community, higher abundances of the phylum Zygomycota, which has parasitic or symbiotic relationships with plants (Read et al., 2000), was in the epiphytic community. *Pilaira anomala*, capable of utilizing amino acids and

thiamine (Fletcher, 1971) was only found in the epiphytic community. Algicolous fungi such as *Aureobasidium*, *Penicillium*, *Rhodotorula* (Godinho et al., 2013; Loque et al., 2009; Suryanarayanan, 2012), *Dothioraceae* (Zuccaro et al., 2003) were also abundant in the epiphytic community. Only 8.1% of OTUs were shared by GE and GW samples. Out of the 365 unique OTUs in the GE samples, seven of them were affiliated with *Agaricomycetes*, which can be saprotrophs, pathogens, and mutualists (Hibbett et al., 2014). Nine OTUs were associated with Ascomycota, the largest phylum of fungi, and its species occur in numerous ecological niches and can breakdown organic substrates and are mutualists, parasites, and pathogens of animals, plants, and other fungi (Schoch et al., 2009). 25 OTUs were affiliated with Chytridiomycota, which is one of the parasitic fungi living on the surface of phytoplankton and seaweed (Ibelings et al., 2004). Five OTUs affiliated with *Cunninghamella*, which is found in soil and plant materials, especially in the Mediterranean and subtropical zones (Asha and Vidyavathi, 2009), showed a high relative abundance. Some unique OTUs were associated with selective genera, such as *Derxomyces* (OTU 341, OTU 505, OTU 1063 and OTU 1579) (Liu et al., 2012b), *Naumovozya* (OTU 352, OTU 381, OTU 1402 and OTU 1593) (Liu et al., 2012a), and *Perenniporia* (OTU 889 and OTU 1183) (Zhao et al., 2012). Also, some unique OTUs were correlated with the production of mycotoxins and medicines, among which more than 20 were affiliated with *Penicillium* (Rundberget et al., 2004); *Wolfiporia* were linked with OTU 502, OTU 620 and OTU 991 (A. Machuca, 2001; Wang et al., 2013).

Since the epiphytic microbial community was found to be distinct from the free-living microbial communities, it represented a divergent and specific seaweed-associated microbial community living on the surface of *G. lemaneiformis*. Differences in community composition were selected by the distinct environments of the seaweed surface and the surrounding water. Seaweed surfaces have more available organic C (Penhale and Smith, 1977; Pregnall, 1983) and other nutrients (Moebus and Johnson, 1974), and provide a protective habitat for epiphytes. Therefore the interaction between the host seaweed and the microbes determines the nature of the

epiphytic community (Egan et al., 2008). The stresses of low nutrients, high UV radiation and predation in water column act as selective pressure on the composition of free-living communities in water column. Thus, the discrepancy in the habitats defines a divergent community of epiphytes (Lam et al., 2008) and plays an important ecological role in *G. lemaneiformis* cultivation ecosystems. The results support our second hypothesis.

5.4.2 Epiphytic and free-living microbial communities have divergent functional profiles

The contrast between the epiphytic and free living microbes poses an essential question as to whether changes in the community structure affect microbial functions and ecosystem services, such as in biogeochemical cycling and contaminant remediation. To address this question, we used a functional gene array GeoChip 5.0 to analyze the functional profile of both epiphytic and free-living microbial communities in *G. lemaneiformis* cultivation ecosystem. The data demonstrated different functional profiles by DCA and dissimilarity test. That is, genes involved in stress, assimilatory N reduction, sulfide and sulfur oxidation, chlorinated solvents, herbicides-related compounds and seaweed-associated functions increased in GE samples. Whereas, most genes involved in C degradation, denitrification, nitrification, nitrogen fixation, P utilization, aromatics degradation, secondary metabolism and viruses were lower in the epiphytic community. The results largely support our third hypothesis.

Seaweed surface is a habitat distinct from the water column. To adapt and sustain life on the seaweed surface, epiphytes may adjust their community composition and function. Biofilm formation is an essential process whereby epiphytic microbes live on the seaweed surface successfully to establish a community (Egan et al., 2013). Genes involved in adhesion, virulence and biofilm formation exhibited higher abundances in the epiphytes. They are essential in the establishment, maintenance and formation of the biofilm matrix on the surface of the host seaweed. For instance, the *amsG* gene functions in exopolysaccharide biosynthesis and was more abundant in GE samples and could, therefore, promote biofilm formation (Geider, 1995;

Whitchurch et al., 2002). Many genes involved in adhesion, virulence and opacity, such as *aae*, *aap* and *oap* (Heilmann et al., 2003; Monteiro et al., 2009; S. J. Vandecasteele, 2003), were only detected in the epiphytic community. These overrepresented genes are expected to play important roles in the symbiosis between *G. lemaneiformis* and the epiphytes.

Most overexpressed genes in epiphytes were related to their adaptation to the *G. lemaneiformis* surface. A high proportion of genes for stress, transcription regulation and organic contaminant degradation are a response and adaptation to the seaweed surface environment. For example, higher abundances of genes involved in heat shock (*dnaK* and *grpE*), desiccation tolerance (*tre-arc*) and osmotic stress (*mtrA*) may be related to the desiccation of the *G. lemaneiformis* thallus when exposed to air at low tide (Burke et al., 2011a; David Straus, 1990; Moker et al., 2004). The overrepresentation of genes associated with heavy metal detoxification (such as *arsC*, *arrA*, *arsM*, *metC*, *chrR*) might help the host seaweed maintain metal homeostasis (Dugas, 2003; Malasarn et al., 2004; Newman et al., 2001; Perez-Jimenez et al., 2005; Sun et al., 2004). Also, genes involved in defense, oxidative stress and ABC transporters in our study were more abundant and could help *G. lemaneiformis* defend itself against other bacteria and biofouling organisms (Veen and Konings, 1998; Wenzel and Muller, 2005). Gene *cqsS* involved in quorum sensing, which is crucial in the interaction between epiphytes and the host seaweed (Henke and Bassler, 2004; Ng et al., 2011), was only detected in the epiphytic community. In addition, the inhibition of genes in viruses protects the host seaweed from pathogens. Finally, epiphytes secrete plant hormones such as IAA, adenine and kinetin, as growth factors and antibiotics, to support normal morphological development of the host and defend it against the secondary colonization by other organisms. For example, a gene encoding an ethylene-inducing protein (*nep*) and subtilin biosynthesis regulatory protein (*spaR*), which may regulate the growth and protect the host seaweed from colonization and biofouling of other organisms (De Castro et al., 2008), showed higher abundances ($P<0.01$) in the GE samples.

Seaweed can secrete various organic compounds that serve as nutrients for

epiphytes in contrast to the free-living community which obtain nutrients from sea water. Although many genes involved in C degradation decreased in epiphytes, some of them still increased due to the special habitat of the seaweed surface. Since cellulose and pectin are important constituents of the cell wall of *G. lemaneiformis* (Cheney and MarCheney, 1986), the epiphytic community may inhibit the genes involved in cellulose and pectin degradation to protect the seaweed surface from the secondary colonization by other organisms. Also, seaweed-detritus serves as a nutrient source and may alter biological processes such as C cycling (Chung et al., 2010; Duarte et al., 2013). In addition, high O₂ concentration can inhibit many anaerobic microbial processes, including N₂ fixation, denitrification, sulfate reduction, methanogenesis and metal reduction (Bernet et al., 2001). The oxygen concentration of *G. lemaneiformis* surface is higher than the surrounding water due to photosynthesis by the seaweed. Thus, the relative abundance of genes involved in denitrification, nitrogen fixation (*nifH*) and P utilization were lower in the epiphytic community. Since NH₄⁺ is the preferential nutrients taken up by *G. lemaneiformis* (Li and Lin, 2005), increases in gene abundances of *nasA*, *nir* and *gdh* in the epiphytic community could enhance assimilatory N reduction and ammonification to utilize NO₃⁻ and organic compounds as NH₄⁺ sources (Simon, 2002; Turpin, 1989). Also, a decrease of *amoA* could inhibit the nitrification of NH₄⁺ (Li and Lin, 2005; Yang et al., 2015), resulting in NH₄⁺ accumulation. The above results demonstrated that genes of epiphytic communities involved in important ecological processes respond and adapt to life on the surface of *G. lemaneiformis*.

5.4.3 Linking the diversity, structure and function in the epiphytic communities

Comprehensive analyses of 16S rRNA, ITS gene amplicon sequencing and GeoChip data were performed to link the diversity, structure and function of the epiphytic communities. Many enriched genera in the epiphytic community had close affiliation with enriched genes in various categories. Most enriched genera were seaweed surface-associated groups, such as *Formosa*, *Maribacter*, *Polaribacter*, *Glaciecola*, *Phaeobacter*, which were isolated from seaweed or have algal polysaccharide abilities.

Those populations can be linked to enriched genes associated with seaweed surface including biofilm formation, drought tolerance, osmotic stress, and nutrient limitation. Other enriched genera also had close associations with enriched genes. *Sulfitobacter* and *Thiopfundum* exhibited higher relative abundances in the epiphytic community capable of sulfite/sulfur oxidation abilities and were linked with enriched genes *fccAB*, *soxA*, *soxB* and *soxV*. *Cellulophaga* showed lower abundances in the epiphytic community which had an association with lower intensity of *cellobiase*, *endoglucanase* and *exoglucanase* genes due to protection of cellulose in seaweed cytoderm. Unique OTUs only detected in the free living community were affiliated with *Pasteuria*, *Ochroconis* and *Phymatotrichopsis*, which show pathogenicity and can be linked with enriched genes in virus. However, such linkages need to be established with more available information in the future.

5.5 Chapter Summary

In summary, our study on epiphytic and free-living microbial communities exhibited significant differences in diversity, structure and function using high-throughput technology. Seaweed-associated genera were more enriched in the epiphytic community compared to the free-living one, such as *Formosa*, *Polaribacter*, *Glaciecola*, which are known to be either isolated from seaweed or able to digest algal polysaccharide. Also, the enriched genes in the epiphytic microbial community were affiliated with adherence, virulence, desiccation stress, defense, quorum sensing and plant hormones, which play crucial roles in biofilm formation, the stabilization of biofilm, defense of other bio-fouling and interaction between epiphyte and seaweed. These results provide valuable insights into interactions between the epiphytic microbial community and its seaweed host, and the importance of microbes in seaweed cultivation ecosystem.

Chapter 6: Conclusions, innovations and prospects

6.1 Conclusions

The major conclusions of this study include:

1. Large-scale cultivation of *G. lemaneiformis* can improve water quality by increasing dissolved oxygen and decreasing nutrients in water column. *Gracilaria* cultivation can significantly influence microbial communities in the water body and sediment through changes in key environmental factors. A conceptual model has been developed to explain the relationships among seaweed processes, microbial communities and their environments in the *Gracilaria* cultivation ecosystem, and can be used in the future studies.

2. Higher abundances of seaweed-associated and algal-degrading microbial groups were observed in water, sediment and epiphytic microbial communities in the *Gracilaria* cultivation zone than in the control zones. The β -diversity of microbial communities increased as the increase of seaweed biomass during *G. lemaneiformis* growth cycle. The increase rate of β -diversity was higher in the *Gracilaria* cultivation zone compared to the control zone. Thus, *G. lemaneiformis* may have selection effects on microbial communities in the *Gracilaria* cultivation ecosystem.

3. The results of this study could help discover molecular mechanisms about how large-scale cultivation of *G. lemaneiformis* influences microbial diversity, composition, structure and function as well as their linkage with environmental factors and ecosystem functioning.

Overall, this study provides important data and scientific evidence to reveal microbial structure and function in mariculture ecosystems.

6.2 Innovations

The innovations of this series of studies are summarized bellows.

1. The effects of large-scale cultivation of *G. lemaneiformis* on structure and function of microbial communities were studied using high-throughput techniques, revealing possible molecular mechanisms in the seaweed-microbe-environment (water and sediment) system.

2. A conceptual model has been developed to explain the relationships among seaweed processes, microbial communities and their environments in *G. lemaneiformis* cultivation ecosystem, and can be used in the future studies.

3. The diversity, structure and function of epiphytic microbial communities on the *Gracilaria* surface were examined by high-throughput technologies including 16S rRNA, ITS gene amplicon sequencing and GeoChip 5.0. The results imply that *G. lemaneiformis* selects its epiphytic communities and possibly determines their functions, indicating possible mechanisms for microbial adaption to the seaweed surface.

6.3 Prospects

Although we have investigated the structure and function of microbial communities in water, sediment and the *G. lemaneiformis* surface in this cultivation ecosystem, many processes and mechanisms are still obscure to uncover the interaction between the seaweed and microbes. In addition, the samples in this study did not cover a whole year, or multiple years, it may cause some difficulties in interpreting the data and explaining the results.

Therefore, more studies need to be done in the future to improve our understanding of the *G. lemaneiformis* cultivation ecosystem, especially the web of food in this complex system.

1. Due to limitation of time and manpower, even though our sampling time across two year, it did not cover one entire year with only eight months of investigation time. In the future study, 3-5 years' monthly sampling should be conducted to understand the temporal turnover of microbial communities at intra-seasonal, seasonal and annual scales, and to explore possible mechanisms related to seaweed-microbe-environment interactions in this *Gracilaria* cultivation ecosystem.

2. Seaweed may have a selection effect to the structure and function of microbial communities even though no direct evidence was found. Further analyses of those datasets may explore if such selection effect exist between seaweed and microbes

using “community assembly approaches”. Of course, more and more bioinformatics (e.g., network analysis) can provide us valuable information when we dig those data more deeply.

3. In this study, we used one seaweed, *G. lemaneiformis*, and its effects on microbial communities. In the future work, we can compare the influence of different seaweeds on microbial communities as well as the divergence of one seaweed at different locations (e.g., southern vs. northern) to reveal the spatial and diverse patterns of water, sediment and epiphytic communities.

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附录 缩写名词对照表

缩写词	英文名	中文名
ANOVA	Analysis of variance	方差分析
CARD-FISH	catalyzed reporter deposition fluorescent <i>in situ</i> hybridization	酶联荧光原位杂交
CCA	Canonical correspondence analysis	典范对应分析
Chl. <i>a</i>	Chlorophyll <i>a</i>	叶绿素 <i>a</i>
CLPP	Community-level physiological profiling	群落水平生理特征
CS	Sediment at GZ	对照区沉积物
CW1m	1m depth of the water at CZ	对照区 1 米深水体
CWB	The bottom water at CZ	对照区底层水体
CWH	Half depth of the water at CZ	对照区水深一半水体
CWS	The surface water at CZ	对照区表层水体
CWT	75% depth of the water at CZ	对照区 75%水深水体
CZ	Control zone	对照区
DGGE	Denatured Gradient Gel Electrophoresis	变性梯度凝胶电泳
DCA	Detrended correspondence analysis	除趋势对应分析
DNA/RNA SIP	DNA/RAN based stable isotope probing	稳定性同位素核酸探针技术
DO	Dissolved oxygen	溶解氧
DOM	Dissolved organic matters	溶解性有机物
FAME	Fatty acid methyl ester	脂肪酸甲酯
FGA	Functional gene array	功能基因芯片
FISH	Fluorescence <i>in situ</i> hybridization	荧光原位杂交
GS	Sediment at GZ	龙须菜栽培区沉积物
GW1m	1m depth of the water at GZ	龙须菜栽培区 1 米深水体
GWB	The bottom water at GZ	龙须菜栽培区底层水体
GWH	Half depth of the water at GZ	龙须菜栽培区水深一半水体
GWS	The surface water at GZ	龙须菜栽培区表层水体

GWT	75% depth of the water at GZ	龙须菜栽培区 75%水深水体
GZ	<i>Gracilaria lemaneiformis</i> cultivation zone	龙须菜栽培区
HAB	Harmful algal blooms	有害藻华爆发
IMTA	Integrated Multi-trophic Aquaculture	多营养级综合养殖
OTU	Operational Taxonomic Unit	分类单元
PCR	Polymerase Chain Reaction	聚合酶链式反应
PERMANOVA	Permutational multivariate analysis of variance	置换多元方差分析
PLFA	Phospholipid fatty acids	磷脂脂肪酸
POM	Particulate organic matters	颗粒性有机物
RDP Classifier	Ribosomal Database Project Classifier	核糖体分类数据库
RFLP	Restriction fragment length polymorphism	限制性片段长度多态性
SEM	Scanning electron microscope	扫描电镜
TC	Total carbon	总碳
TEM	Transmission electron microscope	投射电镜
TN	Total Nitrogen	总氮
TP	Total Phosphorus	总磷
T-RFLP	Terminal restriction fragment length polymorphism	末端限制性片段长度多态性
TS	Total sulfur	总硫
NMDS	Nonmetric multidimensional scaling	非线性多维尺度分析
WC	Water content	含水率

博士研究生期间发表论文及学术交流情况

1. 已发表论文情况

- (1) **Xie, X.**, He, Z.^{*}, Hu, X., Yin, H., Liu, X., Yang, Y.^{*} (2017) Large-scale seaweed cultivation diverges water and sediment microbial communities in the coast of Nan'ao Island, South China Sea. *Science of the Total Environment*, 598: 97-108.
- (2) 解欣斐, 龚映雪, 胡晓娟, 杨宇峰*. 外伶仃海域龙须菜栽培对水质及浮游细菌数量的影响. 暨南大学学报(自然科学与医学版) 2016, 37(3): 185-192.

2. 待发表论文情况

- (1) **Xie, X.**, He, Z., Zhou, J., Yang, Y. Composition and function of the epiphytic microbial community of the red alga *Gracilaria lemaneiformis*. (in preparation)
- (2) **Xie, X.**, He, Z., Zhou, J., Yang, Y. The dynamic of bacterial community assembly in *Gracilaria lemaneiformis* cultivation ecosystem. (in preparation)
- (3) **Xie, X.**, Hu, X., He, Z., Yang, Y. Compositional shifts of bacterial communities associated with mariculture near Nan'ao Island, South China Sea. (in preparation)
- (4) Luo, H., Wang, Q., Nie, X., Ren, H., Shen, Z., **Xie, X.**, Yang, Y. * Heavy metals in the cultivated oyster, *Crassostrea rivularis*, and associated health risks from the typical mariculture zone, South China Sea. *Bulletin of Environmental Contamination and Toxicology* (in revision)

3. 学术交流情况

- (1) 2017 年参加中国海洋湖沼学会藻类学分会第十九次学术讨论会优秀论文评比, 获得“中国藻类学研究优秀论文奖”三等奖.
- (2) 2016 年 8 月参加在加拿大蒙特利尔举办的第十六届国际微生物生态学会会议 (16th ISME), 并做题为“The dynamic of seawater microbial communities in response to seaweed *Gracilaria lemaneiformis* cultivation”的海报展示.
- (3) 2015 年 4 月参加中国环境科学学会海洋环境保护专业委员会与中国海洋湖沼学会水环境分会学术年会, 并做题为“大型海藻龙须菜栽培对细菌群落及水环境的影响”的口头报告, 获得“优秀论文二等奖”.

(4) 2013 年 12 月参加广东省植物学会青年学者学术研讨会，并做题为“汕头市南澳岛养殖海域微生物群落特征”的口头报告，获得“植物生态与环境组”优秀论文一等奖”。

(5) 2015 年 11 月-2016 年 12 月 美国俄克拉荷马大学 环境基因组学研究中心
访问学生

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解欣斐

2017 年 12 月 3 日于暨南园